

# Geotechnical Investigation Report

**Proposed St. Gregory School  
175 Talbot Street, Picton ON**

**Prepared for: Algonquin and Lakeshore Catholic  
District School Board**

**April 20, 2023  
File: 1703.00-101**



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Algonquin and Lakeshore Catholic District School Board – 1 PDF copy

Colbourne & Kembel, Architects Inc. – 1 PDF copy

Malroz Engineering Inc. – 1 PDF copy

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Respectfully submitted,

Malroz Engineering Inc.

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## **1.0 Introduction**

Malroz Engineering Inc. (Malroz) was retained by Colbourne & Kembel, Architects Inc. on behalf of the Algonquin and Lakeshore Catholic District School Board (the Client), to conduct a geotechnical investigation in support of the construction of the proposed new school building at 175 Talbot Street in Picton, Ontario (the Site).

We understand that the proposed development will consist of the construction of a new multi-storey school building. We understand its exact configuration was not known at the time this report was written. We recommend that Malroz be retained to review final designs once they have been developed to confirm that the comments and recommendations outlined in this report have been appropriately implemented and/or accounted for.

This report summarizes the results of the investigation completed by Malroz in support of the proposed development. Work for this investigation was completed in general accordance with Malroz's approved proposal dated October 28, 2022 (ref. 1703.00-100). The agreed scope of work consisted of the following:

1. Advancement of a total of 28 boreholes across the Site within the building addition footprint and within the parking area. A total of six boreholes (within the potential building footprint) would be advanced to 6 m or practical refusal, while all remaining would be advanced to a depth of 3 m or practical refusal;
2. Installation of three PVC piezometer, as well as one return trip to the Site to make groundwater observations;
3. Submittal of all retrieved samples with sufficient quantity for moisture contents testing, as well as up to eight hydrometer grain size analyses, or other index tests as substituted by Malroz at its discretion;
4. Submittal of 10 selected retrieved samples plus one duplicate sample for analysis of environmental soil quality parameters to preliminarily evaluate potential options for excess soil reuse; and
5. Reporting of field and laboratory results.

The recommendations and comments contained herein are based on factual information obtained during the investigation and are intended only for the use of project designers and engineers. They have been prepared with the understanding that the design will be carried out in accordance with applicable codes and standards. The General Conditions and Limitations (Section 7.0 of this report) form an integral part of this report.

## 2.0 Site Description

The Site is presently undeveloped, except for a house present at the northern end of the property, along Talbot Street, and an ancillary structure on the west edge of the Site. The Site is mostly covered in medium to long grasses, having been previously utilized for agricultural purposes. Several other residences and a farm supply store were located adjacent to Talbot Street to the north. The Millennium Trail is located adjacent to the Site's west edge. The Site is surrounded by other agricultural development further to the west and to the east. A new subdivision is being constructed to the south. A large fill pile was observed at the south east corner of the Site.



**Photograph 1 – Partial view of the Site, looking south east towards fill pile  
(December 2022)**

Topography at the Site was locally generally flat, with a slight downward slope to the north near Talbot Street, and upward to the south at its southeast corner.

### 3.0 Method of Investigation

#### 3.1 Fieldwork – Drilling Investigation

A total of 28 boreholes, identified as BH101 through BH128 were advanced between December 1 and December 7, 2022 in the vicinity of the proposed development based on sketches provided to Malroz prior to the field investigation. A Borehole Location Plan (Figure 1) is presented in Appendix A. All boreholes were advanced using a CME-55 drill rig on a rubber tracked carrier, equipped with hollow stem continuous flight augers. Three boreholes locations, BH105, BH109, and BH121, were equipped with a 50 mm diameter PVC monitoring wells to assess stabilized static groundwater levels in the vicinity of the proposed development. Locations and ground surface elevations at each borehole was surveyed with a Trimble R10 Global Navigation Satellite System survey system, connected to the Can-Net Virtual Reference Station network, and are assumed to be accurate within the operating limits of the device. Survey information is summarized in the following table.

**Table 1 – Borehole Locations and Ground Surface Elevations**

| Test Location | Northing  | Easting  | Elevation (mASL) |
|---------------|-----------|----------|------------------|
| BH101         | 4874909.2 | 326770.1 | 101.4            |
| BH102         | 4874865.7 | 326745.1 | 103.1            |
| BH103         | 4874840.9 | 326716.1 | 103.0            |
| BH104         | 4874825.1 | 326732.7 | 102.9            |
| BH105         | 4874801.7 | 326680.7 | 102.7            |
| BH106         | 4874793.6 | 326694.7 | 102.5            |
| BH107         | 4874799.3 | 326703.2 | 102.6            |
| BH108         | 4874811.9 | 326713.1 | 102.7            |
| BH109         | 4874802.4 | 326740.8 | 103.1            |
| BH110         | 4874797.3 | 326738.3 | 103.0            |
| BH111         | 4874807.8 | 326766.8 | 103.0            |
| BH112         | 4874777.3 | 326692.6 | 102.6            |
| BH113         | 4874781.8 | 326704.6 | 102.8            |
| BH114         | 4874778.9 | 326738.0 | 103.5            |
| BH115         | 4874787.5 | 326754.5 | 103.4            |
| BH116         | 4874796.7 | 326764.3 | 103.3            |
| BH117         | 4874791.2 | 326774.8 | 103.6            |
| BH118         | 4874768.3 | 326723.1 | 103.4            |
| BH119         | 4874779.0 | 326755.8 | 103.8            |
| BH120         | 4874766.7 | 326732.2 | 103.8            |

| Test Location | Northing  | Easting  | Elevation<br>(mASL) |
|---------------|-----------|----------|---------------------|
| BH121         | 4874764.3 | 326756.4 | 105.0               |
| BH122         | 4874828.0 | 326753.4 | 103.0               |
| BH123         | 4874791.1 | 326789.5 | 104.1               |
| BH124         | 4874766.9 | 326780.9 | 105.7               |
| BH125         | 4874742.1 | 326773.1 | 107.3               |
| BH126         | 4874751.1 | 326738.8 | 105.2               |
| BH127         | 4874737.8 | 326705.3 | 104.8               |
| BH128         | 4874746.6 | 326672.8 | 103.5               |

Input By: DPH  
Validated By: MS

At each borehole location, soil samples were collected while performing the Standard Penetration Test (SPT) in general accordance with the procedure as described in ASTM D1586. This consisted of freely dropping a 63.5 kg (140 lb) hammer from a vertical distance of 0.76 m (30 in), to drive a 51 mm (2 in) outer diameter split-barrel (split spoon) sampler into the ground. The number of blows of the hammer required to drive the sampler into the relatively undisturbed ground 300 mm (12 in) was recorded as the SPT 'N' value, which correlates to the relative density of non-cohesive soils and is, in certain cases, indicative of the consistency of cohesive soils. Each retrieved sample was placed in a resealable plastic bag.

Soil samples were also collected for analytical testing for the purposes of preliminary assessment of soil quality to support future excess soil management at the Site. Samples collected for analytical testing were placed into laboratory provided containers. Each sample was screened for headspace vapours from resealable plastic bags using an RKI Eagle combustible gas indicator (CGI) and photoionization detector (PID). Vapours were screened from the headspace of the Ziploc bags once the samples had equilibrated to room temperature. Gaseous vapours assessed in full gas response mode in the field and from headspace were generally measured as no response or less than 1% of the lower explosive limit (LEL). Organic vapour concentrations were measured as no response or 1 ppm.

Borehole logs are attached in Appendix B.

### **3.2 Fieldwork – Geophysical Investigation**

In-situ measurement of shear wave velocity at the Site was conducted by a subconsultant (Frontwave Geophysics) retained by Malroz on February 2, 2023. A detailed report outlining the methodology and summarizing the results of the testing is presented in Appendix E.

## **4.0 Subsurface Assessment**

Details of the subsurface conditions encountered are presented in the borehole records presented in Appendix B of this report. We strongly emphasize, however, that the soil types, their sequence, thickness, and physical properties may vary between boreholes, sample and test locations, both vertically and horizontally. This information is solely for general planning purposes and should not be used for detailed quantity take-offs.

Our assessment of subsurface conditions and borehole observations are summarized in the following sections.

### **4.1 Local Geology / Physiography**

The Ontario Geologic Survey characterized bedrock at the subject site as the Lindsay Formation (Simcoe Group) comprising nodular limestone and shale (Armstrong and Dodge, 2007; OGS, 2010a). Surficial geology of the subject site is mapped as glaciolacustrine shallow water deposits comprising massive to bedded sand and silt; however, drumlins comprising silty and sandy till are mapped to the northwest of the site, and deeper water deposits comprising silt and clay are mapped to the southwest (OGS, 2010b & 2011). Physiography of the area is described as sand plains with drumlins and eskers in the vicinity of the site (Chapman and Putnam, 2007).

### **4.2 Surficial Materials**

#### **4.2.1 Topsoil**

While topsoil was not observed during drilling in the agricultural area that makes up most of the work area, it should be expected that organics and disturbed material will be encountered extending to at least 0.3 m below existing site grades. In our experience, organic soils and topsoil thickness can vary significantly between boreholes and across a Site, and are often thicker (in addition to the presence of root balls) in areas where there was prior tree growth.

#### 4.2.2 Native Clayey Silt

A native clayey silt with traces to some sand and in some cases trace gravel was observed at boreholes BH101, BH103 through BH115, BH117 through BH119, BH122 and BH123. This material was visually described as being generally moist. Moisture contents measured on recovered samples of this material varied from 2 to 35% by dry weight. SPT 'N' values of 1 to 54 blows per 300 mm of penetration were measured in this material, which is an estimate of consistency ranging from soft to hard.

#### 4.3 Native Sand to Silty Sand

A brown native sand to silty sand, with traces to some clay and traces to some gravel was observed at all borehole locations. The sand at this Site is generally a bedded sand, overlying a sandy till, however lenses of the bedded sand were observed within the till and clayey soils. This material was visually described as being generally moist. Moisture contents measured in this material ranged from 13 to 33% by dry weight. SPT 'N' values of between 2 and 86 blows per 300 mm of penetration were measured in this material, which is indicative of a relative density ranging from compact to very dense.

#### 4.4 Groundwater

A detailed hydrogeological investigation was not included as a part of this work. However, static groundwater levels were observed in the monitoring wells installed at BH105, BH109, and BH121 on January 30, 2023. The water levels measured from the monitoring wells are summarized in the following table.

**Table 2 – Groundwater Observation (January 30, 2023)**

| Borehole ID | Groundwater Observation<br>(mbg/mASL) |
|-------------|---------------------------------------|
| BH105       | 0.4 / 102.3                           |
| BH109       | 0.5 / 102.6                           |
| BH121       | 0.6 / 104.4                           |

Input By: DPH  
Validated By: MS

Groundwater levels should be expected to fluctuate based on seasonal atmospheric conditions and heavy precipitation events.

## 5.0 Laboratory Testing

### 5.1 Geotechnical Laboratory

Samples recovered during drilling were transported to Malroz's geotechnical laboratory in Kingston, Ontario. Moisture contents were measured on all samples with sufficient available quantity to test. Additionally, a total of eight hydrometer grain size analyses have been submitted for analysis as deeper soils were encountered than initially expected at the Site. The results of the hydrometer grain size analyses are summarized in the following table. Detailed laboratory reports are included in Appendix C.

**Table 3 – Summary of Gradation Testing Results**

| Sample ID | Gravel (%) | Sand (%) | Silt (%) | Clay (%) |
|-----------|------------|----------|----------|----------|
| BH105-SS3 | 0          | 1        | 81       | 18       |
| BH105-SS6 | 6          | 55       | 31       | 8        |
| BH109-SS3 | 0          | 1        | 81       | 18       |
| BH109-SS5 | 3          | 36       | 50       | 11       |
| BH109-SS6 | 5          | 59       | 27       | 9        |
| BH119-SS3 | 20         | 33       | 38       | 9        |
| BH121-SS3 | 2          | 61       | 29       | 8        |
| BH128-SS2 | 9          | 76       | 12       | 3        |

Input By: DPH  
Validated By: TB

Moisture contents are summarized in borehole logs in Appendix B.

### 5.2 Analytical Laboratory

Eleven samples (including one duplicate) were selected for laboratory analysis to provide a preliminary characterization of the environmental soil quality. Samples were selected to provide general lateral coverage of the site with emphasis on providing representativeness from shallow soils within the anticipated building footprint. Duplicate samples were collected for analyses at a rate of approximately 10% for quality assurance/quality control (QA/QC).

Samples were submitted to an external analytical laboratory Paracel Environmental Laboratories (Paracel), for selective analyses of metal, electrical conductivity (EC), sodium adsorption ratio (SAR), petroleum hydrocarbons (PHC F1-F4), and benzene, toluene, ethylbenzene and xylene (BTEX) parameters as described in the Soil Rules.

Due to the agricultural setting of the property, pesticide parameters were also evaluated on several samples. Select samples were assessed for pH for the purpose of confirming applicable reference standards. Soil analytical results for the Site are summarized in Table D1 and discussed further in Section 6.8. Laboratory certificates of analysis are provided in Appendix D.

## **6.0 Discussion and Recommendations**

Based on the results of the field investigation, we anticipate that due to the high groundwater levels observed around the Site, and depending on final building elevations, foundations may be consist of conventional spread, strip or pad foundations constructed on competent native soils, or on pneumatically installed deep foundation elements such as micropiles or caissons. We would recommend against selection of driven elements, both of displacement (e.g. H piles) and non-displacement types (e.g. pipe piles), in addition to helical piers or drilled caissons, which may encounter problems during installation due to obstructions typically observed within the silty to sandy glacial (till) deposits. Any open excavations on the Site may require significant/active dewatering methods due to the high groundwater at the Site, however further study would be necessary to characterize hydrogeological conditions and estimate potential flow rates at the Site.

### **6.1 Excavation and Temporary Shoring**

Excavations up to a depth of approximately 1.5 to 2.5 m from the existing site grades may be necessary for construction of foundations, or for buried utilities. We would expect that sufficient area would be available to backslope excavations to this depth without the use of shoring at this Site. Once foundations are constructed, excavations should not extend within a zone extending 10 horizontal to 7 vertical from the bottom outside edge of these foundations, to prevent them from becoming undermined. Groundwater must be lowered to at least 0.5 m below the base of the lowest excavation to ensure a stable working surface.

All excavations and construction of any shoring should be carried out in accordance with the latest edition of the OHSa and Regulations for Construction Projects. The OHSa regulations require that if workers must enter an excavation deeper than 1.2 m, the excavation must be suitably sloped and/or braced in accordance with OHSa requirements. OHSa specifies the maximum slope of excavations into four broad soil types, summarized as follows:

**Table 4 – OHSA Soil Types**

| <b>Soil Type</b> | <b>Maximum Slope Inclination</b> | <b>Base of Slope Location</b>      |
|------------------|----------------------------------|------------------------------------|
| Type 1           | 1 horizontal to 1 vertical       | Within 1.2 m of base of excavation |
| Type 2           | 1 horizontal to 1 vertical       | Within 1.2 m of base of excavation |
| Type 3           | 1 horizontal to 1 vertical       | From bottom of excavation          |
| Type 4           | 3 horizontal to 1 vertical       | From bottom of excavation          |

At this Site, the upper native silty clay to clayey silt and silty sand can be considered as Type 3, while lower native silty sand glacial deposits can be considered Type 2, provided they are unaffected by seepage. Any soils affected by seepage must be considered as Type 4, and the lowest soil type in any given excavation shall govern its side slopes.

Excavations into overburden soils should be relatively easy using conventional excavating equipment; however, contractors should be prepared to manage erratic large particles up to boulder sized material that could be found within the overburden (particularly within the glacial silty clayey sand at this Site).

Stockpiles of excavated materials should be kept away from the edges of open excavations by a distance at least equivalent to the depth of the excavation to avoid slope instability. Care should be taken to avoid overloading any underground services/structures from any construction stockpiles. It should be noted that this distance is also applicable to the passage of heavy machinery near excavations. This condition should be respected at all times, unless specific studies are conducted for individual cases.

A shoring system will be required anywhere appropriate back sloping is not possible, (e.g. potentially in utility trench excavations). Shoring systems must be designed by a professional engineer licensed in the province of Ontario, in accordance with relevant codes, standards and regulations such as the latest version of the Canadian Foundation Engineering Manual and the Ontario Occupational Health and Safety Act (OHSA) Regulations for Construction Projects. The system should be designed to resist full earth and hydrostatic pressures, as well as surcharges due to construction and highway

traffic loadings. The following parameters may be used in the design of temporary shoring:

**Table 5 – Lateral Earth Pressure Parameters**

| Soil Type                      | Bulk Unit Weight (kN/m <sup>3</sup> ) | Angle of Internal Friction (deg.) | Coefficient of Lateral Earth Pressure |                |                |
|--------------------------------|---------------------------------------|-----------------------------------|---------------------------------------|----------------|----------------|
|                                |                                       |                                   | K <sub>a</sub>                        | K <sub>o</sub> | K <sub>p</sub> |
| Native Silty Clay              | 19.0                                  | 29                                | 0.35                                  | 0.52           | 2.88           |
| Native Silty Sand              | 21.0                                  | 32                                | 0.31                                  | 0.47           | 3.25           |
| Native Glacial Deposits (Till) | 21.0                                  | 34                                | 0.28                                  | 0.44           | 3.54           |
| Compacted Gran. 'A'            | 21.0                                  | 34                                | 0.28                                  | 0.44           | 3.54           |
| Compacted Gran. 'B' Type II    | 21.5                                  | 35                                | 0.27                                  | 0.43           | 3.69           |

At rest parameters should be utilized for non-yielding shoring elements. We recommend shoring plans be reviewed by a geotechnical engineer prior to the start of construction.

## 6.2 Dewatering

Based on the observed groundwater levels between 0.4 and 0.6 mbg in the installed monitoring wells, groundwater infiltration into nearly all open excavations should be expected. Shallow excavations, e.g. those up to 1 m or so may be controllable with one or more conventional filtered sumps and construction pumps and/or passive relief wells installed below the lowest point of the excavation. Deeper excavations may produce higher quantity of infiltration into open excavations, and necessitate use of active dewatering techniques such as vacuum well points. Use of cut-off methods can also be considered, e.g. installation of tightly interlocking sheeting (though obstructions in glacial soils may cause installation issues with this method), particularly if any excavation is required to be shored if adequate area to backslope is not possible. Carrying out excavations in dry seasonal conditions will reduce dewatering requirements. Further in-situ testing would be necessary to quantify groundwater flow volumes into open excavations.

## **6.3 Foundations**

### **6.3.1 Conventional Shallow Foundations**

Conventional spread, strip or pad foundations constructed on competent native soils at an elevation of 101.1 or below, can be designed for a factored geotechnical bearing reaction at the ultimate limit state (ULS) of 300 kPa and a geotechnical bearing reaction 200 kPa at the serviceability limit state (SLS). This assumes that there will be no load eccentricities and the foundations will be constructed on flat and level ground. If higher bearing reactions are required, excavations could be carried deeper into the deeper glacial soils, however as mentioned above groundwater control may become problematic.

The native soils at this Site are susceptible to disturbance (e.g. due, but not limited to freezing, and repeated loading from construction traffic). Contractors should make provisions to ensure the subgrade is not exposed to these conditions. All subgrade surfaces should be inspected by a geotechnical engineer or their designate prior to placement of foundations.

### **6.3.2 Micropiles**

The use of pneumatically installed micropiles, while more costly than conventional shallow foundations, may eliminate the need for significant dewatering while still providing adequate and potentially improved bearing reactions. Grout to soil bond capacity of between 90 to 200 kPa would likely be achievable at this Site. Exact values and sealed shop drawings would be obtained by specialist geostructural contractors who would perform the installation of these deep foundation elements. A load testing program prior to the start of construction may provide improved capacities beyond those available using empirical data only. Verification load testing at the time of construction should be conducted, including creep testing.

### **6.3.3 Design for Earthquakes**

In accordance with the Ontario Building Code 2012 (OBC), foundations must be designed to withstand a minimum earthquake force. Based on the measured average in-situ shear wave velocity of 794 m/s, the Site Classification for Seismic Site Response

would be C (unless base of foundations or top of pile caps are located within 3 m of bedrock surface in which case Class B could be considered, however this is not likely practicable at this Site).

#### **6.3.4 Liquefaction**

Based on the encountered subsurface conditions and in-situ and laboratory testing conducted, summarized in sections above, the Site soils would not be considered liquefiable.

#### **6.3.5 Foundation Dewatering**

Generally, provided surficial slabs are located 0.2 m above existing site grades, we would not consider under slab drainage a necessity. However, given the close proximity of the observed static groundwater levels at this Site, we would recommend consideration of an underslab drainage network. Perimeter drainage around the structure and any below grade levels should be provided, tied to a frost-free outlet. If a basement is to be constructed, complete structural waterproofing such as a blindside system or a 'tanked' construction would likely be necessary. Surface grades should be sloped away from the building, and landscaped areas directly adjacent to the building should be capped with impermeable soils, if practicable.

#### **6.3.6 Frost Protection**

Design frost depth in the Picton area should be considered 1.4 m below unprotected surface grades. If thermal insulation is utilized as opposed to soil cover, some manufacturers recommend placement of foundations directly over top of insulating boards.

## **6.4 Backfilling**

Backfill around the building should consist of non-frost susceptible, free draining granular material. Below hardscaped areas, all backfill material should be compacted to a minimum of 98% of its Standard Proctor Maximum Dry Density (SPMDD) to 1 m below the hardscape base or pavement subbase, and 100% thereafter. Elsewhere, all backfill should be compacted to 95% of its SPMDD.

Earth pressure parameters provided in Table 5 above can be used for any walls subjected to such pressures (e.g., for sump pits, etc.).

## **6.5 Slabs on Grade**

A modulus of subgrade reaction of 30 MPa/m, based on a uniformly loaded circular plate with diameter 0.76 m (30 in.) loaded to a deflection of 1.25 mm, at an elevation of approximately 102.8. Malroz can provide updated values based on final slab elevations. We recommend that a minimum of 150 mm of new Ontario Provincial Standard Specifications (OPSS) Granular 'A' be placed under any slabs on grade.

## **6.6 Pavement Design**

For any bus access laneways we recommend considering a pavement section consisting of 50 mm of HL3 (or SP12.5 Cat. B) and 60 mm of HL8 (or SP19.0 Cat. B), 200 mm of OPSS Granular 'A' and 300 mm of OPSS Granular 'B'. A pavement consisting of 60 mm of HL3 (or SP12.5 Cat. A), overlying 150 mm of Granular 'A' and 250 mm of Granular 'B' can be considered for light vehicle parking areas.

### **6.6.1 Materials and Construction Considerations**

For Site preparation, surface vegetation, tree roots, topsoil and organics should be stripped under the pavement areas, and the underlying subgrade surface proof rolled. Some limited amount of organics in these areas may be able to remain, subject to review by a geotechnical engineer or their designate (e.g. pavement areas would not require all organics to be necessarily removed, provided the Client understands there may be some longer term maintenance risk, albeit likely minor following a review and

successful proof roll). New fill under pavement areas should be placed as an Engineered Fill operation as described in the section below.

Paving work should be completed in accordance with the requirements of applicable OPSS and municipal standards. All asphalt mix designs should be reviewed prior to the commencement of construction.

HMA used in this project should meet the minimum requirements of OPSS 1150/1151 (depending on whether Marshall or Superpave mixes are utilized). Asphalt cements should be minimum grade of PG 58-28 and meet the requirements of OPSS 1101.

Tack coat should be applied between any vertical surfaces or joints including curbs, abutting and walls, etc., butt and lap joints and at all tie-ins to other existing asphalt. SS-1 emulsified asphalts used for this purpose should meet requirements of applicable OPSS.

## **6.7 Engineered Fill**

Engineered Fill application may be required on this project to raise subgrade elevations under foundations and/or during construction of slabs on grade. For any operation to be considered Engineered Fill, the following criteria must be satisfied:

- Materials used as Engineered Fill must be uniform and homogenous. The material should be free of deleterious materials and organics;
- Prior to the placement of Engineered Fill, it must be assessed in a geotechnical laboratory for, at a minimum, gradation and Standard Proctor analyses;
- The material must be within +/- 2% of its optimum moisture content, as determined through laboratory testing;
- Engineered Fill operations must take place under the supervision of a geotechnical engineer or their designate;
- Suitable compaction equipment must be selected for the operation, based on the material to be compacted;
- Materials should be placed in lifts which are suitable for the compaction equipment utilized, but generally not greater than 0.2 m loose lifts;

- Density testing must be taken on each lift of Engineered Fill. Any Engineered Fill which is tested and found to be outside of the specified density range shall be either removed, reworked, or retested; and
- Under no circumstances shall frozen material be placed in any Engineered Fill operation.
- Site activities may be carried out in the winter months and periods of cold weather. The following procedures are recommended for the placement of engineered or structural fill during cold weather conditions:
  - The fill placement must be inspected by qualified field personnel on a full time basis under the supervision of a geotechnical engineer, with the authority to stop the fill operations at any time when conditions are considered to be unfavorable.
  - The intended area of fill must be clearly identified in the field prior to commencing the work.
  - Ramps for access must be constructed outside of the limits of intended fill.
  - The fill material must be comprised of OPSS Granular 'B' - Type 1 or 2, or other approved equivalent granular material when temperatures are below 0° C.
  - Imported fill containing ice, snow or any frozen material cannot be accepted for use and should be returned to the source.
  - Overnight frost penetration into the fill mantle in areas below intended structure construction, parking lots, and on side slopes of the fill mantle within this area must be prevented by using insulation blankets. Any frozen fill must be completely removed prior to placing subsequent lifts. Breaking the frost and frozen fill in situ will not be accepted. Following removal of frozen materials, the remaining exposed surface should be compacted to a minimum of 98% of SPMDD prior to placement of new fill.
  - During periods of cold where ambient temperatures are -5° C or less, placement of engineered fill shall stop, and the existing fill materials must be protected from frost penetration.

It should be noted that the placement of engineered or structural fill materials during cold weather conditions requires extra effort beyond that typical in better climatic

2conditions. At any time where conditions are deemed unfavourable, the Engineered Fill operation must be suspended.

## **6.8 Preliminary Soil Quality Characterization Summary**

The soil quality characterization program was undertaken to obtain preliminary information to support future excess soil management activities. Additional excess soil planning activities (e.g. an Assessment of Past Use and Sampling and Analysis Plan) and further characterization may need to be completed if the project will require registration on the Excess Soil Registry. The Project Leader should evaluate whether registration will be required once additional design details regarding the anticipated volume of excess soil are known.

Applicable environmental standards were selected by referencing the excess soil quality standards published by the MECP in the Soil Rules (2022), pursuant to O. Reg. 406/19. Potential excess soil reuse sites have not been identified at the time of issue of this report. However, based on anticipated property uses for potential receiving sites, sample results were compared to the following standards:

- Table 1 background standards for residential/parkland/institutional/industrial/commercial/community property use,
- Table 2.1 volume independent standards for residential/parkland/institutional (R/P/I) property use, and
- Table 2.1 volume independent standards for industrial/commercial/community (I/C/C) property use.

In general, Table 2.1 I/C/C standards are the least stringent of the foregoing standards, and the Table 1 standards are the most stringent. Once one or more reuse sites have been identified, the Project Leader must confirm appropriate reuse standards in consultation with a Qualified Person (QP), as defined by O. Reg. 406/19. This may require comparison of data to additional/alternate reference standards.

Soil analytical results for the Site are compared to the volume independent standards in Table D-1 (Appendix D). The pH results for the submitted samples was approximate 7.3, which falls within the acceptable range for surface and subsurface soil for the

application of the generic site condition standards (MECP, 2022). Results generally met the Table 1 background standards with the following exception:

- Silver exceeded the Table 1 standards at BH111-S01.

One duplicate sample was submitted to Paracel for the suite of analyses described in Section 5.2. The duplicate sample was retrieved from BH103. Results of the duplicate sample analyses were generally in agreement for the analyzed parameters. Based on these results, we are of the opinion that these results may be relied upon.

Soils from the characterized area are generally expected to be suitable for beneficial reuse at sites where the Table 2.1 R/P/I or I/C/C standards apply. However, if excess soils will be generated from the property, further characterization of soils should be considered and a QP should be consulted, particularly if excess soils will be generated from the area of the residence or near other potentially contaminating activities.

Where excess soil will be reused, the Project Leader and contractor must ensure that excess soils are adequately represented by the samples as identified in the above soil management cells and ensure that soil types and conditions are consistent with the observations provided in this Preliminary Soil Quality Characterization Summary. The soil quality, quantity, and use must be compatible with the reuse site setting and any applicable site instruments (e.g. permits, licenses) and should be confirmed by a QP prior to transferring the soil to the reuse site. Written consent for reuse of the soils must be obtained from the reuse site by the Project Leader.

Throughout the work, the contractor should take all necessary precautions to keep soil separated from asphaltic materials and any other non-soil materials encountered during the work.

If evidence of contamination is encountered during excavation (e.g., stained or discoloured earth, garbage or waste, oily residue on soils, drums or containers, sheen/product/film, cinders/ash, etc.), excavation must be halted and the contract authority should be immediately notified so that soils may be re-evaluated by Malroz or an alternate QP. If evidence of contamination is encountered, appropriate efforts should be undertaken to mitigate potential contaminant migration from erosion or

sedimentation, including but not limited to covering stockpiles with an impermeable barrier (e.g., polyethylene sheeting).

## **7.0 General Conditions and Limitations**

This report was completed for the specific needs of The Algonquin and Lakeshore Catholic District School Board and is based on a specific scope of work which is defined in the mutually agreed upon workplan. The scope of work has limitations as described throughout the report and in the notice to reader. Data, tables, charts, and interpretive illustrations presented in this document are instruments of service for this mandate and can only be properly evaluated when reviewed together with the accompanying report. Reference to this report should only be made to the complete signed document.

By issuing this report, Malroz is the Geotechnical Engineer of Record for this project. It is recommended that Malroz be retained during construction of all foundations, for earthwork operations and for paving. The intent of this requirement is to verify conditions encountered during construction are consistent with the findings in the report and, that inherent knowledge developed as a part of our study is correctly carried forward to construction phases. We should be retained to review whether our recommendations have been applied appropriately, once drawings and specifications are complete. Without this review, Malroz will not be liable for any misunderstanding of our recommendations or their application and adaptation into final designs.

The work performed in this report was carried out in accordance with the terms and conditions made as a part of our proposal and/or contract pursuant to which this report was issued, in a manner consistent with that level of care and skill ordinarily exercised by members of the Geotechnical Engineering profession currently practicing under similar conditions in the same locality. The conclusions presented in the report are based solely upon the scope of services, governed by the time and budgetary considerations to which this work was subject.

The factual data, recommendations and comments in this report pertain to the specific project as described in the report, and are not applicable to any other project or location. If the project is conceptually modified or changes location, or if it is not initiated within twelve months of the date of this report, Malroz should be given an opportunity to confirm that the information in this report is still valid and/or applicable.

The comments in this report are intended only for the guidance of project designers and engineers. Contractors bidding on or undertaking the work should rely on their own

investigations, as well as their own interpretations of the factual borehole and in-situ test information, and how subsurface conditions may affect their work.

This report must be read as a whole, as sections taken out of context can be misleading. Drafts and working copies, whether or not marked as “draft”, “for discussion purposes” or otherwise, do not necessarily reflect Malroz’s final opinion following consideration of all matters which are subject to the study giving rise thereto; they are issued for comment and information purposes only, and are subject to change and should not be relied upon in any way or for any purpose.

It is important to emphasize that a soil investigation is, in fact, a random sampling of a site and the comments included in this report are based solely on the results obtained at the borehole locations only. Soil and groundwater conditions between and beyond the borehole locations may differ both horizontally and vertically from those encountered at the borehole locations and may become apparent during construction, which could not be detected or anticipated at the time of our investigation. Should any conditions at the site be encountered which differ from those found during this investigation, we request that we be notified immediately in order to permit a reassessment of our comment and recommendations. If changed conditions are identified during construction, no matter how minor, the recommendations in this report shall be considered invalid until sufficient review and written assessment of said conditions by Malroz has been completed.

## 8.0 Closure

We trust that this report meets your present requirements. Please do not hesitate to contact us should there be any further questions or comments.

Respectfully Submitted,

Malroz Engineering Inc.,



A handwritten signature in black ink, appearing to read "D. Hill".

per: Dylan Hill, P. Eng.  
Geotechnical Engineer, Project Manager

A handwritten signature in black ink, appearing to read "D. Hodgson".

reviewed: David Hodgson, P.Eng.  
Senior Engineer, Principal

## 9.0 References

Armstrong, D.K. and Dodge, J.E.P. (2007). Paleozoic geology of southern Ontario; Ontario Geological Survey, Miscellaneous Release Data 219.

Canadian Geotechnical Society. (2006). Canadian Foundation Engineering Manual, 4<sup>th</sup> Edition.

Chapman, L.J. and Putnam, D.F. (2007). Physiography of southern Ontario; Ontario Geological Survey. Miscellaneous Release Data 228.

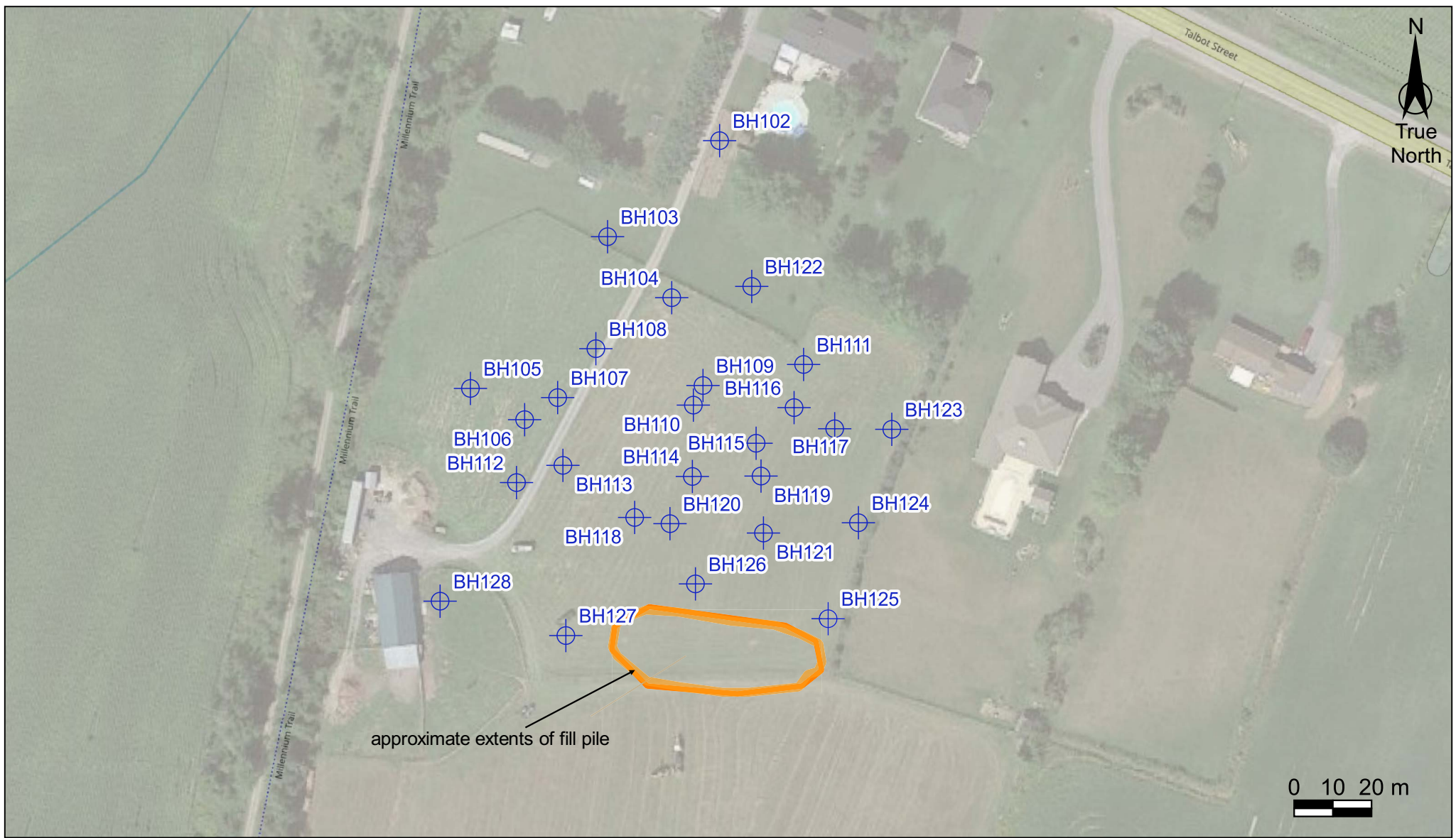
Ontario Geological Survey (2010a). Bedrock geology of Ontario; Ontario Geological Survey. Miscellaneous Release Data 126 revisions 1, scale 1:250,000.

Ontario Geological Survey (2010b). Surficial geology of Southern Ontario; Ontario Geological Survey. Miscellaneous Release Data 128.

Ontario Ministry of the Environment, Conservation and Parks (2022). Rules for Soil Management and Excess Soil Quality Standards.

## **Appendix A**

### **Figures**



**Legend**

 BH Locations

 Fill Pile

|                                                    |            |             |                                                                                       |      |
|----------------------------------------------------|------------|-------------|---------------------------------------------------------------------------------------|------|
| D0                                                 | 2023-02-01 | for issue   | CM                                                                                    | DPH  |
| Rev                                                | Date       | Description | By                                                                                    | Chkd |
| Site Layout                                        |            |             |                                                                                       |      |
| St Gregory School<br>175 Talbot Street, Picton, ON |            |             |                                                                                       |      |
| File: 1703.00-101                                  |            | Figure<br>1 |  |      |

**Appendix B**  
**Record of Boreholes**

# BOREHOLE LOG

Enclosure No.: 1

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Borehole Number: <b>BH101</b>          |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 7, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 7, 2022</b> |  | Elevation: <b>101.4 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326770.1 m</b>             |  | Northing: <b>4874909.2 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |

| Depth BGS (m) | Elevation (m) | Stratigraphy | Description                                                                       | Sample Type | Sample Number | Recovery (%) | Penetration Index / RQD | Shear Strength (kPa) |     |     |     |
|---------------|---------------|--------------|-----------------------------------------------------------------------------------|-------------|---------------|--------------|-------------------------|----------------------|-----|-----|-----|
|               |               |              |                                                                                   |             |               |              |                         | 50                   | 100 | 150 | 200 |
| 0.2           |               |              | <b>CLAYEY SILT</b><br>some sand, trace rootlets, brown, moist, hard.              |             |               |              |                         |                      |     |     |     |
| 0.4           |               |              |                                                                                   |             |               |              |                         |                      |     |     |     |
| 0.6           | 100.8         |              | <b>SAND</b><br>some silt, trace gravel, trace clay, brown, moist to wet, compact. |             |               |              |                         |                      |     |     |     |
| 0.8           |               |              |                                                                                   |             |               |              |                         |                      |     |     |     |
| 1.0           |               |              |                                                                                   |             |               |              |                         |                      |     |     |     |
| 1.2           | 100.2         |              | <b>SILTY SAND</b><br>trace gravel, trace clay, brown, wet, very dense.            |             |               |              |                         |                      |     |     |     |
| 1.4           |               |              |                                                                                   |             |               |              |                         |                      |     |     |     |
| 1.6           |               |              |                                                                                   |             |               |              |                         |                      |     |     |     |
| 1.8           |               |              |                                                                                   |             |               |              |                         |                      |     |     |     |
| 2.0           |               |              |                                                                                   |             |               |              |                         |                      |     |     |     |
| 2.2           |               |              |                                                                                   |             |               |              |                         |                      |     |     |     |
| 2.4           |               |              |                                                                                   |             |               |              |                         |                      |     |     |     |
| 2.6           |               |              |                                                                                   |             |               |              |                         |                      |     |     |     |
| 2.8           | 98.7          |              | End of borehole at 2.7 mbg by sampler refusal on inferred cobbles and boulders.   |             |               |              |                         |                      |     |     |     |
| 3.0           |               |              |                                                                                   |             |               |              |                         |                      |     |     |     |
| 3.2           |               |              |                                                                                   |             |               |              |                         |                      |     |     |     |

NOTES: Borehole open upon completion

# BOREHOLE LOG

Enclosure No.: 2

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Borehole Number: <b>BH102</b>          |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 7, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 7, 2022</b> |  | Elevation: <b>103.1 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326745.1 m</b>             |  | Northing: <b>4874865.7 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |

| Depth BGS (m) | Elevation (m) | Stratigraphy | Description                                                                        | Sample Type | Sample Number | Recovery (%) | Penetration Index / RQD | Shear Strength (kPa)                                                               |      |     |      |
|---------------|---------------|--------------|------------------------------------------------------------------------------------|-------------|---------------|--------------|-------------------------|------------------------------------------------------------------------------------|------|-----|------|
|               |               |              |                                                                                    |             |               |              |                         | 50                                                                                 | 100  | 150 | 200  |
|               |               |              |                                                                                    |             |               |              |                         |  |      |     |      |
| 0.2           |               |              | <b>SILTY SAND</b><br>trace clay, trace rootlets, dark brown, moist, loose.         |             |               |              |                         |                                                                                    |      |     |      |
| 0.4           |               |              |                                                                                    |             | SS1           | 84           | 5                       | ● 5                                                                                | ○ 18 |     |      |
| 0.6           |               |              |                                                                                    |             |               |              |                         |                                                                                    |      |     |      |
| 0.8           | 102.3         |              | <b>SILTY SAND</b><br>trace gravel, trace clay, grey, moist, compact to very dense. |             |               |              |                         |                                                                                    |      |     |      |
| 1.0           |               |              |                                                                                    |             | SS2           | 100          | 29                      | ○ 8                                                                                | ● 29 |     |      |
| 1.2           |               |              |                                                                                    |             |               |              |                         |                                                                                    |      |     |      |
| 1.4           |               |              |                                                                                    |             |               |              |                         |                                                                                    |      |     |      |
| 1.6           |               |              |                                                                                    |             |               |              |                         |                                                                                    |      |     |      |
| 1.8           |               |              |                                                                                    |             | SS3           | 100          | 39                      | ○ 5                                                                                | ● 39 |     |      |
| 2.0           |               |              |                                                                                    |             |               |              |                         |                                                                                    |      |     |      |
| 2.2           |               |              |                                                                                    |             |               |              |                         |                                                                                    |      |     |      |
| 2.4           |               |              |                                                                                    |             |               |              |                         |                                                                                    |      |     |      |
| 2.6           |               |              |                                                                                    |             |               |              |                         |                                                                                    |      |     |      |
| 2.8           |               |              |                                                                                    |             | SS4           | 100          | 86                      | ○ 7                                                                                |      |     | ● 86 |
| 3.0           | 100.1         |              | End of borehole at target depth of 3.0 mbg.                                        |             |               |              |                         |                                                                                    |      |     |      |
| 3.2           |               |              |                                                                                    |             |               |              |                         |                                                                                    |      |     |      |

NOTES: Borehole open upon completion



# BOREHOLE LOG

Enclosure No.: 3

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Borehole Number: <b>BH103</b>          |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 7, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 7, 2022</b> |  | Elevation: <b>103.0 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326716.1 m</b>             |  | Northing: <b>4874840.9 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |

| Depth BGS (m) | Elevation (m) | Stratigraphy | Description                                                                         | Sample Type | Sample Number | Recovery (%) | Penetration Index / RQD | Shear Strength (kPa) |     |     |     |
|---------------|---------------|--------------|-------------------------------------------------------------------------------------|-------------|---------------|--------------|-------------------------|----------------------|-----|-----|-----|
|               |               |              |                                                                                     |             |               |              |                         | 50                   | 100 | 150 | 200 |
| 0.2           | 102.0         |              | <b>CLAYEY SILT</b><br>some sand, trace rootlets, dark brown, moist, stiff.          |             | SS1           | 75           | 5                       |                      |     |     |     |
| 0.4           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 0.6           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 0.8           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 1.0           | 101.7         |              | <b>SILTY SAND</b><br>some clay, trace gravel, brown, wet, loose.                    |             | SS2           | 75           | 3                       |                      |     |     |     |
| 1.2           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 1.4           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 1.6           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 1.8           | 101.7         |              | <b>SILTY SAND</b><br>trace gravel, trace clay, brown, moist, compact to very dense. |             | SS3           | 100          | 14                      |                      |     |     |     |
| 2.0           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 2.2           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 2.4           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 2.6           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 2.8           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 3.0           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 3.2           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 3.4           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 3.6           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 3.8           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 4.0           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 4.2           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 4.4           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 4.6           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 4.8           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 5.0           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 5.2           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 5.4           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 5.6           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 5.8           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 6.0           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 6.2           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 6.4           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 6.6           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |
| 6.8           |               |              |                                                                                     |             |               |              |                         |                      |     |     |     |

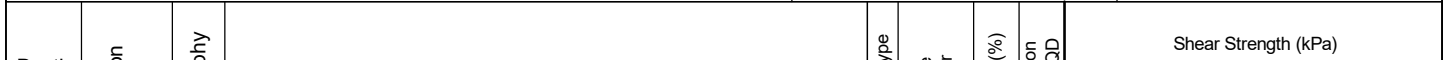
NOTES: Borehole open upon completion

BOREHOLE LOG 1703.GPJ MALROZ ENGINEERING.GDT 2/10/23

# BOREHOLE LOG

Enclosure No.: 4

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Borehole Number: <b>BH104</b>          |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 6, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 6, 2022</b> |  | Elevation: <b>102.9 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326732.7 m</b>             |  | Northing: <b>4874825.1 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |

| Depth BGS (m) | Elevation (m) | Stratigraphy | Description                                                                                | Sample Type | Sample Number | Recovery (%) | Penetration Index / RQD | Shear Strength (kPa)                                                               |     |     |     |
|---------------|---------------|--------------|--------------------------------------------------------------------------------------------|-------------|---------------|--------------|-------------------------|------------------------------------------------------------------------------------|-----|-----|-----|
|               |               |              |                                                                                            |             |               |              |                         | 50                                                                                 | 100 | 150 | 200 |
|               |               |              |                                                                                            |             |               |              |                         |  |     |     |     |
| 0.2           |               |              | <b>CLAYEY SILT</b><br>some sand, trace rootlets, dark brown, moist, soft to stiff.         |             |               |              |                         |                                                                                    |     |     |     |
| 0.4           |               |              |                                                                                            |             | SS1           | 100          | 2                       | 2                                                                                  |     | O22 |     |
| 0.6           |               |              |                                                                                            |             |               |              |                         |                                                                                    |     |     |     |
| 0.8           |               |              |                                                                                            |             |               |              |                         |                                                                                    |     |     |     |
| 1.0           |               |              |                                                                                            |             | SS2           | 100          | 6                       | 6                                                                                  |     | O19 |     |
| 1.2           |               |              |                                                                                            |             |               |              |                         |                                                                                    |     |     |     |
| 1.4           |               |              |                                                                                            |             |               |              |                         |                                                                                    |     |     |     |
| 1.6           | 101.4         |              | <b>SILTY SAND</b><br>trace gravel, trace clay, brown, moist to wet, compact to very dense. |             |               |              |                         |                                                                                    |     |     |     |
| 1.8           |               |              |                                                                                            |             | SS3           | 100          | 27                      |                                                                                    | O12 | 27  |     |
| 2.0           |               |              |                                                                                            |             |               |              |                         |                                                                                    |     |     |     |
| 2.2           |               |              |                                                                                            |             |               |              |                         |                                                                                    |     |     |     |
| 2.4           |               |              |                                                                                            |             |               |              |                         |                                                                                    |     |     |     |
| 2.6           | 100.3         |              |                                                                                            |             | SS4           | 100          | R                       |                                                                                    | O9  |     |     |
| 2.8           |               |              | End of borehole at 2.7 mbg by sampler refusal on inferred cobbles and boulders.            |             |               |              |                         |                                                                                    |     |     |     |
| 3.0           |               |              |                                                                                            |             |               |              |                         |                                                                                    |     |     |     |
| 3.2           |               |              |                                                                                            |             |               |              |                         |                                                                                    |     |     |     |

NOTES: Borehole caved to 1.8 mbg upon completion

|                                                                                    |  |  |                                        |                                    |  |
|------------------------------------------------------------------------------------|--|--|----------------------------------------|------------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |                                        | Project Number: <b>1703.00-100</b> |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  | Well Number: <b>BH105</b>              | Page: <b>1 of 1</b>                |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  | Date Started: <b>December 6, 2022</b>  | Datum: <b>NAD 83, Zone 18T</b>     |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  | Date Finished: <b>December 6, 2022</b> | Elevation: <b>102.7 m</b>          |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  | Easting: <b>326680.7 m</b>             | Northing: <b>4874801.7 m</b>       |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  | Logged By: <b>C. Meyers</b>            | Checked By: <b>D. Hill</b>         |  |

| Depth BGS (m) | Elevation (m) | Stratigraphy | Description                                                                         | Monitoring Well | Sample Type | Sample Number | Recovery (%) | Penetration Index / RQD | Shear Strength (kPa)                                                                |     |     |     |
|---------------|---------------|--------------|-------------------------------------------------------------------------------------|-----------------|-------------|---------------|--------------|-------------------------|-------------------------------------------------------------------------------------|-----|-----|-----|
|               |               |              |                                                                                     |                 |             |               |              |                         | 50                                                                                  | 100 | 150 | 200 |
|               |               |              |                                                                                     |                 |             |               |              |                         |  |     |     |     |
| 0.2           |               |              | <b>CLAYEY SILT</b><br>trace sand, trace rootlets, dark brown, moist, firm to stiff. |                 |             |               |              |                         |                                                                                     |     |     |     |
| 0.4           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 0.6           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 0.8           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 1.0           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 1.2           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 1.4           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 1.6           |               |              | G: 0%, S: 1%, SI: 81%, CL: 18%.                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 1.8           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 2.0           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 2.2           | 100.4         |              | <b>SILTY SAND</b><br>trace clay, trace gravel, brown, moist, compact to very dense. |                 |             |               |              |                         |                                                                                     |     |     |     |
| 2.4           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 2.6           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 2.8           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 3.0           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 3.2           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 3.4           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 3.6           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 3.8           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 4.0           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 4.2           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 4.4           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 4.6           |               |              | G: 6%, S: 55%, SI: 31%, CL: 8%.                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 4.8           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 5.0           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 5.2           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 5.4           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 5.6           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 5.8           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 6.0           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 6.2           | 96.3          |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 6.4           |               |              | End of borehole at 6.3 mbg by sampler refusal on inferred cobbles and boulders.     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 6.6           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |
| 6.8           |               |              |                                                                                     |                 |             |               |              |                         |                                                                                     |     |     |     |

|                                                                                                                                                         |  |  |  |                                                                             |  |  |  |                               |  |  |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|-----------------------------------------------------------------------------|--|--|--|-------------------------------|--|--|--|
| <b>NOTES:</b> Well Construction Details<br>-0.67 m stick-up with aluminum monument<br>1016 mm Sch. 40 PVC<br>slotted screen<br>3.05 m screen<br>#1 sand |  |  |  | <b>Groundwater Monitoring Details</b><br>monitored on<br>DTW: 3.0 m<br>DTB: |  |  |  | Borehole open upon completion |  |  |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|-----------------------------------------------------------------------------|--|--|--|-------------------------------|--|--|--|

# BOREHOLE LOG

Enclosure No.: 6

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Borehole Number: <b>BH106</b>          |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 6, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 6, 2022</b> |  | Elevation: <b>102.5 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326694.7 m</b>             |  | Northing: <b>4874793.6 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |

| Depth BGS (m) | Elevation (m) | Stratigraphy | Description                                                                     | Sample Type | Sample Number | Recovery (%) | Penetration Index / RQD | Shear Strength (kPa)                    |     |     |     |
|---------------|---------------|--------------|---------------------------------------------------------------------------------|-------------|---------------|--------------|-------------------------|-----------------------------------------|-----|-----|-----|
|               |               |              |                                                                                 |             |               |              |                         | 50                                      | 100 | 150 | 200 |
|               |               |              |                                                                                 |             |               |              |                         | ■ DCPT<br>● SPT-N<br>○ MC<br>PL      LL |     |     |     |
| 0.2           |               |              | <b>CLAYEY SILT</b><br>some sand, trace rootlets, brown, moist, soft.            |             |               |              |                         |                                         |     |     |     |
| 0.4           |               |              | Increasing clay content.                                                        |             | SS1           | 100          | 2                       | ●                                       | O22 |     |     |
| 0.6           |               |              |                                                                                 |             |               |              |                         |                                         |     |     |     |
| 0.8           | 101.8         |              | <b>SILTY SAND</b><br>trace clay, brown, moist to wet, loose.                    |             |               |              |                         |                                         |     |     |     |
| 1.0           |               |              |                                                                                 |             | SS2           | 100          | 7                       | ●                                       | O14 |     |     |
| 1.2           |               |              |                                                                                 |             |               |              |                         |                                         |     |     |     |
| 1.4           |               |              |                                                                                 |             |               |              |                         |                                         |     |     |     |
| 1.6           | 101.0         |              | <b>SILTY SAND</b><br>trace gravel, trace clay, brown, moist, dense.             |             |               |              |                         |                                         |     |     |     |
| 1.8           |               |              |                                                                                 |             | SS3           | 100          | 35                      |                                         | O9  | ●   | 35  |
| 2.0           |               |              |                                                                                 |             |               |              |                         |                                         |     |     |     |
| 2.2           |               |              |                                                                                 |             |               |              |                         |                                         |     |     |     |
| 2.4           |               |              |                                                                                 |             |               |              |                         |                                         |     |     |     |
| 2.6           |               |              |                                                                                 |             | SS4           | 100          | R                       |                                         | O8  |     |     |
| 2.8           | 99.8          |              | End of borehole at 2.7 mbg by sampler refusal on inferred cobbles and boulders. |             |               |              |                         |                                         |     |     |     |
| 3.0           |               |              |                                                                                 |             |               |              |                         |                                         |     |     |     |
| 3.2           |               |              |                                                                                 |             |               |              |                         |                                         |     |     |     |

NOTES: Borehole caved to 2.9 mbg upon completion

# BOREHOLE LOG

Enclosure No.: 7

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Borehole Number: <b>BH107</b>          |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 6, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 6, 2022</b> |  | Elevation: <b>102.6 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326703.2 m</b>             |  | Northing: <b>4874799.3 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |

| Depth BGS (m) | Elevation (m) | Stratigraphy | Description                                                                     | Sample Type | Sample Number | Recovery (%) | Penetration Index / RQD | Shear Strength (kPa) |     |     |     |
|---------------|---------------|--------------|---------------------------------------------------------------------------------|-------------|---------------|--------------|-------------------------|----------------------|-----|-----|-----|
|               |               |              |                                                                                 |             |               |              |                         | 50                   | 100 | 150 | 200 |
| 0.2           |               |              | <b>CLAYEY SILT</b><br>some sand, dark brown, moist, soft to firm.               |             |               |              |                         |                      |     |     |     |
| 0.4           |               |              |                                                                                 |             |               |              |                         |                      |     |     |     |
| 0.6           |               |              |                                                                                 |             |               |              |                         |                      |     |     |     |
| 0.8           |               |              |                                                                                 |             |               |              |                         |                      |     |     |     |
| 1.0           |               |              | Increasing clay content, trace sand, brown, moist to wet.                       |             |               |              |                         |                      |     |     |     |
| 1.2           | 101.4         |              |                                                                                 |             |               |              |                         |                      |     |     |     |
| 1.4           |               |              | <b>SILTY SAND</b><br>trace clay, brown, moist to wet, loose.                    |             |               |              |                         |                      |     |     |     |
| 1.6           | 101.1         |              |                                                                                 |             |               |              |                         |                      |     |     |     |
| 1.8           |               |              | <b>SILTY SAND</b><br>trace gravel, trace clay, brown, moist, very dense.        |             |               |              |                         |                      |     |     |     |
| 2.0           |               |              |                                                                                 |             |               |              |                         |                      |     |     |     |
| 2.2           |               |              |                                                                                 |             |               |              |                         |                      |     |     |     |
| 2.4           |               |              |                                                                                 |             |               |              |                         |                      |     |     |     |
| 2.6           |               |              |                                                                                 |             |               |              |                         |                      |     |     |     |
| 2.8           | 99.9          |              | End of borehole at 2.8 mbg by sampler refusal on inferred cobbles and boulders. |             |               |              |                         |                      |     |     |     |
| 3.0           |               |              |                                                                                 |             |               |              |                         |                      |     |     |     |
| 3.2           |               |              |                                                                                 |             |               |              |                         |                      |     |     |     |

NOTES: Borehole caved to 2.3 mbg upon completion

# BOREHOLE LOG

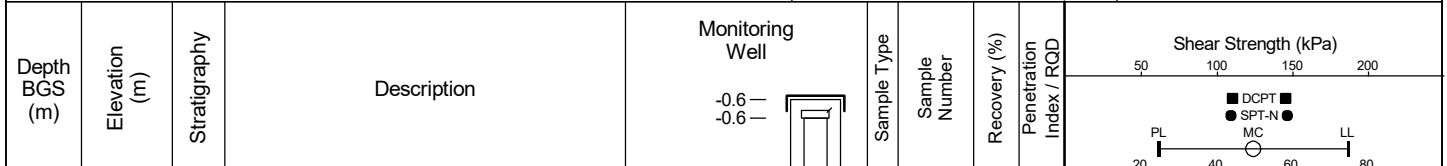
Enclosure No.: 8

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Borehole Number: <b>BH108</b>          |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 6, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 6, 2022</b> |  | Elevation: <b>102.7 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326713.1 m</b>             |  | Northing: <b>4874811.9 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |

| Depth BGS (m) | Elevation (m) | Stratigraphy | Description                                                               | Sample Type | Sample Number | Recovery (%) | Penetration Index / RQD | Shear Strength (kPa)                         |     |     |     |
|---------------|---------------|--------------|---------------------------------------------------------------------------|-------------|---------------|--------------|-------------------------|----------------------------------------------|-----|-----|-----|
|               |               |              |                                                                           |             |               |              |                         | 50                                           | 100 | 150 | 200 |
|               |               |              |                                                                           |             |               |              |                         | ■ DCPT ■<br>● SPT-N ●<br>○ MC ○<br>PL ——— LL |     |     |     |
| 0.2           |               |              | <b>CLAYEY SILT</b><br>some sand, trace rootlets, dark brown, moist, soft. |             |               |              |                         |                                              |     |     |     |
| 0.4           |               |              |                                                                           |             |               |              |                         |                                              |     |     |     |
| 0.6           |               |              |                                                                           |             |               |              |                         |                                              |     |     |     |
| 0.8           |               |              |                                                                           |             |               |              |                         |                                              |     |     |     |
| 1.0           |               |              | Increasing clay content, brown, moist.                                    |             |               |              |                         |                                              |     |     |     |
| 1.2           |               |              |                                                                           |             |               |              |                         |                                              |     |     |     |
| 1.4           |               |              |                                                                           |             |               |              |                         |                                              |     |     |     |
| 1.6           | 101.1         |              | <b>SILTY SAND</b><br>trace clay, trace gravel, brown, moist, compact.     |             |               |              |                         |                                              |     |     |     |
| 1.8           |               |              |                                                                           |             |               |              |                         |                                              |     |     |     |
| 2.0           |               |              |                                                                           |             |               |              |                         |                                              |     |     |     |
| 2.2           |               |              |                                                                           |             |               |              |                         |                                              |     |     |     |
| 2.4           |               |              |                                                                           |             |               |              |                         |                                              |     |     |     |
| 2.6           |               |              | Becoming moist to wet.                                                    |             |               |              |                         |                                              |     |     |     |
| 2.8           |               |              |                                                                           |             |               |              |                         |                                              |     |     |     |
| 3.0           | 99.6          |              |                                                                           |             |               |              |                         |                                              |     |     |     |
| 3.2           |               |              | End of borehole at target depth of 3.0 mbg.                               |             |               |              |                         |                                              |     |     |     |

NOTES: Borehole open upon completion

|                                                                                    |  |  |                                        |                                    |  |
|------------------------------------------------------------------------------------|--|--|----------------------------------------|------------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |                                        | Project Number: <b>1703.00-100</b> |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  | Well Number: <b>BH109</b>              | Page: <b>1 of 1</b>                |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  | Date Started: <b>December 6, 2022</b>  | Datum: <b>NAD 83, Zone 18T</b>     |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  | Date Finished: <b>December 6, 2022</b> | Elevation: <b>103.1 m</b>          |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  | Easting: <b>326740.8 m</b>             | Northing: <b>4874802.4 m</b>       |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  | Logged By: <b>C. Meyers</b>            | Checked By: <b>D. Hill</b>         |  |

| Depth BGS (m) | Elevation (m) | Stratigraphy | Description                                                                        | Monitoring Well | Sample Type | Sample Number | Recovery (%) | Penetration Index / RQD | Shear Strength (kPa)                                                               |     |     |     |
|---------------|---------------|--------------|------------------------------------------------------------------------------------|-----------------|-------------|---------------|--------------|-------------------------|------------------------------------------------------------------------------------|-----|-----|-----|
|               |               |              |                                                                                    |                 |             |               |              |                         | 50                                                                                 | 100 | 150 | 200 |
|               |               |              |                                                                                    |                 |             |               |              |                         |  |     |     |     |
| 0.2           |               |              | <b>CLAYEY SILT</b><br>some sand, trace rootlets, dark brown, moist, soft to stiff. |                 |             |               |              |                         |                                                                                    |     |     |     |
| 0.4           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 0.6           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 0.8           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 1.0           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 1.2           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 1.4           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 1.6           |               |              | G: 0%, S: 1%, SI: 81%, CL: 18%.                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 1.8           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 2.0           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 2.2           | 100.8         |              | <b>SANDY SILT</b><br>some clay, trace gravel, brown, moist, compact.               |                 |             |               |              |                         |                                                                                    |     |     |     |
| 2.4           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 2.6           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 2.8           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 3.0           |               |              | Becoming grey, wet                                                                 |                 |             |               |              |                         |                                                                                    |     |     |     |
| 3.2           |               |              | G: 3%, S: 36%, SI: 50%, CL: 9%.                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 3.4           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 3.6           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 3.8           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 4.0           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 4.2           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 4.4           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 4.6           | 98.5          |              | <b>SILTY SAND</b><br>trace clay, trace gravel, grey, moist, dense                  |                 |             |               |              |                         |                                                                                    |     |     |     |
| 4.8           |               |              | G: 5%, S: 59%, SI: 27%, CL: 9%.                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 5.0           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 5.2           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 5.4           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 5.6           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 5.8           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 6.0           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 6.2           | 96.7          |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 6.4           |               |              | End of borehole at 6.3 mbg by sampler refusal on inferred cobbles and boulders.    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 6.6           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |
| 6.8           |               |              |                                                                                    |                 |             |               |              |                         |                                                                                    |     |     |     |

|                                                                                                                                                         |  |  |  |                                                                             |  |  |  |                               |  |  |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|-----------------------------------------------------------------------------|--|--|--|-------------------------------|--|--|--|
| <b>NOTES:</b> Well Construction Details<br>-0.62 m stick-up with aluminum monument<br>1016 mm Sch. 40 PVC<br>slotted screen<br>3.05 m screen<br>#1 sand |  |  |  | <b>Groundwater Monitoring Details</b><br>monitored on<br>DTW: 3.0 m<br>DTB: |  |  |  | Borehole open upon completion |  |  |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|-----------------------------------------------------------------------------|--|--|--|-------------------------------|--|--|--|

# BOREHOLE LOG

Enclosure No.: 10

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Borehole Number: <b>BH110</b>          |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 6, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 6, 2022</b> |  | Elevation: <b>103.0 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326738.3 m</b>             |  | Northing: <b>4874797.3 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |

| Depth BGS (m) | Elevation (m) | Stratigraphy                                                                       | Description                                                                               | Sample Type | Sample Number | Recovery (%) | Penetration Index / RQD | Shear Strength (kPa)                                                                                                         |     |      |     |
|---------------|---------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------|---------------|--------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------|-----|------|-----|
|               |               |                                                                                    |                                                                                           |             |               |              |                         | 50                                                                                                                           | 100 | 150  | 200 |
|               |               |                                                                                    |                                                                                           |             |               |              |                         | <div> <div> <div>■</div> <div>●</div> <div>○</div> </div> <div> <div>DCPT</div> <div>SPT-N</div> <div>MC</div> </div> </div> |     |      |     |
| 0.2           | 102.7         |   | <b>FILL</b><br>clayey silt, some sand, possibly reworked native, dark brown, moist, soft. |             |               |              |                         |                                                                                                                              |     |      |     |
| 0.4           |               |   | <b>SILTY SAND</b><br>some clay, trace rootlets, brown, moist, very loose.                 |             | SS1           | 67           | 2                       | 2                                                                                                                            |     | O 18 |     |
| 0.6           |               |                                                                                    |                                                                                           |             |               |              |                         |                                                                                                                              |     |      |     |
| 0.8           | 102.3         |  | <b>CLAYEY SILT</b><br>trace sand, brown, moist, stiff.                                    |             |               |              |                         |                                                                                                                              |     |      |     |
| 1.0           |               |                                                                                    |                                                                                           |             | SS2           | 92           | 14                      | 14                                                                                                                           |     | O 24 |     |
| 1.2           |               |                                                                                    |                                                                                           |             |               |              |                         |                                                                                                                              |     |      |     |
| 1.4           |               |                                                                                    |                                                                                           |             |               |              |                         |                                                                                                                              |     |      |     |
| 1.6           |               |                                                                                    |                                                                                           |             |               |              |                         |                                                                                                                              |     |      |     |
| 1.8           |               |                                                                                    |                                                                                           |             | SS3           | 100          | 14                      | 14                                                                                                                           |     | O 25 |     |
| 2.0           |               |                                                                                    |                                                                                           |             |               |              |                         |                                                                                                                              |     |      |     |
| 2.2           |               |                                                                                    |                                                                                           |             |               |              |                         |                                                                                                                              |     |      |     |
| 2.4           |               |                                                                                    |                                                                                           |             |               |              |                         |                                                                                                                              |     |      |     |
| 2.6           |               |                                                                                    |                                                                                           |             |               |              |                         |                                                                                                                              |     |      |     |
| 2.8           |               |                                                                                    |                                                                                           |             | SS4           | 100          | 10                      | 10                                                                                                                           |     | O 26 |     |
| 3.0           | 100.0         |                                                                                    | End of borehole at target depth of 3.0 mbg.                                               |             |               |              |                         |                                                                                                                              |     |      |     |
| 3.2           |               |                                                                                    |                                                                                           |             |               |              |                         |                                                                                                                              |     |      |     |

NOTES: Borehole open upon completion



# BOREHOLE LOG

Enclosure No.: 11

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Borehole Number: <b>BH111</b>          |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 6, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 6, 2022</b> |  | Elevation: <b>103.0 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326766.8 m</b>             |  | Northing: <b>4874807.8 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |

| Depth BGS (m) | Elevation (m) | Stratigraphy | Description                                                                             | Sample Type | Sample Number | Recovery (%) | Penetration Index / RQD | Shear Strength (kPa)                                                                    |     |     |     |
|---------------|---------------|--------------|-----------------------------------------------------------------------------------------|-------------|---------------|--------------|-------------------------|-----------------------------------------------------------------------------------------|-----|-----|-----|
|               |               |              |                                                                                         |             |               |              |                         | 50                                                                                      | 100 | 150 | 200 |
|               |               |              |                                                                                         |             |               |              |                         | <div>■ DCPT ■</div> <div>● SPT-N ●</div> <div>○ MC ○</div> <div>PL 20 40 60 80 LL</div> |     |     |     |
| 0.2           |               |              | <b>CLAYEY SILT</b><br>some sand, trace rootlets, dark brown, moist, soft to very stiff. |             | SS1           | 84           | 3                       | 3                                                                                       |     | O30 |     |
| 0.4           |               |              |                                                                                         |             |               |              |                         |                                                                                         |     |     |     |
| 0.6           |               |              |                                                                                         |             |               |              |                         |                                                                                         |     |     |     |
| 0.8           |               |              |                                                                                         |             |               |              |                         |                                                                                         |     |     |     |
| 1.0           |               |              | Increasing clay content, trace sand, brown, moist.                                      |             | SS2           | 100          | 8                       | 8                                                                                       |     | O20 |     |
| 1.2           |               |              |                                                                                         |             |               |              |                         |                                                                                         |     |     |     |
| 1.4           |               |              |                                                                                         |             |               |              |                         |                                                                                         |     |     |     |
| 1.6           |               |              |                                                                                         |             |               |              |                         |                                                                                         |     |     |     |
| 1.8           |               |              |                                                                                         |             | SS3           | 100          | 23                      |                                                                                         |     | O14 | 23  |
| 2.0           |               |              |                                                                                         |             |               |              |                         |                                                                                         |     |     |     |
| 2.2           |               |              |                                                                                         |             |               |              |                         |                                                                                         |     |     |     |
| 2.4           |               |              |                                                                                         |             |               |              |                         |                                                                                         |     |     |     |
| 2.6           |               |              |                                                                                         |             | SS4           | 100          | 9                       | 9                                                                                       |     | O30 |     |
| 2.8           |               |              |                                                                                         |             |               |              |                         |                                                                                         |     |     |     |
| 3.0           | 99.9          |              |                                                                                         |             |               |              |                         |                                                                                         |     |     |     |
| 3.2           |               |              | End of borehole at target depth of 3.0 mbg.                                             |             |               |              |                         |                                                                                         |     |     |     |

NOTES: Borehole open upon completion

# BOREHOLE LOG

Enclosure No.: 12

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Borehole Number: <b>BH112</b>          |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 5, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 5, 2022</b> |  | Elevation: <b>102.6 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326692.6 m</b>             |  | Northing: <b>4874777.3 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |

| Depth BGS (m) | Elevation (m) | Stratigraphy | Description                                                                        | Sample Type | Sample Number | Recovery (%) | Penetration Index / RQD | Shear Strength (kPa)                                                                                                                                         |     |     |     |
|---------------|---------------|--------------|------------------------------------------------------------------------------------|-------------|---------------|--------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|
|               |               |              |                                                                                    |             |               |              |                         | 50                                                                                                                                                           | 100 | 150 | 200 |
|               |               |              |                                                                                    |             |               |              |                         | <div> <div>OCPT</div> <div>SPT-N</div> <div>MC</div> <div>LL</div> </div> <div> <div>PL</div> <div>20</div> <div>40</div> <div>60</div> <div>80</div> </div> |     |     |     |
| 0.2           |               |              | <b>CLAYEY SILT</b><br>trace sand, brown, moist, firm to very stiff.                |             | SS1           | 75           | 6                       | 6                                                                                                                                                            | O21 |     |     |
| 0.4           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 0.6           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 0.8           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 1.0           |               |              |                                                                                    |             | SS2           | 84           | 7                       | 7                                                                                                                                                            | O26 |     |     |
| 1.2           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 1.4           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 1.6           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 1.8           |               |              |                                                                                    |             | SS3           | 100          | 7                       | 7                                                                                                                                                            | O23 |     |     |
| 2.0           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 2.2           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 2.4           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 2.6           |               |              |                                                                                    |             | SS4           | 100          | 21                      | 21                                                                                                                                                           | O20 |     |     |
| 2.8           | 99.8          |              | <b>SILTY SAND</b><br>trace gravel, trace clay, grey, moist to wet, loose to dense. |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 3.0           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 3.2           |               |              |                                                                                    |             | SS5           | 100          | 7                       | 7                                                                                                                                                            | O10 |     |     |
| 3.4           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 3.6           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 3.8           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 4.0           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 4.2           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 4.4           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 4.6           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 4.8           |               |              |                                                                                    |             | SS6           | 100          | 43                      | O9                                                                                                                                                           | 43  |     |     |
| 5.0           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 5.2           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 5.4           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 5.6           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 5.8           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 6.0           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 6.2           | 96.4          |              | End of borehole at 6.2 mbg by sampler refusal on inferred cobbles and boulders.    |             | SS7           | 100          | R                       | O13                                                                                                                                                          |     |     |     |
| 6.4           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 6.6           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |
| 6.8           |               |              |                                                                                    |             |               |              |                         |                                                                                                                                                              |     |     |     |

NOTES: Borehole open upon completion

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Borehole Number: <b>BH113</b>          |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 5, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 5, 2022</b> |  | Elevation: <b>102.8 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326704.6 m</b>             |  | Northing: <b>4874781.8 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |

| Depth BGS (m) | Elevation (m) | Stratigraphy | Description                                                                          | Sample Type | Sample Number | Recovery (%) | Penetration Index / RQD | Shear Strength (kPa)                                                              |     |     |     |
|---------------|---------------|--------------|--------------------------------------------------------------------------------------|-------------|---------------|--------------|-------------------------|-----------------------------------------------------------------------------------|-----|-----|-----|
|               |               |              |                                                                                      |             |               |              |                         | 50                                                                                | 100 | 150 | 200 |
|               |               |              |                                                                                      |             |               |              |                         | <div> <div>■</div> DCPT <div>●</div> SPT-N <div>○</div> MC <div>■</div> LL </div> |     |     |     |
|               |               |              |                                                                                      |             |               |              |                         | 20                                                                                | 40  | 60  | 80  |
| 0.2           |               |              | <b>SANDY SILT</b><br>some clay, trace rootlets, dark brown, moist, loose.            |             |               |              |                         |                                                                                   |     |     |     |
| 0.4           |               |              |                                                                                      |             | SS1           | 100          | 4                       | ●                                                                                 |     | O28 |     |
| 0.6           |               |              |                                                                                      |             |               |              |                         |                                                                                   |     |     |     |
| 0.8           | 102.1         |              | <b>CLAYEY SILT</b><br>trace sand, brown, moist, stiff to very stiff.                 |             |               |              |                         |                                                                                   |     |     |     |
| 1.0           |               |              |                                                                                      |             | SS2           | 100          | 9                       | ●                                                                                 |     | O20 |     |
| 1.2           |               |              |                                                                                      |             |               |              |                         |                                                                                   |     |     |     |
| 1.4           |               |              |                                                                                      |             |               |              |                         |                                                                                   |     |     |     |
| 1.6           |               |              |                                                                                      |             |               |              |                         |                                                                                   |     |     |     |
| 1.8           |               |              |                                                                                      |             | SS3           | 100          | 15                      | ●                                                                                 |     | O22 |     |
| 2.0           |               |              |                                                                                      |             |               |              |                         |                                                                                   |     |     |     |
| 2.2           |               |              |                                                                                      |             |               |              |                         |                                                                                   |     |     |     |
| 2.4           | 100.4         |              | <b>SILTY SAND</b><br>trace gravel, trace clay, brown to grey, moist to wet, compact. |             |               |              |                         |                                                                                   |     |     |     |
| 2.6           |               |              |                                                                                      |             | SS4           | 100          | 12                      | ●                                                                                 |     | O15 |     |
| 2.8           |               |              |                                                                                      |             |               |              |                         |                                                                                   |     |     |     |
| 3.0           | 99.8          |              |                                                                                      |             |               |              |                         |                                                                                   |     |     |     |
| 3.2           |               |              | End of borehole at target depth of 3.0 mbg.                                          |             |               |              |                         |                                                                                   |     |     |     |

NOTES: Borehole open upon completion

# BOREHOLE LOG

Enclosure No.: 14

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Borehole Number: <b>BH114</b>          |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 5, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 5, 2022</b> |  | Elevation: <b>103.5 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326738.0 m</b>             |  | Northing: <b>4874778.9 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |

| Depth BGS (m) | Elevation (m) | Stratigraphy | Description                                                                    | Sample Type | Sample Number | Recovery (%) | Penetration Index / RQD | Shear Strength (kPa)                                                                |     |     |     |
|---------------|---------------|--------------|--------------------------------------------------------------------------------|-------------|---------------|--------------|-------------------------|-------------------------------------------------------------------------------------|-----|-----|-----|
|               |               |              |                                                                                |             |               |              |                         | 50                                                                                  | 100 | 150 | 200 |
|               |               |              |                                                                                |             |               |              |                         |  |     |     |     |
| 0.2           |               |              | <b>SILTY SAND</b><br>some clay, trace rootlets, dark brown, moist, very loose. |             |               |              |                         |                                                                                     |     |     |     |
| 0.4           |               |              |                                                                                |             |               |              |                         |                                                                                     |     |     |     |
| 0.6           |               |              |                                                                                |             |               |              |                         |                                                                                     |     |     |     |
| 0.8           | 102.7         |              | <b>CLAYEY SILT</b><br>trace sand, light brown, moist, soft to stiff.           |             |               |              |                         |                                                                                     |     |     |     |
| 1.0           |               |              |                                                                                |             |               |              |                         |                                                                                     |     |     |     |
| 1.2           |               |              |                                                                                |             |               |              |                         |                                                                                     |     |     |     |
| 1.4           |               |              |                                                                                |             |               |              |                         |                                                                                     |     |     |     |
| 1.6           |               |              |                                                                                |             |               |              |                         |                                                                                     |     |     |     |
| 1.8           |               |              |                                                                                |             |               |              |                         |                                                                                     |     |     |     |
| 2.0           |               |              |                                                                                |             |               |              |                         |                                                                                     |     |     |     |
| 2.2           |               |              |                                                                                |             |               |              |                         |                                                                                     |     |     |     |
| 2.4           | 101.0         |              | <b>SILTY SAND</b><br>trace boulder, trace gravel, trace clay, dense.           |             |               |              |                         |                                                                                     |     |     |     |
| 2.6           |               |              |                                                                                |             |               |              |                         |                                                                                     |     |     |     |
| 2.8           |               |              |                                                                                |             |               |              |                         |                                                                                     |     |     |     |
| 3.0           | 100.4         |              |                                                                                |             |               |              |                         |                                                                                     |     |     |     |
| 3.2           |               |              | End of borehole at target depth of 3.0 mbg.                                    |             |               |              |                         |                                                                                     |     |     |     |

|                                                         |  |  |  |  |  |  |  |  |  |  |  |
|---------------------------------------------------------|--|--|--|--|--|--|--|--|--|--|--|
| <b>NOTES:</b> Borehole caved to 2.6 mbg upon completion |  |  |  |  |  |  |  |  |  |  |  |
|---------------------------------------------------------|--|--|--|--|--|--|--|--|--|--|--|



# BOREHOLE LOG

Enclosure No.: 15

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Borehole Number: <b>BH115</b>          |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 5, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 5, 2022</b> |  | Elevation: <b>103.4 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326754.5 m</b>             |  | Northing: <b>4874787.5 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |

| Depth BGS (m) | Elevation (m) | Stratigraphy | Description                                                               | Sample Type | Sample Number | Recovery (%) | Penetration Index / RQD | Shear Strength (kPa)                                                                    |     |     |     |
|---------------|---------------|--------------|---------------------------------------------------------------------------|-------------|---------------|--------------|-------------------------|-----------------------------------------------------------------------------------------|-----|-----|-----|
|               |               |              |                                                                           |             |               |              |                         | 50                                                                                      | 100 | 150 | 200 |
|               |               |              |                                                                           |             |               |              |                         | <div>■ DCPT ■</div> <div>● SPT-N ●</div> <div>○ MC ○</div> <div>PL 20 40 60 80 LL</div> |     |     |     |
| 0.2           |               |              | <b>SILTY SAND</b><br>some clay, trace rootlets, dark brown, moist, loose. |             | SS1           | 84           | 3                       | 3                                                                                       |     | O29 |     |
| 0.4           |               |              |                                                                           |             |               |              |                         |                                                                                         |     |     |     |
| 0.6           |               |              |                                                                           |             |               |              |                         |                                                                                         |     |     |     |
| 0.8           | 102.6         |              | <b>CLAYEY SILT</b><br>trace sand, light brown, moist, firm to very stiff. |             | SS2           | 100          | 4                       | 4                                                                                       |     | O27 |     |
| 1.0           |               |              |                                                                           |             |               |              |                         |                                                                                         |     |     |     |
| 1.2           |               |              |                                                                           |             |               |              |                         |                                                                                         |     |     |     |
| 1.4           |               |              |                                                                           |             |               |              |                         |                                                                                         |     |     |     |
| 1.6           |               |              |                                                                           |             |               |              |                         |                                                                                         |     |     |     |
| 1.8           |               |              |                                                                           |             | SS3           | 100          | 6                       | 6                                                                                       |     | O23 |     |
| 2.0           |               |              |                                                                           |             |               |              |                         |                                                                                         |     |     |     |
| 2.2           |               |              |                                                                           |             |               |              |                         |                                                                                         |     |     |     |
| 2.4           |               |              |                                                                           |             |               |              |                         |                                                                                         |     |     |     |
| 2.6           |               |              |                                                                           |             |               |              |                         |                                                                                         |     |     |     |
| 2.8           |               |              | Sand seam (2.7-2.9 mbg).                                                  |             | SS4           | 100          | 15                      | 15                                                                                      |     | O22 |     |
| 3.0           | 100.3         |              |                                                                           |             |               |              |                         |                                                                                         |     |     |     |
| 3.2           |               |              | End of borehole at target depth of 3.0 mbg.                               |             |               |              |                         |                                                                                         |     |     |     |

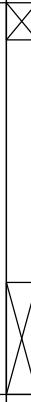
NOTES: Borehole open upon completion

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Borehole Number: <b>BH116</b>          |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 5, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 5, 2022</b> |  | Elevation: <b>103.3 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326764.3 m</b>             |  | Northing: <b>4874796.7 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |

| Depth BGS (m) | Elevation (m) | Stratigraphy | Description                                                                           | Sample Type | Sample Number | Recovery (%) | Penetration Index / RQD | Shear Strength (kPa)            |     |     |     |
|---------------|---------------|--------------|---------------------------------------------------------------------------------------|-------------|---------------|--------------|-------------------------|---------------------------------|-----|-----|-----|
|               |               |              |                                                                                       |             |               |              |                         | 50                              | 100 | 150 | 200 |
|               |               |              |                                                                                       |             |               |              |                         | ■ DCPT<br>● SPT-N<br>○ MC<br>LL |     |     |     |
|               |               |              |                                                                                       |             |               |              |                         | 20                              | 40  | 60  | 80  |
| 0.2           |               |              | <b>SILTY SAND</b><br>some clay, dark brown, moist, very loose.                        |             |               |              |                         |                                 |     |     |     |
| 0.4           |               |              |                                                                                       |             | SS1           | 67           | 2                       | 2                               |     | O26 |     |
| 0.6           |               |              |                                                                                       |             |               |              |                         |                                 |     |     |     |
| 0.8           | 102.5         |              | <b>SILTY SAND</b><br>trace gravel, trace clay, brown, moist to wet, loose to compact. |             |               |              |                         |                                 |     |     |     |
| 1.0           |               |              |                                                                                       |             | SS2           | 100          | 8                       | 8                               |     | O19 |     |
| 1.2           |               |              |                                                                                       |             |               |              |                         |                                 |     |     |     |
| 1.4           |               |              |                                                                                       |             |               |              |                         |                                 |     |     |     |
| 1.6           |               |              |                                                                                       |             |               |              |                         |                                 |     |     |     |
| 1.8           |               |              |                                                                                       |             | SS3           | 100          | 16                      | 16                              |     | O24 |     |
| 2.0           |               |              |                                                                                       |             |               |              |                         |                                 |     |     |     |
| 2.2           |               |              |                                                                                       |             |               |              |                         |                                 |     |     |     |
| 2.4           |               |              |                                                                                       |             |               |              |                         |                                 |     |     |     |
| 2.6           |               |              |                                                                                       |             |               |              |                         |                                 |     |     |     |
| 2.8           |               |              | Becoming grey (2.7 mbg).                                                              |             | SS4           | 100          | 6                       | 6                               |     | O19 |     |
| 3.0           | 100.2         |              |                                                                                       |             |               |              |                         |                                 |     |     |     |
| 3.2           |               |              | End of borehole at target depth of 3.0 mbg.                                           |             |               |              |                         |                                 |     |     |     |

|                                      |
|--------------------------------------|
| NOTES: Borehole open upon completion |
|--------------------------------------|

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Borehole Number: <b>BH117</b>          |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 5, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 5, 2022</b> |  | Elevation: <b>103.6 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326774.8 m</b>             |  | Northing: <b>4874791.2 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |

| Depth<br>BGS<br>(m) | Elevation<br>(m) | Stratigraphy                                                                        | Description                                                                     | Sample Type                                                                          | Sample<br>Number | Recovery (%) | Penetration<br>Index / RQD | Shear Strength (kPa)                                                                                                                                    |     |     |     |  |  |
|---------------------|------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------------|--------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|--|--|
|                     |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            | 50                                                                                                                                                      | 100 | 150 | 200 |  |  |
|                     |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            | <div><div>■ DCPT</div><div>● SPT-N</div><div>○ MC</div></div> <div><div>PL</div><div>20</div><div>40</div><div>60</div><div>80</div><div>LL</div></div> |     |     |     |  |  |
| 0.2                 | 102.8            |    | <b>SILTY SAND</b><br>trace clay, trace rootlets, dark brown, moist, very loose. |    | SS1              | 75           | 2                          | 2                                                                                                                                                       | O28 |     |     |  |  |
| 0.4                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
| 0.6                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
| 0.8                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
| 1.0                 | 101.3            |   | <b>CLAYEY SILT</b><br>native, trace sand, light brown, moist, firm to stiff.    |   | SS2              | 100          | 7                          | 7                                                                                                                                                       | O26 |     |     |  |  |
| 1.2                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
| 1.4                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
| 1.6                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
| 1.8                 | 101.3            |  | <b>SILTY SAND</b><br>some clay, light brown, moist to wet, compact.             |  | SS3              | 100          | 11                         | 11                                                                                                                                                      | O23 |     |     |  |  |
| 2.0                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
| 2.2                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
| 2.4                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
| 2.6                 | 99.0             |  | <b>SILTY SAND</b><br>some clay, light brown, moist to wet, compact.             |  | SS4              | 50           | 19                         | O19                                                                                                                                                     |     |     |     |  |  |
| 2.8                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
| 3.0                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
| 3.2                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
| 3.4                 | 99.0             |  | Trace gravel, trace clay, brown, moist to wet.                                  |  | SS5              | 100          | 22                         | O11                                                                                                                                                     | 22  |     |     |  |  |
| 3.6                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
| 3.8                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
| 4.0                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
| 4.2                 | 99.0             |  | <b>SILTY SAND</b><br>trace gravel, trace clay, brown, moist, very dense.        |  | SS6              | 100          | R                          | O8                                                                                                                                                      |     |     |     |  |  |
| 4.4                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
| 4.6                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
| 4.8                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
| 5.0                 | 96.9             |  | <b>SILTY SAND</b><br>trace gravel, trace clay, brown, moist, very dense.        |  | SS7              | 100          | 68                         | O9                                                                                                                                                      |     |     | 68  |  |  |
| 5.2                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
| 5.4                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
| 5.6                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
| 5.8                 | 96.9             |  | <b>SILTY SAND</b><br>trace gravel, trace clay, brown, moist, very dense.        |  | SS8              | 100          | 68                         | O9                                                                                                                                                      |     |     |     |  |  |
| 6.0                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
| 6.2                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
| 6.4                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
| 6.6                 | 96.9             |  | <b>SILTY SAND</b><br>trace gravel, trace clay, brown, moist, very dense.        |  | SS9              | 100          | 68                         | O9                                                                                                                                                      |     |     |     |  |  |
| 6.8                 |                  |                                                                                     |                                                                                 |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |
|                     |                  |                                                                                     | End of borehole at target depth of 6.7 mbg.                                     |                                                                                      |                  |              |                            |                                                                                                                                                         |     |     |     |  |  |

NOTES: Borehole open upon completion



# BOREHOLE LOG

Enclosure No.: 18

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Borehole Number: <b>BH118</b>          |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 5, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 5, 2022</b> |  | Elevation: <b>103.4 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326723.1 m</b>             |  | Northing: <b>4874768.3 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |

| Depth BGS (m) | Elevation (m) | Stratigraphy | Description                                                                    | Sample Type | Sample Number | Recovery (%) | Penetration Index / RQD | Shear Strength (kPa)                                       |     |     |     |
|---------------|---------------|--------------|--------------------------------------------------------------------------------|-------------|---------------|--------------|-------------------------|------------------------------------------------------------|-----|-----|-----|
|               |               |              |                                                                                |             |               |              |                         | 50                                                         | 100 | 150 | 200 |
|               |               |              |                                                                                |             |               |              |                         | <div>■ DCPT ■<br/>● SPT-N ●<br/>○ MC ○<br/>PL ——— LL</div> |     |     |     |
|               |               |              |                                                                                |             |               |              |                         | 20                                                         | 40  | 60  | 80  |
| 0.2           | 103.1         |              | <b>SILTY SAND</b><br>some clay, trace rootlets, dark brown, moist, very loose. |             | SS1           | 84           | 2                       | 2                                                          | O23 |     |     |
| 0.4           |               |              | <b>CLAYEY SILT</b><br>trace sand, brown, moist, very soft to stiff.            |             |               |              |                         |                                                            |     |     |     |
| 0.6           |               |              |                                                                                |             |               |              |                         |                                                            |     |     |     |
| 0.8           |               |              |                                                                                |             |               |              |                         |                                                            |     |     |     |
| 1.0           |               |              |                                                                                |             | SS2           | 100          | 5                       | 5                                                          | O21 |     |     |
| 1.2           |               |              |                                                                                |             |               |              |                         |                                                            |     |     |     |
| 1.4           |               |              |                                                                                |             |               |              |                         |                                                            |     |     |     |
| 1.6           |               |              |                                                                                |             |               |              |                         |                                                            |     |     |     |
| 1.8           |               |              |                                                                                |             | SS3           | 100          | 11                      | 11                                                         | O22 |     |     |
| 2.0           |               |              |                                                                                |             |               |              |                         |                                                            |     |     |     |
| 2.2           |               |              |                                                                                |             |               |              |                         |                                                            |     |     |     |
| 2.4           |               |              |                                                                                |             |               |              |                         |                                                            |     |     |     |
| 2.6           |               |              |                                                                                |             | SS4           | 100          | 6                       | 6                                                          | O20 |     |     |
| 2.8           |               |              |                                                                                |             |               |              |                         |                                                            |     |     |     |
| 3.0           |               |              |                                                                                |             |               |              |                         |                                                            |     |     |     |
| 3.2           |               |              |                                                                                |             |               |              |                         |                                                            |     |     |     |
|               | 100.3         |              | End of borehole at target depth of 3.0 mbg.                                    |             |               |              |                         |                                                            |     |     |     |

NOTES: Borehole caved to 2.7 mbg upon completion

BOREHOLE LOG 1703.GPJ MALROZ ENGINEERING.GDT 2/10/23

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Borehole Number: <b>BH119</b>          |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 5, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 5, 2022</b> |  | Elevation: <b>103.8 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326755.8 m</b>             |  | Northing: <b>4874779.0 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |

| Depth BGS (m) | Elevation (m) | Stratigraphy | Description                                                                                                                                          | Sample Type | Sample Number | Recovery (%) | Penetration Index / RQD | Shear Strength (kPa)                   |     |     |     |
|---------------|---------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------|--------------|-------------------------|----------------------------------------|-----|-----|-----|
|               |               |              |                                                                                                                                                      |             |               |              |                         | 50                                     | 100 | 150 | 200 |
|               |               |              |                                                                                                                                                      |             |               |              |                         | ■ DCPT<br>● SPT-N<br>○ MC<br>— PL — LL |     |     |     |
| 0.2           |               |              | <b>SILTY SAND</b><br>some clay, trace rootlets, dark brown, moist, very loose.                                                                       |             |               |              |                         |                                        |     |     |     |
| 0.4           |               |              |                                                                                                                                                      |             |               |              |                         |                                        |     |     |     |
| 0.6           |               |              |                                                                                                                                                      |             |               |              |                         |                                        |     |     |     |
| 0.8           | 103.1         |              | <b>CLAYEY SILT</b><br>trace sand, light brown, moist, firm.                                                                                          |             |               |              |                         |                                        |     |     |     |
| 1.0           |               |              |                                                                                                                                                      |             |               |              |                         |                                        |     |     |     |
| 1.2           |               |              |                                                                                                                                                      |             |               |              |                         |                                        |     |     |     |
| 1.4           |               |              |                                                                                                                                                      |             |               |              |                         |                                        |     |     |     |
| 1.6           | 102.3         |              | <b>SANDY SILT</b><br>trace clay, possible cobbles and boulders inferred from auger grinding, brown, wet, compact<br>G: 20%, S: 33%, SI: 38%, CL: 9%. |             |               |              |                         |                                        |     |     |     |
| 1.8           |               |              |                                                                                                                                                      |             |               |              |                         |                                        |     |     |     |
| 2.0           |               |              |                                                                                                                                                      |             |               |              |                         |                                        |     |     |     |
| 2.2           |               |              |                                                                                                                                                      |             |               |              |                         |                                        |     |     |     |
| 2.4           |               |              |                                                                                                                                                      |             |               |              |                         |                                        |     |     |     |
| 2.6           |               |              |                                                                                                                                                      |             |               |              |                         |                                        |     |     |     |
| 2.8           |               |              |                                                                                                                                                      |             |               |              |                         |                                        |     |     |     |
| 3.0           | 100.8         |              |                                                                                                                                                      |             |               |              |                         |                                        |     |     |     |
| 3.2           |               |              | End of borehole at target depth of 3.0 mbg.                                                                                                          |             |               |              |                         |                                        |     |     |     |

|                                                         |  |  |  |  |  |  |  |  |  |  |  |
|---------------------------------------------------------|--|--|--|--|--|--|--|--|--|--|--|
| <b>NOTES:</b> Borehole caved to 2.1 mbg upon completion |  |  |  |  |  |  |  |  |  |  |  |
|---------------------------------------------------------|--|--|--|--|--|--|--|--|--|--|--|



# BOREHOLE LOG

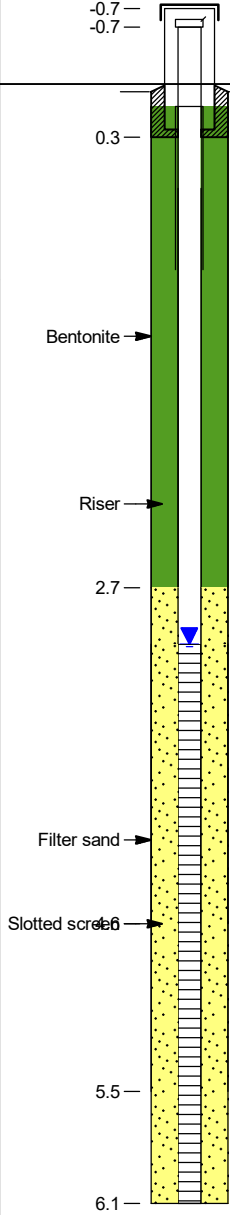
Enclosure No.: 20

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Borehole Number: <b>BH120</b>          |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 5, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 5, 2022</b> |  | Elevation: <b>103.8 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326732.2 m</b>             |  | Northing: <b>4874766.7 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |

| Depth BGS (m) | Elevation (m) | Stratigraphy | Description                                                                             | Sample Type | Sample Number | Recovery (%) | Penetration Index / RQD | Shear Strength (kPa)                                                                                    |     |     |     |
|---------------|---------------|--------------|-----------------------------------------------------------------------------------------|-------------|---------------|--------------|-------------------------|---------------------------------------------------------------------------------------------------------|-----|-----|-----|
|               |               |              |                                                                                         |             |               |              |                         | 50                                                                                                      | 100 | 150 | 200 |
|               |               |              |                                                                                         |             |               |              |                         | <div>■ DCPT</div> <div>● SPT-N</div> <div>○ MC</div> <div>LL</div> <div>PL</div> <div>20 40 60 80</div> |     |     |     |
| 0.2           |               |              | <b>SILTY SAND</b><br>some clay, trace gravel, dark brown, moist, very loose to compact. |             |               |              |                         |                                                                                                         |     |     |     |
| 0.4           |               |              |                                                                                         |             | SS1           | 84           | 2                       | 2                                                                                                       |     | O14 |     |
| 0.6           |               |              |                                                                                         |             |               |              |                         |                                                                                                         |     |     |     |
| 0.8           |               |              | Becoming brown.                                                                         |             |               |              |                         |                                                                                                         |     |     |     |
| 1.0           | 102.7         |              |                                                                                         |             | SS2           | 100          | 10                      | 10                                                                                                      |     | O27 |     |
| 1.2           |               |              | <b>SILTY SAND</b><br>trace gravel, trace clay, brown, moist, loose to dense.            |             |               |              |                         |                                                                                                         |     |     |     |
| 1.4           |               |              |                                                                                         |             |               |              |                         |                                                                                                         |     |     |     |
| 1.6           |               |              |                                                                                         |             |               |              |                         |                                                                                                         |     |     |     |
| 1.8           |               |              |                                                                                         |             | SS3           | 100          | 34                      |                                                                                                         | O14 | 34  |     |
| 2.0           |               |              |                                                                                         |             |               |              |                         |                                                                                                         |     |     |     |
| 2.2           |               |              |                                                                                         |             |               |              |                         |                                                                                                         |     |     |     |
| 2.4           |               |              | Becoming grey.                                                                          |             |               |              |                         |                                                                                                         |     |     |     |
| 2.6           |               |              |                                                                                         |             | SS4           | 100          | R                       |                                                                                                         | O8  |     |     |
| 2.8           | 101.0         |              | End of borehole at 2.8 mbg by sampler refusal on inferred cobbles and boulders.         |             |               |              |                         |                                                                                                         |     |     |     |
| 3.0           |               |              |                                                                                         |             |               |              |                         |                                                                                                         |     |     |     |
| 3.2           |               |              |                                                                                         |             |               |              |                         |                                                                                                         |     |     |     |

NOTES: Borehole open upon completion

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Well Number: <b>BH121</b>              |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 1, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 1, 2022</b> |  | Elevation: <b>105.0 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326756.4 m</b>             |  | Northing: <b>4874764.3 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |

| Depth BGS (m) | Elevation (m) | Stratigraphy                                                                       | Description                                                                                                            | Monitoring Well                                                                    | Sample Type | Sample Number | Recovery (%) | Penetration Index / RQD | Shear Strength (kPa) |     |     |     |
|---------------|---------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------|---------------|--------------|-------------------------|----------------------|-----|-----|-----|
|               |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         | 50                   | 100 | 150 | 200 |
| 0.2           | 104.0         |  | <b>SILTY SAND</b><br>trace rootlets, dark brown, moist, very loose to compact.                                         |  | X           | SS1           | 84           | 2                       | ●                    | O14 |     |     |
| 0.4           |               |                                                                                    |                                                                                                                        |                                                                                    |             | SS2           | 100          | 16                      |                      | O9  | 16  |     |
| 0.6           |               |                                                                                    |                                                                                                                        |                                                                                    |             | SS3           | 100          | 35                      |                      | O9  | 35  |     |
| 0.8           |               |                                                                                    |                                                                                                                        |                                                                                    |             | SS4           | 100          | 67                      |                      | O7  |     | 67  |
| 1.0           |               |                                                                                    |                                                                                                                        |                                                                                    |             | SS5           | 100          | R                       |                      | O8  |     |     |
| 1.2           |               |                                                                                    | <b>SILTY SAND</b><br>trace gravel, trace clay, brown, moist, compact to very dense.<br>G: 2%, S: 61%, SI: 29%, CL: 8%. |                                                                                    |             | SS6           | 100          | R                       |                      | O7  |     |     |
| 1.4           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 1.6           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 1.8           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 2.0           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 2.2           | 98.8          |  |                                                                                                                        |                                                                                    | X           |               |              |                         |                      |     |     |     |
| 2.4           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 2.6           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 2.8           |               |                                                                                    | Becoming wet.                                                                                                          |                                                                                    |             |               |              |                         |                      |     |     |     |
| 3.0           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 3.2           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 3.4           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 3.6           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 3.8           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 4.0           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 4.2           | 98.8          |  |                                                                                                                        |                                                                                    | X           |               |              |                         |                      |     |     |     |
| 4.4           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 4.6           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 4.8           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 5.0           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 5.2           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 5.4           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 5.6           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 5.8           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 6.0           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 6.2           | 98.8          |  |                                                                                                                        |                                                                                    | X           | SS7           | 100          | R                       |                      | O6  |     |     |
| 6.4           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 6.6           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 6.8           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 6.8           |               |                                                                                    | End of borehole at 6.3 mbg by sampler refusal on inferred cobbles and boulders.                                        |                                                                                    |             |               |              |                         |                      |     |     |     |
| 6.8           |               |                                                                                    |                                                                                                                        |                                                                                    |             |               |              |                         |                      |     |     |     |

|                                                                                                                                                         |  |  |  |                                                                             |  |  |  |                               |  |  |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|-----------------------------------------------------------------------------|--|--|--|-------------------------------|--|--|--|
| <b>NOTES:</b> Well Construction Details<br>-0.67 m stick-up with aluminum monument<br>1016 mm Sch. 40 PVC<br>slotted screen<br>3.05 m screen<br>#1 sand |  |  |  | <b>Groundwater Monitoring Details</b><br>monitored on<br>DTW: 3.0 m<br>DTB: |  |  |  | Borehole open upon completion |  |  |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|-----------------------------------------------------------------------------|--|--|--|-------------------------------|--|--|--|



# BOREHOLE LOG

Enclosure No.: 22

|                                                                                    |  |  |                                        |                                    |  |
|------------------------------------------------------------------------------------|--|--|----------------------------------------|------------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |                                        | Project Number: <b>1703.00-100</b> |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  | Borehole Number: <b>BH122</b>          | Page: <b>1 of 1</b>                |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  | Date Started: <b>December 6, 2022</b>  | Datum: <b>NAD 83, Zone 18T</b>     |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  | Date Finished: <b>December 6, 2022</b> | Elevation: <b>103.0 m</b>          |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  | Easting: <b>326753.4 m</b>             | Northing: <b>4874828.0 m</b>       |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  | Logged By: <b>C. Meyers</b>            | Checked By: <b>D. Hill</b>         |  |

| Depth<br>BGS<br>(m) | Elevation<br>(m) | Stratigraphy | Description                                                                    | Sample Type | Sample Number | Recovery (%) | Penetration Index / RQD | Shear Strength (kPa)                                                                                   |     |     |     |
|---------------------|------------------|--------------|--------------------------------------------------------------------------------|-------------|---------------|--------------|-------------------------|--------------------------------------------------------------------------------------------------------|-----|-----|-----|
|                     |                  |              |                                                                                |             |               |              |                         | 50                                                                                                     | 100 | 150 | 200 |
|                     |                  |              |                                                                                |             |               |              |                         | <div>■ DCPT ■</div> <div>● SPT-N ●</div> <div>○ MC ○</div> <div>PL ——— LL</div> <div>20 40 60 80</div> |     |     |     |
| 0.2                 |                  |              | <b>CLAYEY SILT</b><br>trace sand, dark brown, wet, very soft to hard.          |             |               |              |                         |                                                                                                        |     |     |     |
| 0.4                 |                  |              |                                                                                |             | SS1           |              | 1                       | 1                                                                                                      |     |     |     |
| 0.6                 |                  |              | Sand seam (0.3-0.5 mbg).                                                       |             |               |              |                         |                                                                                                        |     |     |     |
| 0.8                 |                  |              |                                                                                |             |               |              |                         |                                                                                                        |     |     |     |
| 1.0                 |                  |              |                                                                                |             | SS2           |              | 30                      |                                                                                                        |     |     |     |
| 1.2                 |                  |              |                                                                                |             |               |              |                         |                                                                                                        |     |     |     |
| 1.4                 |                  |              |                                                                                |             |               |              |                         |                                                                                                        |     |     |     |
| 1.6                 |                  |              |                                                                                |             |               |              |                         |                                                                                                        |     |     |     |
| 1.8                 | 101.2            |              |                                                                                |             | SS3           |              | 24                      |                                                                                                        |     |     |     |
| 2.0                 |                  |              | <b>SILTY SAND</b><br>trace gravel, trace clay, brown, moist, compact to dense. |             |               |              |                         |                                                                                                        |     |     |     |
| 2.2                 |                  |              |                                                                                |             |               |              |                         |                                                                                                        |     |     |     |
| 2.4                 |                  |              |                                                                                |             |               |              |                         |                                                                                                        |     |     |     |
| 2.6                 |                  |              |                                                                                |             |               |              |                         |                                                                                                        |     |     |     |
| 2.8                 |                  |              |                                                                                |             | SS4           |              | 32                      |                                                                                                        |     |     |     |
| 3.0                 | 99.9             |              |                                                                                |             |               |              |                         |                                                                                                        |     |     |     |
| 3.2                 |                  |              | End of borehole at target depth of 3.0 mbg.                                    |             |               |              |                         |                                                                                                        |     |     |     |

NOTES: Borehole caved to 2.7 mbg upon completion



# BOREHOLE LOG

Enclosure No.: 23

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Borehole Number: <b>BH123</b>          |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 1, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 1, 2022</b> |  | Elevation: <b>104.1 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326789.5 m</b>             |  | Northing: <b>4874791.1 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |

| Depth<br>BGS<br>(m) | Elevation<br>(m) | Stratigraphy | Description                                                                     | Sample Type | Sample<br>Number | Recovery (%) | Penetration<br>Index / RQD | Shear Strength (kPa)                                                             |     |     |     |  |  |
|---------------------|------------------|--------------|---------------------------------------------------------------------------------|-------------|------------------|--------------|----------------------------|----------------------------------------------------------------------------------|-----|-----|-----|--|--|
|                     |                  |              |                                                                                 |             |                  |              |                            | 50                                                                               | 100 | 150 | 200 |  |  |
|                     |                  |              |                                                                                 |             |                  |              |                            | <div>■ DCPT</div> <div>● SPT-N</div> <div>○ MC</div> <div>PL</div> <div>LL</div> |     |     |     |  |  |
|                     |                  |              |                                                                                 |             |                  |              |                            |                                                                                  |     |     |     |  |  |
| 0.2                 |                  |              | <b>SILTY SAND</b><br>trace clay, trace rootlets, dark brown, moist, very loose. |             |                  |              |                            |                                                                                  |     |     |     |  |  |
| 0.4                 |                  |              |                                                                                 |             | SS1              | 67           | 2                          | 2                                                                                |     | O27 |     |  |  |
| 0.6                 |                  |              |                                                                                 |             |                  |              |                            |                                                                                  |     |     |     |  |  |
| 0.8                 | 103.3            |              | <b>CLAYEY SILT</b><br>some sand, brown, moist to wet, very soft to stiff.       |             |                  |              |                            |                                                                                  |     |     |     |  |  |
| 1.0                 |                  |              |                                                                                 |             | SS2              | 84           | 2                          | 2                                                                                |     | O33 |     |  |  |
| 1.2                 |                  |              |                                                                                 |             |                  |              |                            |                                                                                  |     |     |     |  |  |
| 1.4                 |                  |              |                                                                                 |             |                  |              |                            |                                                                                  |     |     |     |  |  |
| 1.6                 |                  |              |                                                                                 |             |                  |              |                            |                                                                                  |     |     |     |  |  |
| 1.8                 |                  |              |                                                                                 |             | SS3              |              | 8                          | 8                                                                                |     | O30 |     |  |  |
| 2.0                 |                  |              |                                                                                 |             |                  |              |                            |                                                                                  |     |     |     |  |  |
| 2.2                 |                  |              |                                                                                 |             |                  |              |                            |                                                                                  |     |     |     |  |  |
| 2.4                 | 101.6            |              | <b>SILTY SAND</b><br>some clay, brown, wet, loose.                              |             |                  |              |                            |                                                                                  |     |     |     |  |  |
| 2.6                 |                  |              |                                                                                 |             | SS4              |              | 6                          | 6                                                                                |     | O13 |     |  |  |
| 2.8                 |                  |              |                                                                                 |             |                  |              |                            |                                                                                  |     |     |     |  |  |
| 3.0                 | 101.0            |              | End of borehole at target depth of 3.0 mbg.                                     |             |                  |              |                            |                                                                                  |     |     |     |  |  |
| 3.2                 |                  |              |                                                                                 |             |                  |              |                            |                                                                                  |     |     |     |  |  |

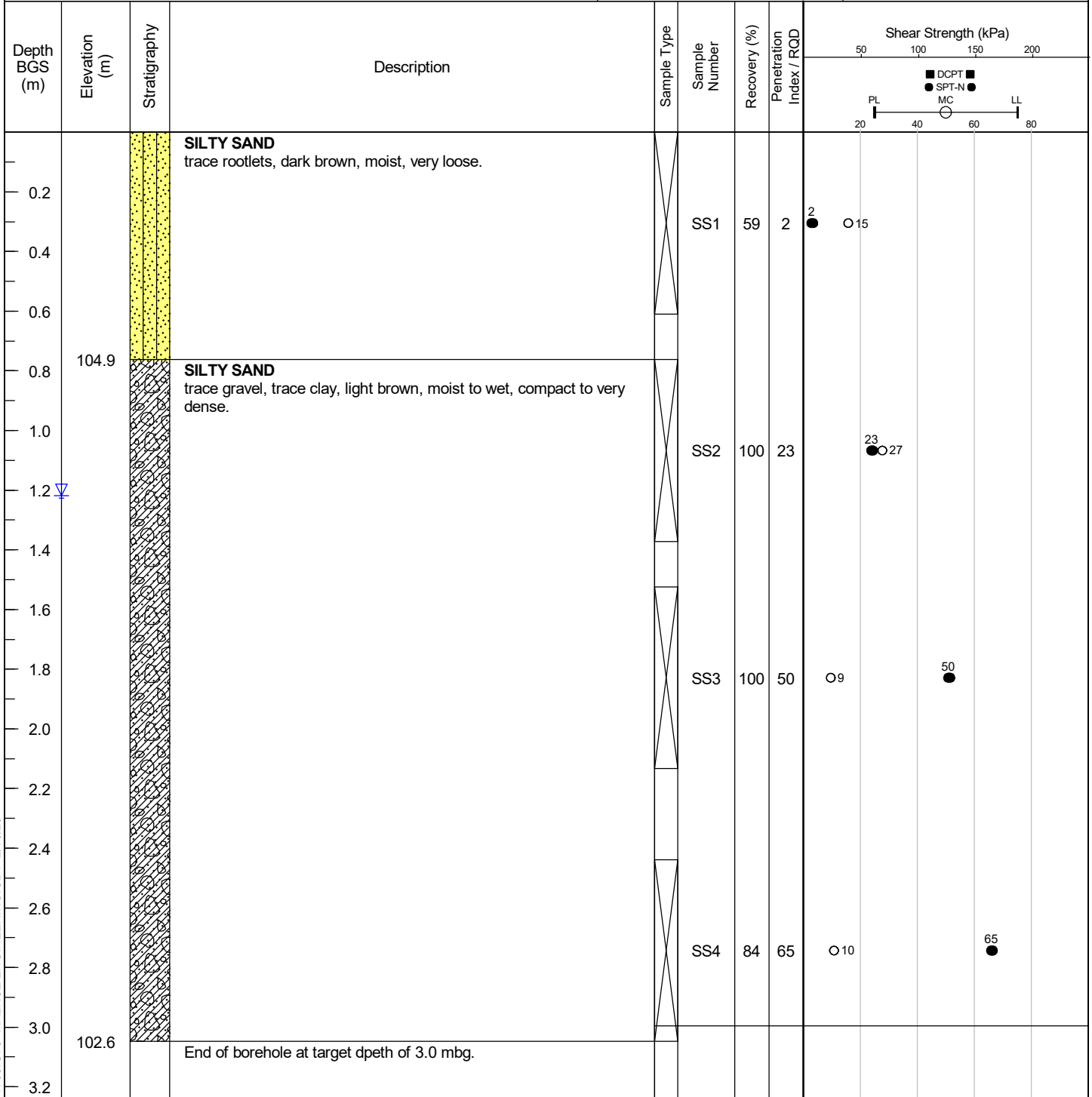
NOTES: Borehole caved to 2.4 mbg upon completion



# BOREHOLE LOG

Enclosure No.: 24

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Borehole Number: <b>BH124</b>          |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 1, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 1, 2022</b> |  | Elevation: <b>105.7 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326780.9 m</b>             |  | Northing: <b>4874766.9 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |



NOTES: Borehole caved to 2.7 mbg upon completion

# BOREHOLE LOG

Enclosure No.: 25

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Borehole Number: <b>BH125</b>          |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 1, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 1, 2022</b> |  | Elevation: <b>107.3 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326773.1 m</b>             |  | Northing: <b>4874742.1 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |

| Depth BGS (m) | Elevation (m) | Stratigraphy | Description                                                                       | Sample Type | Sample Number | Recovery (%) | Penetration Index / RQD | Shear Strength (kPa)            |      |      |     |
|---------------|---------------|--------------|-----------------------------------------------------------------------------------|-------------|---------------|--------------|-------------------------|---------------------------------|------|------|-----|
|               |               |              |                                                                                   |             |               |              |                         | 50                              | 100  | 150  | 200 |
|               |               |              |                                                                                   |             |               |              |                         | ■ DCPT<br>● SPT-N<br>○ MC<br>LL |      |      |     |
|               |               |              |                                                                                   |             |               |              |                         | 20                              | 40   | 60   | 80  |
| 0.2           |               |              | <b>SILTY SAND</b><br>trace rootlets, dark brown, moist, very loose.               |             |               |              |                         |                                 |      |      |     |
| 0.4           |               |              |                                                                                   |             | SS1           | 75           | 2                       | ● 2                             | ○ 13 |      |     |
| 0.6           |               |              |                                                                                   |             |               |              |                         |                                 |      |      |     |
| 0.8           | 106.5         |              | <b>SILTY SAND</b><br>trace gravel, trace clay, brown, moist, dense to very dense. |             |               |              |                         |                                 |      |      |     |
| 1.0           |               |              |                                                                                   |             | SS2           | 100          | 35                      | ○ 10                            | ● 35 |      |     |
| 1.2           |               |              |                                                                                   |             |               |              |                         |                                 |      |      |     |
| 1.4           |               |              |                                                                                   |             |               |              |                         |                                 |      |      |     |
| 1.6           |               |              |                                                                                   |             |               |              |                         |                                 |      |      |     |
| 1.8           |               |              |                                                                                   |             | SS3           | 100          | 68                      | ○ 8                             |      | ● 68 |     |
| 2.0           |               |              |                                                                                   |             |               |              |                         |                                 |      |      |     |
| 2.2           |               |              |                                                                                   |             |               |              |                         |                                 |      |      |     |
| 2.4           |               |              |                                                                                   |             |               |              |                         |                                 |      |      |     |
| 2.6           |               |              |                                                                                   |             |               |              |                         |                                 |      |      |     |
| 2.8           |               |              |                                                                                   |             | SS4           | 100          | 61                      | ○ 8                             |      | ● 61 |     |
| 3.0           | 104.2         |              |                                                                                   |             |               |              |                         |                                 |      |      |     |
| 3.2           |               |              | End of borehole at target depth of 3.0 mbg.                                       |             |               |              |                         |                                 |      |      |     |

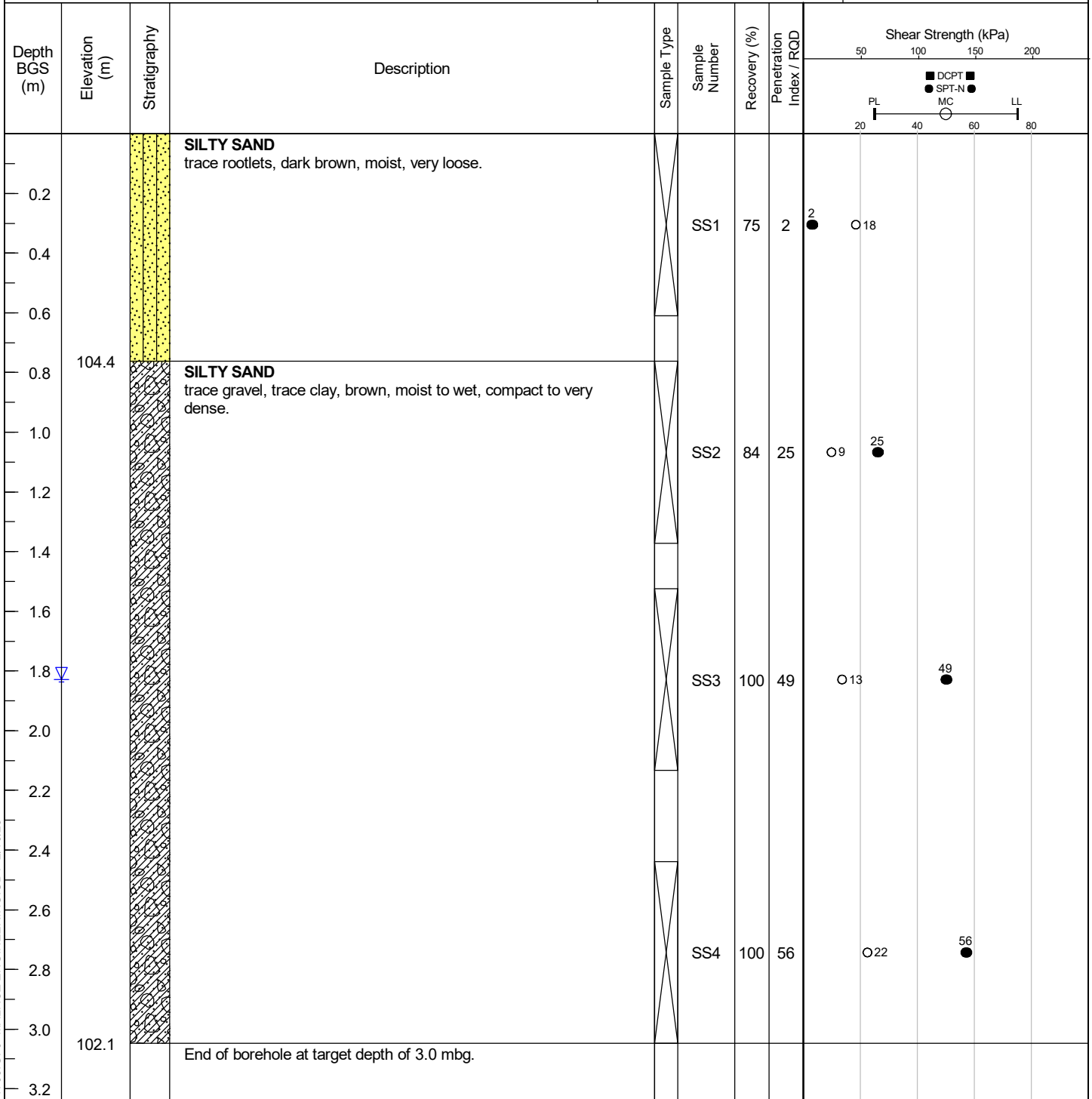
NOTES: Borehole open upon completion



# BOREHOLE LOG

Enclosure No.: 26

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Borehole Number: <b>BH126</b>          |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 1, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 1, 2022</b> |  | Elevation: <b>105.2 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326738.8 m</b>             |  | Northing: <b>4874751.1 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |



NOTES: Borehole caved to 2.4 mbg upon completion

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Borehole Number: <b>BH127</b>          |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 1, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 1, 2022</b> |  | Elevation: <b>104.8 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326705.3 m</b>             |  | Northing: <b>4874737.8 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |

| Depth BGS (m) | Elevation (m) | Stratigraphy | Description                                                                          | Sample Type | Sample Number | Recovery (%) | Penetration Index / RQD | Shear Strength (kPa)                                                                                                                                                                                                                                                                                                                                                      |     |     |     |
|---------------|---------------|--------------|--------------------------------------------------------------------------------------|-------------|---------------|--------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|
|               |               |              |                                                                                      |             |               |              |                         | 50                                                                                                                                                                                                                                                                                                                                                                        | 100 | 150 | 200 |
|               |               |              |                                                                                      |             |               |              |                         | DCPT <br>SPT-N <br>MC <br>LL  |     |     |     |
| 0.2           |               |              | <b>SILTY SAND</b><br>trace clay, trace rootlets, dark brown, loose.                  |             |               |              |                         |                                                                                                                                                                                                                                                                                                                                                                           |     |     |     |
| 0.4           |               |              |                                                                                      |             |               |              |                         |                                                                                                                                                                                                                                                                                                                                                                           |     |     |     |
| 0.6           |               |              |                                                                                      |             |               |              |                         |                                                                                                                                                                                                                                                                                                                                                                           |     |     |     |
| 0.8           | 104.0         |              | <b>SILTY SAND</b><br>trace gravel, trace clay, light brown, moist, compact to dense. |             |               |              |                         |                                                                                                                                                                                                                                                                                                                                                                           |     |     |     |
| 1.0           |               |              |                                                                                      |             |               |              |                         |                                                                                                                                                                                                                                                                                                                                                                           |     |     |     |
| 1.2           |               |              |                                                                                      |             |               |              |                         |                                                                                                                                                                                                                                                                                                                                                                           |     |     |     |
| 1.4           |               |              |                                                                                      |             |               |              |                         |                                                                                                                                                                                                                                                                                                                                                                           |     |     |     |
| 1.6           |               |              | Becoming moist to wet (1.5 mbg).                                                     |             |               |              |                         |                                                                                                                                                                                                                                                                                                                                                                           |     |     |     |
| 1.8           |               |              |                                                                                      |             |               |              |                         |                                                                                                                                                                                                                                                                                                                                                                           |     |     |     |
| 2.0           |               |              |                                                                                      |             |               |              |                         |                                                                                                                                                                                                                                                                                                                                                                           |     |     |     |
| 2.2           |               |              |                                                                                      |             |               |              |                         |                                                                                                                                                                                                                                                                                                                                                                           |     |     |     |
| 2.4           |               |              |                                                                                      |             |               |              |                         |                                                                                                                                                                                                                                                                                                                                                                           |     |     |     |
| 2.6           |               |              |                                                                                      |             |               |              |                         |                                                                                                                                                                                                                                                                                                                                                                           |     |     |     |
| 2.8           | 102.0         |              | End of borehole at 2.8 mbg by sampler refusal on inferred cobbles and boulders.      |             |               |              |                         |                                                                                                                                                                                                                                                                                                                                                                           |     |     |     |
| 3.0           |               |              |                                                                                      |             |               |              |                         |                                                                                                                                                                                                                                                                                                                                                                           |     |     |     |
| 3.2           |               |              |                                                                                      |             |               |              |                         |                                                                                                                                                                                                                                                                                                                                                                           |     |     |     |

|                                                         |  |  |  |  |  |  |  |  |  |  |  |
|---------------------------------------------------------|--|--|--|--|--|--|--|--|--|--|--|
| <b>NOTES:</b> Borehole caved to 1.8 mbg upon completion |  |  |  |  |  |  |  |  |  |  |  |
|---------------------------------------------------------|--|--|--|--|--|--|--|--|--|--|--|



# BOREHOLE LOG

Enclosure No.: 28

|                                                                                    |  |  |  |                                        |  |                                |  |
|------------------------------------------------------------------------------------|--|--|--|----------------------------------------|--|--------------------------------|--|
| Project Name: <b>Geotechnical Investigation</b>                                    |  |  |  | Project Number: <b>1703.00-100</b>     |  |                                |  |
| Client: <b>Algonquin and Lakeshore Catholic School Board</b>                       |  |  |  | Borehole Number: <b>BH128</b>          |  | Page: <b>1 of 1</b>            |  |
| Location: <b>175 Talbot Street, Picton ON</b>                                      |  |  |  | Date Started: <b>December 1, 2022</b>  |  | Datum: <b>NAD 83, Zone 18T</b> |  |
| Drilling Contractor: <b>G.E.T. Drilling Ltd.</b>                                   |  |  |  | Date Finished: <b>December 1, 2022</b> |  | Elevation: <b>103.5 m</b>      |  |
| Drilling Equipment: <b>CME 55</b>                                                  |  |  |  | Easting: <b>326672.8 m</b>             |  | Northing: <b>4874746.6 m</b>   |  |
| Drilling and Sampling Method: <b>200 mm hollow-stem augers, 50 mm split spoons</b> |  |  |  | Logged By: <b>C. Meyers</b>            |  | Checked By: <b>D. Hill</b>     |  |

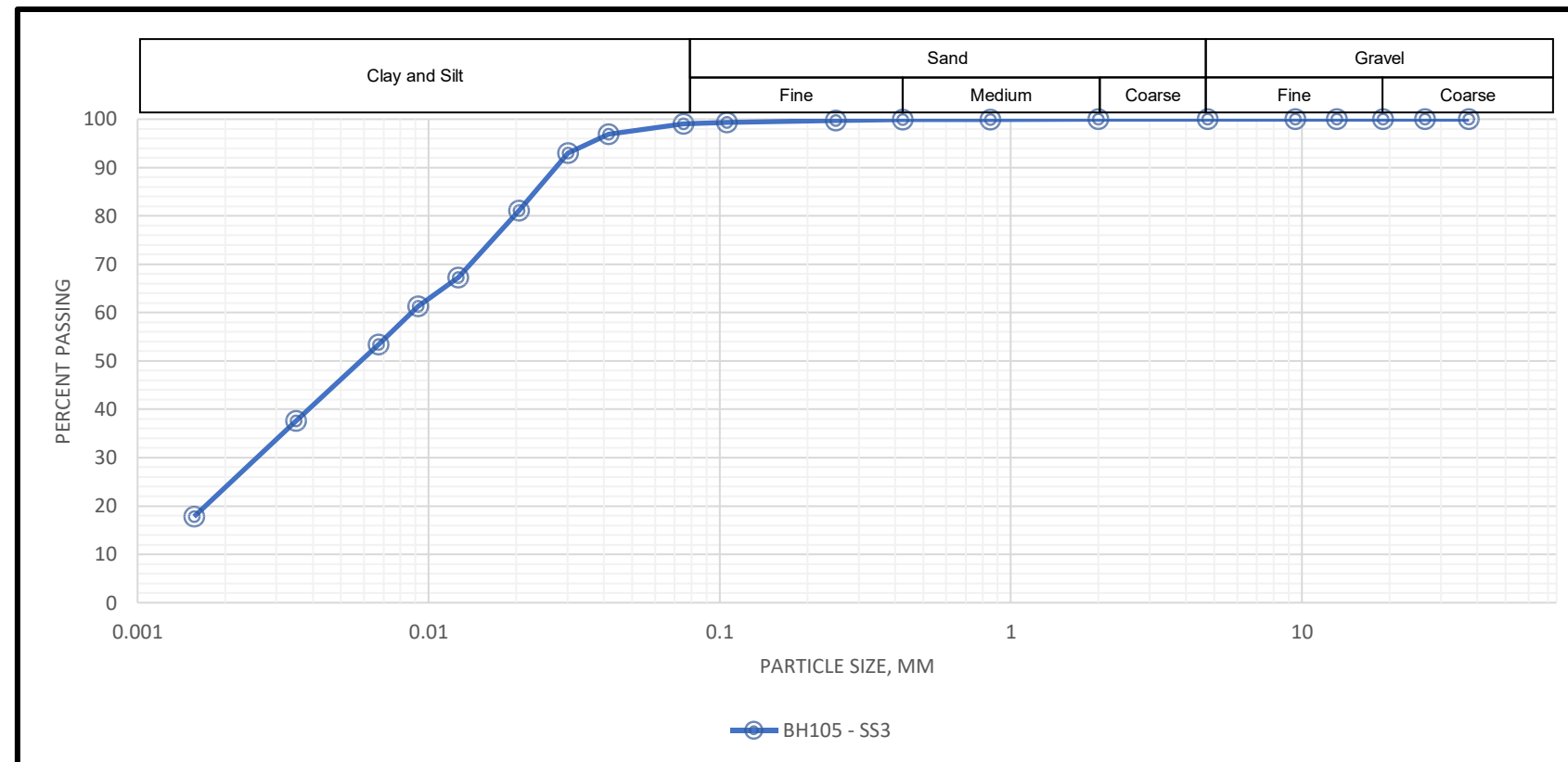
| Depth<br>BGS<br>(m) | Elevation<br>(m) | Stratigraphy | Description                                                                     | Sample Type | Sample Number | Recovery (%) | Penetration Index / RQD | Shear Strength (kPa)                                                                                                                     |     |     |     |
|---------------------|------------------|--------------|---------------------------------------------------------------------------------|-------------|---------------|--------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|
|                     |                  |              |                                                                                 |             |               |              |                         | 50                                                                                                                                       | 100 | 150 | 200 |
|                     |                  |              |                                                                                 |             |               |              |                         | <div>■ DCPT</div> <div>● SPT-N</div> <div>○ MC</div> <div>LL</div> <div>PL</div> <div>20</div> <div>40</div> <div>60</div> <div>80</div> |     |     |     |
| 0.2                 |                  |              | <b>SILTY SAND</b><br>trace gravel, trace rootlets, dark brown, moist, loose.    |             | SS1           | 67           | 7                       | 7                                                                                                                                        | 12  |     |     |
| 0.4                 |                  |              |                                                                                 |             |               |              |                         |                                                                                                                                          |     |     |     |
| 0.6                 |                  |              |                                                                                 |             |               |              |                         |                                                                                                                                          |     |     |     |
| 0.8                 |                  |              |                                                                                 |             |               |              |                         |                                                                                                                                          |     |     |     |
| 1.0                 |                  |              | Becoming brown, moist to wet<br>G: 9%, S: 76%, SI: 12%, CL: 3%.                 |             | SS2           | 92           | 4                       | 4                                                                                                                                        | 17  |     |     |
| 1.2                 |                  |              |                                                                                 |             |               |              |                         |                                                                                                                                          |     |     |     |
| 1.4                 |                  |              |                                                                                 |             |               |              |                         |                                                                                                                                          |     |     |     |
| 1.6                 | 101.9            |              |                                                                                 |             |               |              |                         |                                                                                                                                          |     |     |     |
| 1.8                 |                  |              | <b>SILTY SAND</b><br>trace gravel, trace clay, brown, wet, compact.             |             | SS3           | 100          | 17                      | 17                                                                                                                                       | 5   |     |     |
| 2.0                 |                  |              |                                                                                 |             |               |              |                         |                                                                                                                                          |     |     |     |
| 2.2                 |                  |              |                                                                                 |             |               |              |                         |                                                                                                                                          |     |     |     |
| 2.4                 |                  |              |                                                                                 |             |               |              |                         |                                                                                                                                          |     |     |     |
| 2.6                 |                  |              |                                                                                 |             | SS4           | 59           | R                       |                                                                                                                                          | 9   |     |     |
| 2.8                 | 100.7            |              |                                                                                 |             |               |              |                         |                                                                                                                                          |     |     |     |
| 3.0                 |                  |              | End of borehole at 2.8 mbg by sampler refusal on inferred cobbles and boulders. |             |               |              |                         |                                                                                                                                          |     |     |     |
| 3.2                 |                  |              |                                                                                 |             |               |              |                         |                                                                                                                                          |     |     |     |

NOTES: Borehole caved to 1.5 mbg upon completion

## **Appendix C**

### **Geotechnical Laboratory Results**

### Particle Size Analysis (LS-702)

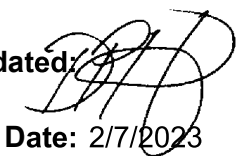


| Clay (%)                    | Silt (%) | Sand (%) | Gravel (%) |
|-----------------------------|----------|----------|------------|
| 18                          | 81       | 1        | 0          |
| SILT, some clay, trace sand |          |          |            |

**Lab No.:** 406A  
**Project No.:** 1703  
**BH:** BH105 - SS3  
**Location:** 175 Talbot Street, Picton ON  
**Client:** ALCDSB  
**Client Ref.:** -

**Tested:** DS  
**Date:** Feb 3rd, 2023

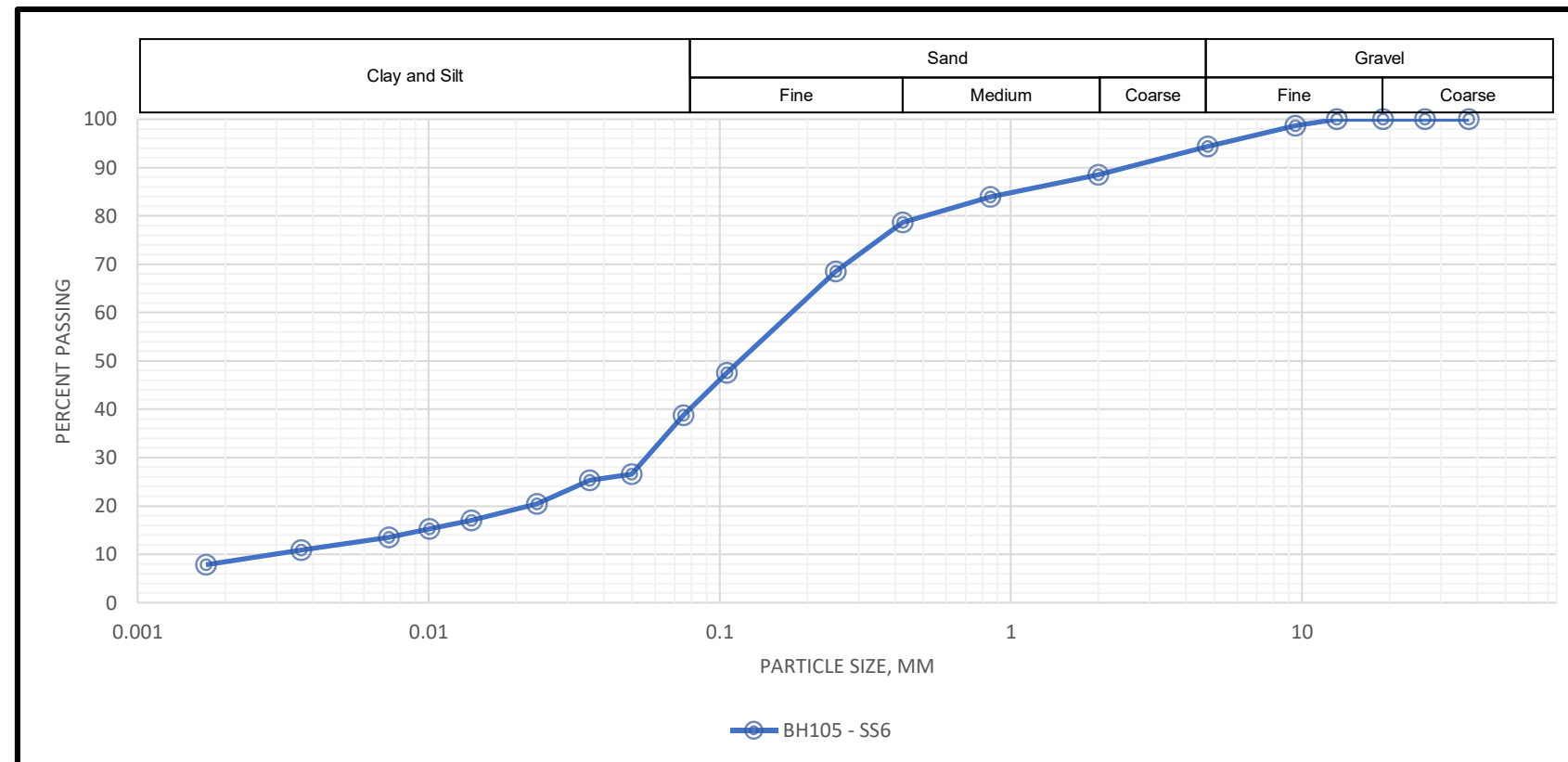
**Notes:** Additional information available upon request

**Validated:**   
**Date:** 2/7/2023

#### Supplementary Data:

|                                          |       |                                          |      |
|------------------------------------------|-------|------------------------------------------|------|
| Volume of bulb, $V_B$ (cm <sup>3</sup> ) | 57    | Specific Gravity, $G_S$ (assumed):       | 2.7  |
| Length of Bulb, $L_2$ (cm)               | 13.60 | Dispersing Agent (g/L):                  | 40   |
| Scale Dimension, $h_s$ (cm/Div.)         | 0.18  | Area of Cylinder, $A$ (cm <sup>2</sup> ) | 28.3 |
| Maximum Particle Size (mm):              | 19    | 0' to Top of Bulb, $L_1$ (cm)            | 11.0 |

### Particle Size Analysis (LS-702)

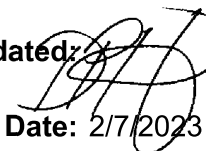


| Clay (%)                             | Silt (%) | Sand (%) | Gravel (%) |
|--------------------------------------|----------|----------|------------|
| 8                                    | 31       | 55       | 6          |
| SILTY SAND, trace clay, trace gravel |          |          |            |

**Lab No.:** 406B  
**Project No.:** 1703  
**BH:** BH105 - SS6  
**Location:** 175 Talbot Street, Picton ON  
**Client:** ALCDSB  
**Client Ref.:** -

**Tested:** DS  
**Date:** Feb 1st, 2023

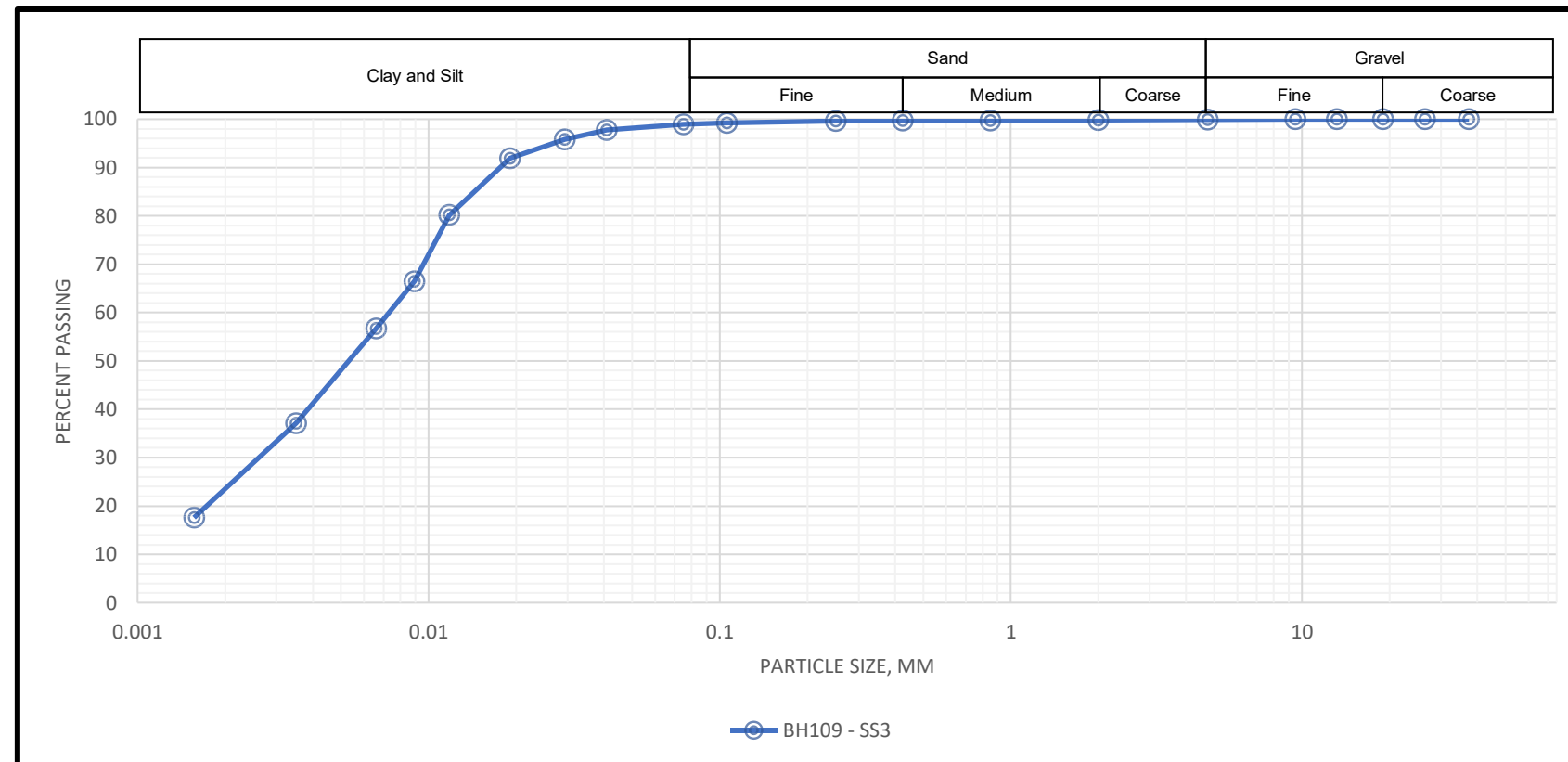
**Notes:** Additional information available upon request

**Validated:**   
**Date:** 2/7/2023

#### Supplementary Data:

|                                          |       |                                          |      |
|------------------------------------------|-------|------------------------------------------|------|
| Volume of bulb, $V_B$ (cm <sup>3</sup> ) | 57    | Specific Gravity, $G_S$ (assumed):       | 2.7  |
| Length of Bulb, $L_2$ (cm)               | 13.60 | Dispersing Agent (g/L):                  | 24   |
| Scale Dimension, $h_s$ (cm/Div.)         | 0.18  | Area of Cylinder, $A$ (cm <sup>2</sup> ) | 28.3 |
| Maximum Particle Size (mm):              | 19    | 0' to Top of Bulb, $L_1$ (cm)            | 11.0 |

### Particle Size Analysis (LS-702)

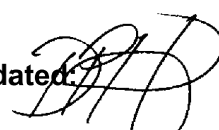


| Clay (%)                    | Silt (%) | Sand (%) | Gravel (%) |
|-----------------------------|----------|----------|------------|
| 18                          | 81       | 1        | 0          |
| SILT, some clay, trace sand |          |          |            |

**Lab No.:** 406C  
**Project No.:** 1703  
**BH:** BH109 - SS3  
**Location:** 175 Talbot Street, Picton ON  
**Client:** ALCDSB  
**Client Ref.:** -

**Tested:** DS  
**Date:** Feb 3rd, 2023

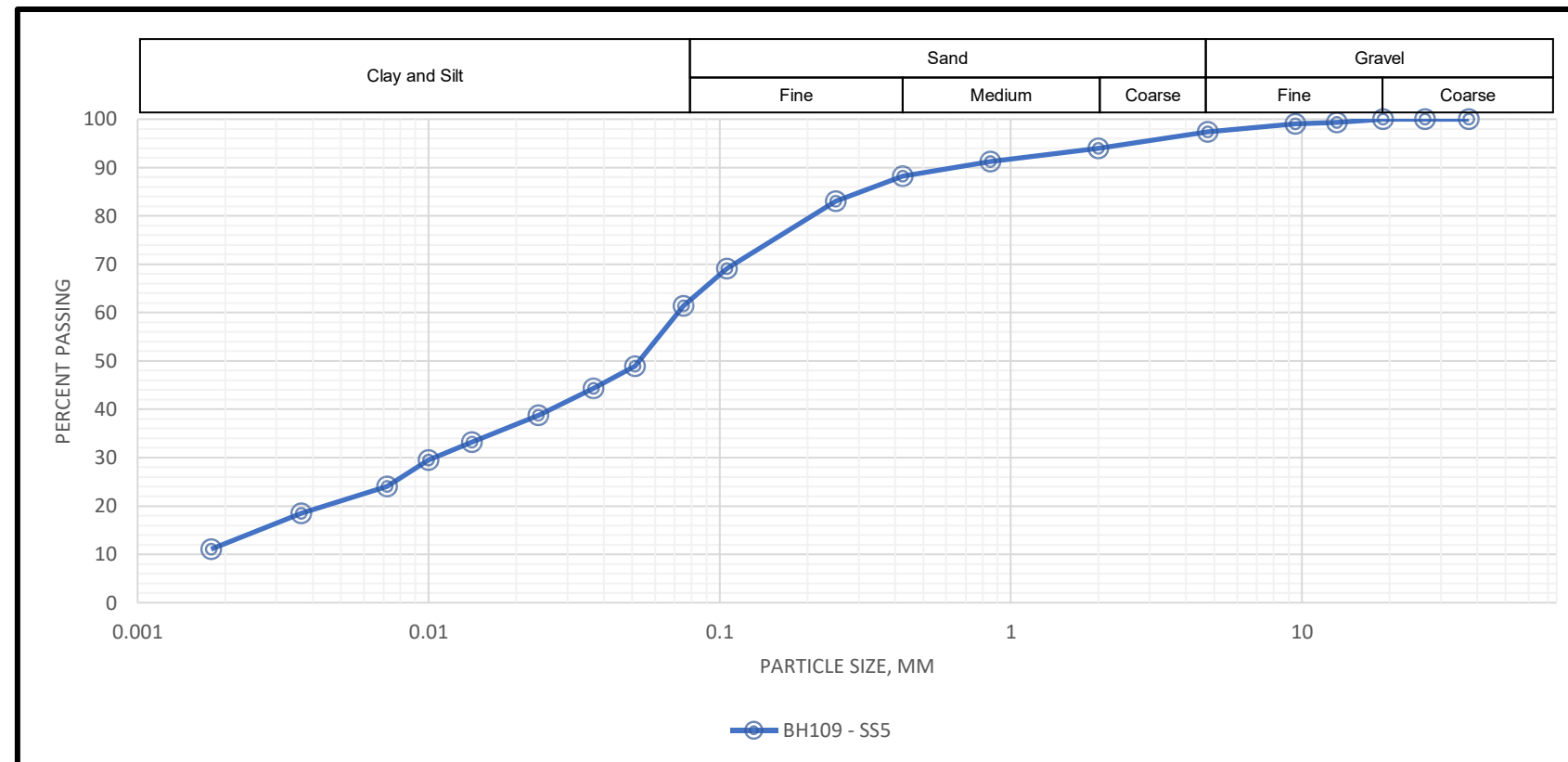
**Notes:** Additional information available upon request

**Validated:**   
**Date:** 2/7/2023

#### Supplementary Data:

|                                          |       |                                          |      |
|------------------------------------------|-------|------------------------------------------|------|
| Volume of bulb, $V_B$ (cm <sup>3</sup> ) | 57    | Specific Gravity, $G_S$ (assumed):       | 2.7  |
| Length of Bulb, $L_2$ (cm)               | 13.60 | Dispersing Agent (g/L):                  | 40   |
| Scale Dimension, $h_s$ (cm/Div.)         | 0.18  | Area of Cylinder, $A$ (cm <sup>2</sup> ) | 28.3 |
| Maximum Particle Size (mm):              | 19    | 0' to Top of Bulb, $L_1$ (cm)            | 11.0 |

### Particle Size Analysis (LS-702)

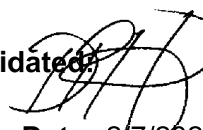


| Clay (%)                               | Silt (%) | Sand (%) | Gravel (%) |
|----------------------------------------|----------|----------|------------|
| 11                                     | 50       | 36       | 3          |
| SILT and SAND, some clay, trace gravel |          |          |            |

**Lab No.:** 406D  
**Project No.:** 1703  
**BH:** BH109 - SS5  
**Location:** 175 Talbot Street, Picton ON  
**Client:** ALCDSB  
**Client Ref.:** -

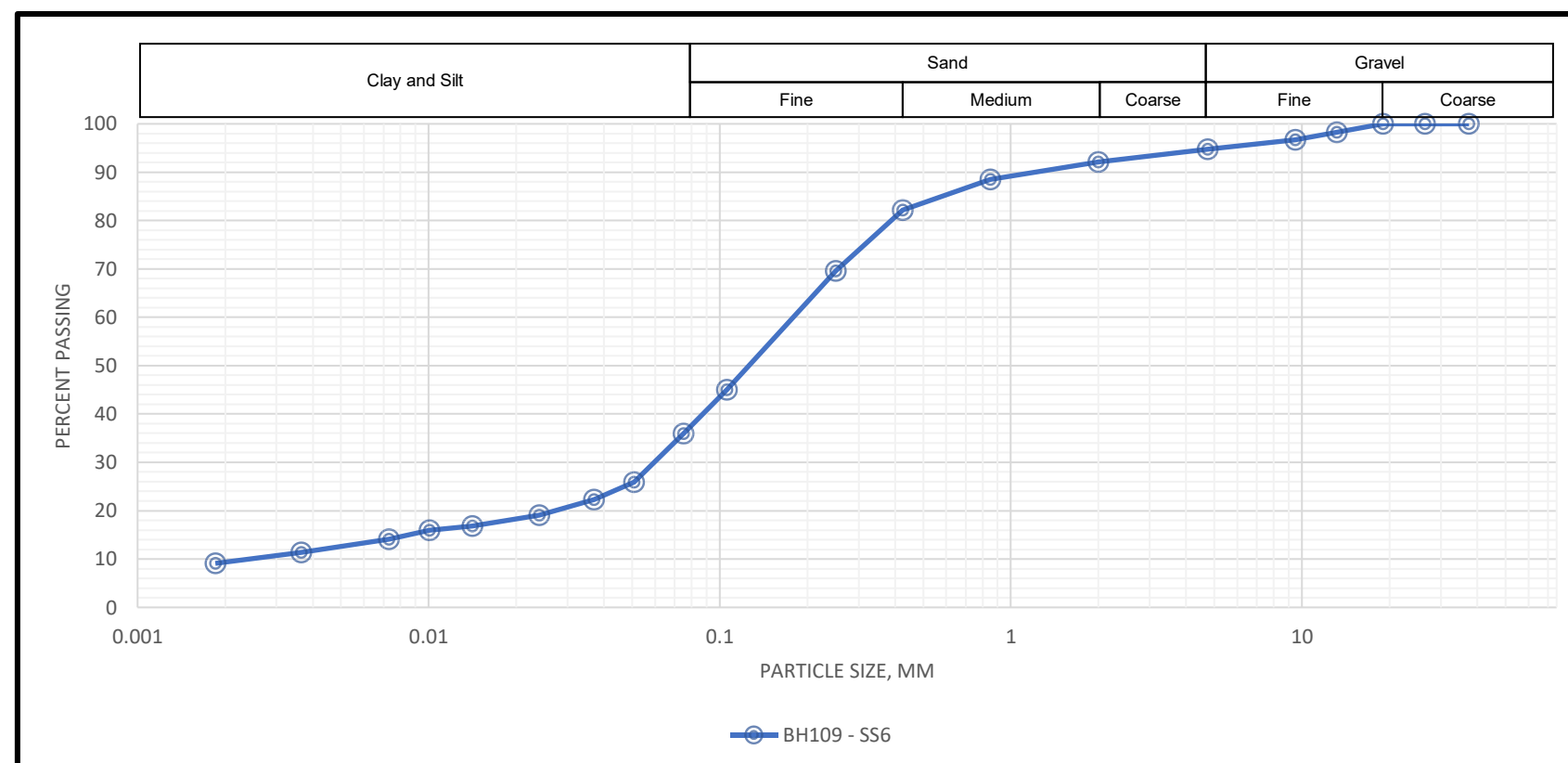
**Tested:** DS  
**Date:** Feb 2nd, 2023

**Notes:** Additional information available upon request

**Validated:**   
**Date:** 2/7/2023

| <u>Supplementary Data:</u>               |       |                                          |      |
|------------------------------------------|-------|------------------------------------------|------|
| Volume of bulb, $V_B$ (cm <sup>3</sup> ) | 57    | Specific Gravity, $G_S$ (assumed):       | 2.7  |
| Length of Bulb, $L_2$ (cm)               | 13.60 | Dispersing Agent (g/L):                  | 40   |
| Scale Dimension, $h_s$ (cm/Div.)         | 0.18  | Area of Cylinder, $A$ (cm <sup>2</sup> ) | 28.3 |
| Maximum Particle Size (mm):              | 19    | 0' to Top of Bulb, $L_1$ (cm)            | 11.0 |

### Particle Size Analysis (LS-702)

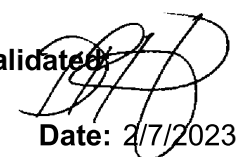


| Clay (%)                             | Silt (%) | Sand (%) | Gravel (%) |
|--------------------------------------|----------|----------|------------|
| 9                                    | 27       | 59       | 5          |
| SILTY SAND, trace clay, trace gravel |          |          |            |

Lab No.: 406E  
Project No.: 1703  
BH: BH109 - SS6  
Location: 175 Talbot Street, Picton ON  
Client: ALCDSB  
Client Ref.: -

Tested: DS  
Date: Feb 1st, 2023

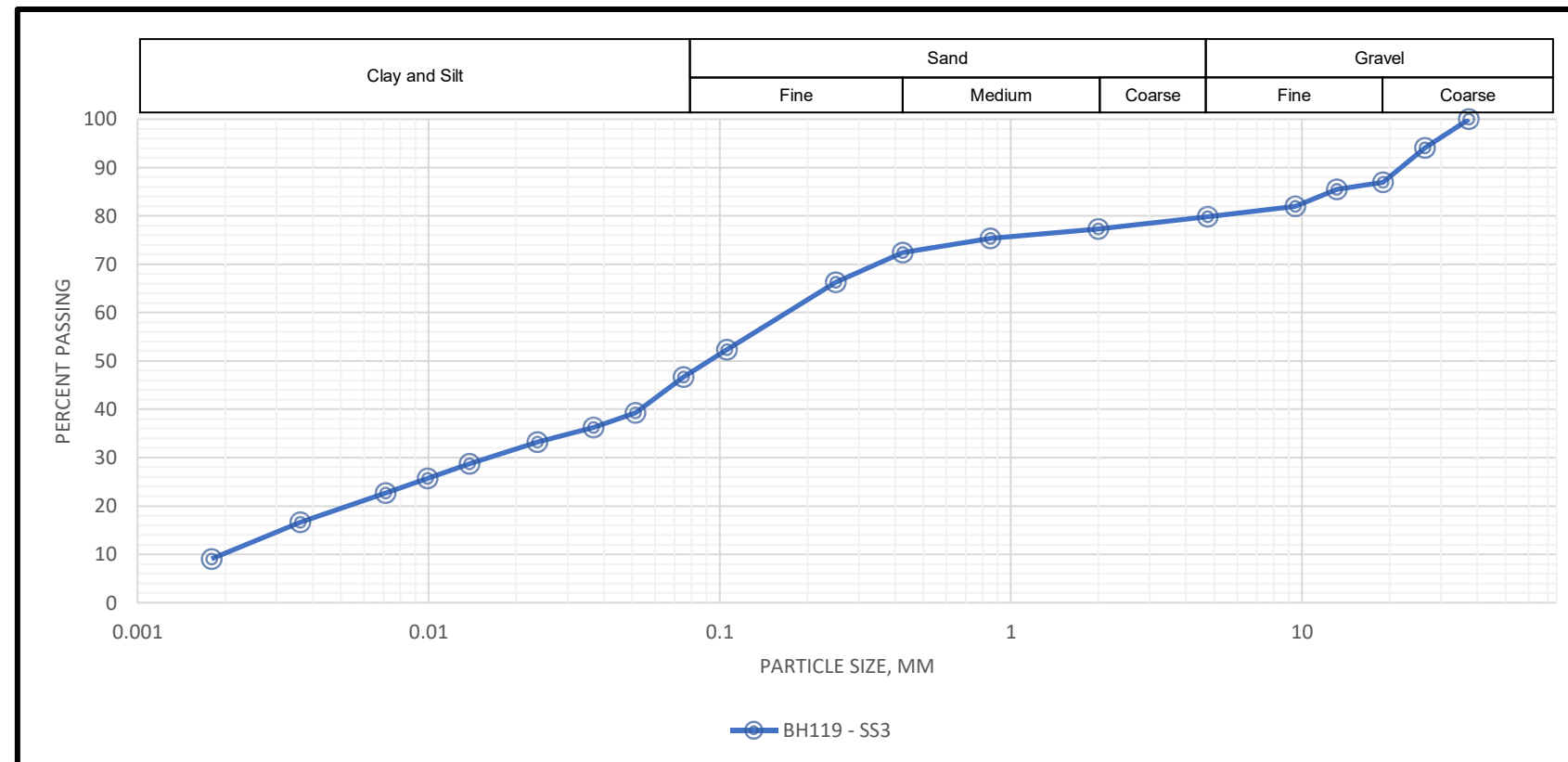
Notes: Additional information available upon request

Validated:   
Date: 2/7/2023

#### Supplementary Data:

|                                          |       |                                          |      |
|------------------------------------------|-------|------------------------------------------|------|
| Volume of bulb, $V_B$ (cm <sup>3</sup> ) | 57    | Specific Gravity, $G_S$ (assumed):       | 2.7  |
| Length of Bulb, $L_2$ (cm)               | 13.60 | Dispersing Agent (g/L):                  | 40   |
| Scale Dimension, $h_s$ (cm/Div.)         | 0.18  | Area of Cylinder, $A$ (cm <sup>2</sup> ) | 28.3 |
| Maximum Particle Size (mm):              | 19    | 0' to Top of Bulb, $L_1$ (cm)            | 11.0 |

### Particle Size Analysis (LS-702)

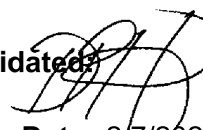


| Clay (%)                        | Silt (%) | Sand (%) | Gravel (%) |
|---------------------------------|----------|----------|------------|
| 9                               | 38       | 33       | 20         |
| SANDY GRAVELLY SILT, trace clay |          |          |            |

**Lab No.:** 406F  
**Project No.:** 1703  
**BH:** BH119 - SS3  
**Location:** 175 Talbot Street, Picton ON  
**Client:** ALCDSB  
**Client Ref.:** -

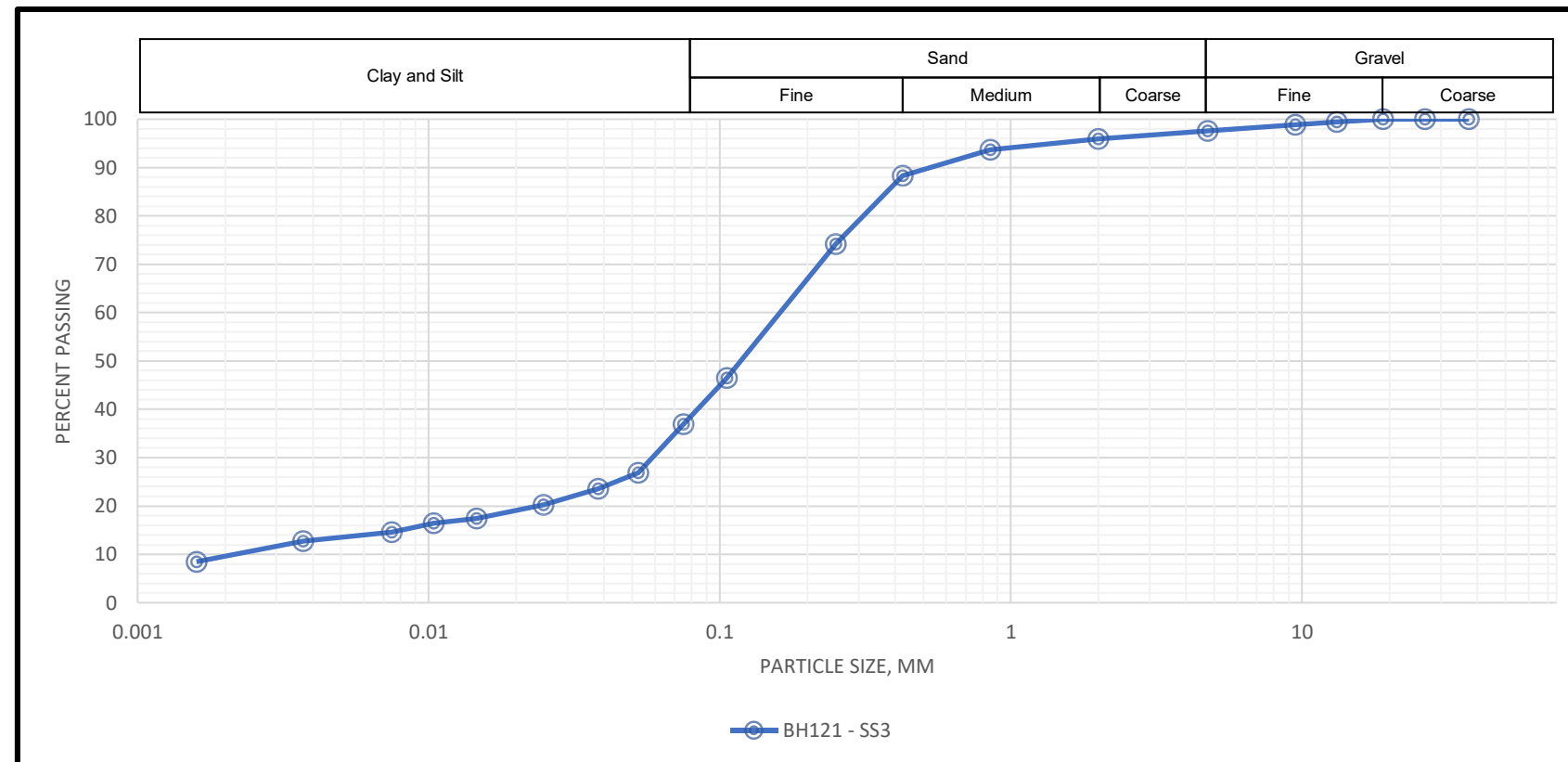
**Tested:** DS  
**Date:** Feb 2nd, 2023

**Notes:** Additional information available upon request

**Validated:**   
**Date:** 2/7/2023

| Supplementary Data:                      |       |                                          |      |
|------------------------------------------|-------|------------------------------------------|------|
| Volume of bulb, $V_B$ (cm <sup>3</sup> ) | 57    | Specific Gravity, $G_S$ (assumed):       | 2.7  |
| Length of Bulb, $L_2$ (cm)               | 13.60 | Dispersing Agent (g/L):                  | 40   |
| Scale Dimension, $h_s$ (cm/Div.)         | 0.18  | Area of Cylinder, $A$ (cm <sup>2</sup> ) | 28.3 |
| Maximum Particle Size (mm):              | 19    | 0' to Top of Bulb, $L_1$ (cm)            | 11.0 |

### Particle Size Analysis (LS-702)

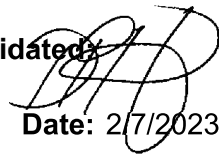


| Clay (%)                             | Silt (%) | Sand (%) | Gravel (%) |
|--------------------------------------|----------|----------|------------|
| 8                                    | 29       | 61       | 2          |
| SILTY SAND, trace clay, trace gravel |          |          |            |

Lab No.: 406G  
Project No.: 1703  
BH: BH121 - SS3  
Location: 175 Talbot Street, Picton ON  
Client: ALCDSB  
Client Ref.: -

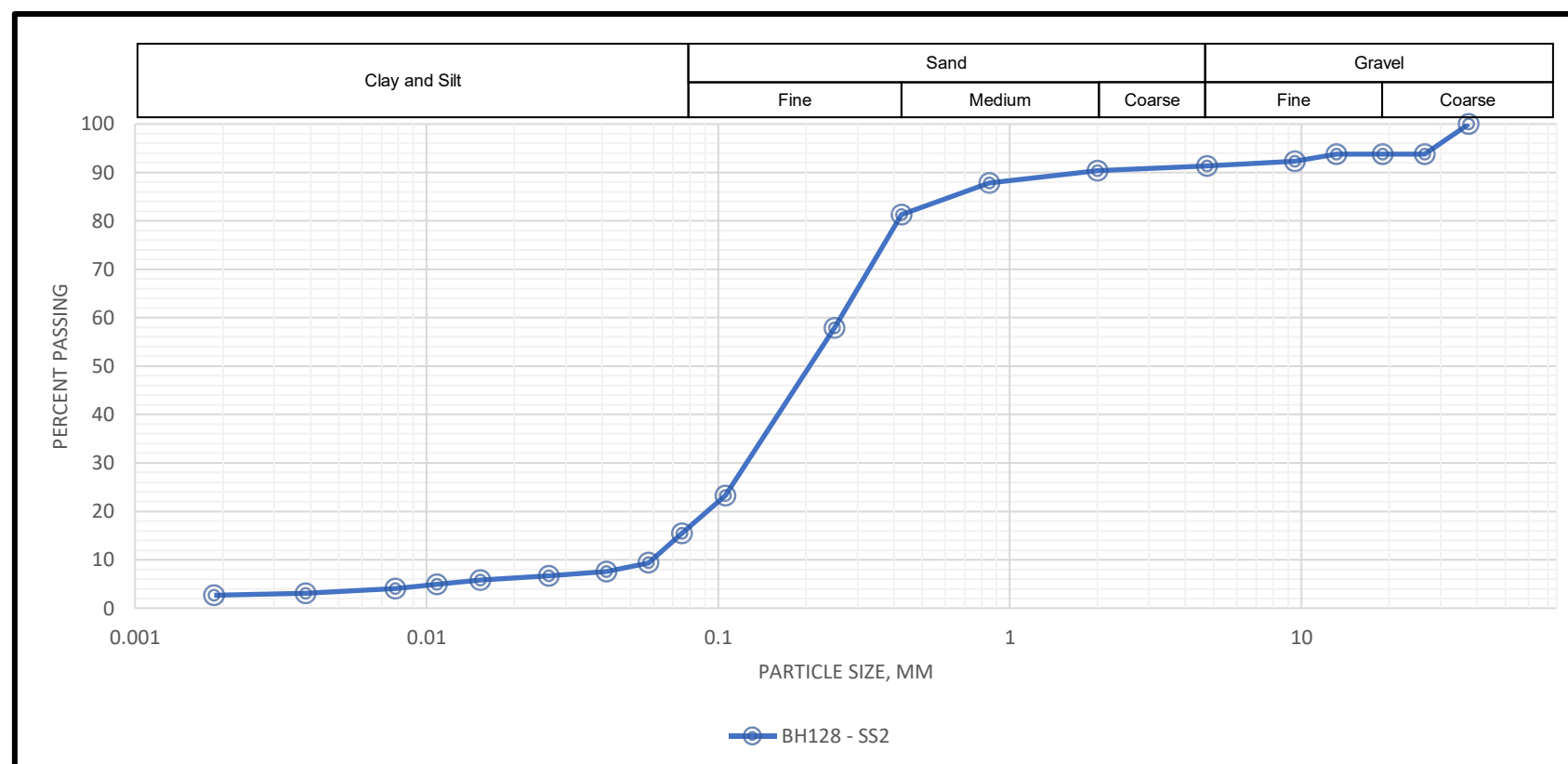
Tested: DS  
Date: Feb 3rd, 2023

Notes: Additional information available upon request

Validated:   
Date: 2/7/2023

| Supplementary Data:                      |       |                                          |      |
|------------------------------------------|-------|------------------------------------------|------|
| Volume of bulb, $V_B$ (cm <sup>3</sup> ) | 57    | Specific Gravity, $G_S$ (assumed):       | 2.7  |
| Length of Bulb, $L_2$ (cm)               | 13.60 | Dispersing Agent (g/L):                  | 24   |
| Scale Dimension, $h_s$ (cm/Div.)         | 0.18  | Area of Cylinder, $A$ (cm <sup>2</sup> ) | 28.3 |
| Maximum Particle Size (mm):              | 19    | 0' to Top of Bulb, $L_1$ (cm)            | 11.0 |

### Particle Size Analysis (LS-702)

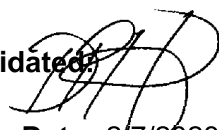


| Clay (%)                                  | Silt (%) | Sand (%) | Gravel (%) |
|-------------------------------------------|----------|----------|------------|
| 3                                         | 12       | 76       | 9          |
| SAND, some silt, trace gravel, trace clay |          |          |            |

**Lab No.:** 406H  
**Project No.:** 1703  
**BH:** BH128 - SS2  
**Location:** 175 Talbot Street, Picton ON  
**Client:** ALCDSB  
**Client Ref.:** -

**Tested:** DS  
**Date:** Feb 1st, 2023

**Notes:** Additional information available upon request

**Validated:**   
**Date:** 2/7/2023

#### Supplementary Data:

|                                          |       |                                          |      |
|------------------------------------------|-------|------------------------------------------|------|
| Volume of bulb, $V_B$ (cm <sup>3</sup> ) | 57    | Specific Gravity, $G_S$ (assumed):       | 2.7  |
| Length of Bulb, $L_2$ (cm)               | 13.60 | Dispersing Agent (g/L):                  | 24   |
| Scale Dimension, $h_s$ (cm/Div.)         | 0.18  | Area of Cylinder, $A$ (cm <sup>2</sup> ) | 28.3 |
| Maximum Particle Size (mm):              | 19    | 0' to Top of Bulb, $L_1$ (cm)            | 11.0 |

## **Appendix D**

### **Analytical Laboratory Results**

Table D1  
Soil Analytical Results

| Parameter                                        | Location<br>BH ID          |           | 175 Talbot Street |             |            |             |             |             |             |            |            |            |                                          | 2020 "Table 2.1":<br>Volume Independent,<br>Industrial / Commercial /<br>Community Property Use | 2020 "Table 2.1":<br>Volume Independent,<br>Residential / Parkland /<br>Institutional Property Use | 2011 "TABLE 1":<br>Residential / Parkland /<br>Institutional / Industrial /<br>Commercial / Community<br>Property Use |
|--------------------------------------------------|----------------------------|-----------|-------------------|-------------|------------|-------------|-------------|-------------|-------------|------------|------------|------------|------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
|                                                  | Sample ID                  |           | BH102             | BH103       | BH106      | BH110       | BH111       | BH112       | BH117       | BH118      | BH121      | BH126      | QA/QC                                    |                                                                                                 |                                                                                                    |                                                                                                                       |
|                                                  | Date Sampled               |           | BH102-S01         | BH103-S01   | BH106-S01  | BH110-S01   | BH111-S01   | BH112-S01   | BH117-S01   | BH118-S01  | BH121-S01  | BH126-S01  | BH103-S02 -<br>Duplicate of<br>BH103-S01 |                                                                                                 |                                                                                                    |                                                                                                                       |
|                                                  | Date Analyzed <sup>a</sup> |           | 2022-12-07        | 2022-12-07  | 2022-12-06 | 2022-12-06  | 2022-12-06  | 2022-12-05  | 2022-12-05  | 2022-12-05 | 2022-12-01 | 2022-12-01 | 2022-12-07                               |                                                                                                 |                                                                                                    |                                                                                                                       |
|                                                  | Lab Report ID              |           | 2250463-01        | 2250463-02  | 2250463-04 | 2250463-05  | 2250463-06  | 2250463-07  | 2250463-08  | 2250463-09 | 2250463-10 | 2250463-11 | 2250463-03                               |                                                                                                 |                                                                                                    |                                                                                                                       |
| Depth (m)                                        |                            | 0.76-1.37 | 1.07-1.37         | 0.30-0.61   | 0.30-0.61  | 0.61-1.37   | 0.76-1.37   | 0.76-1.37   | 0.30-0.61   | 0.76-1.07  | 0.76-1.37  | 1.07-1.37  |                                          |                                                                                                 |                                                                                                    |                                                                                                                       |
| Soil Type                                        |                            | Sand Till | Silty Sand        | Clayey Silt | Silty Sand | Clayey Silt | Clayey Silt | Clayey Silt | Clayey Silt | Sandy Silt | Sand Till  | Silty Sand |                                          |                                                                                                 |                                                                                                    |                                                                                                                       |
| Units                                            | MDL                        |           |                   |             |            |             |             |             |             |            |            |            |                                          |                                                                                                 |                                                                                                    |                                                                                                                       |
| PHCs                                             |                            |           |                   |             |            |             |             |             |             |            |            |            |                                          |                                                                                                 |                                                                                                    |                                                                                                                       |
| PHC F1(C6-C10)                                   | µg/g                       | 7         | <                 | <           | <          | <           | <           | <           | <           | <          | <          | <          | <                                        | 25                                                                                              | 25                                                                                                 | 25                                                                                                                    |
| PHC F2(>C10-C16)                                 | µg/g                       | 4         | <                 | <           | <          | <           | <           | <           | <           | <          | <          | <          | <                                        | 26                                                                                              | 10                                                                                                 | 10                                                                                                                    |
| PHC F3(>C16-C34)                                 | µg/g                       | 8         | <                 | <           | <          | <           | <           | <           | <           | <          | <          | <          | <                                        | 240                                                                                             | 240                                                                                                | 240                                                                                                                   |
| PHC F4(>C34-C50)                                 | µg/g                       | 6         | <                 | <           | <          | <           | <           | <           | <           | <          | <          | <          | <                                        | 3300                                                                                            | 2800                                                                                               | 120                                                                                                                   |
| BTEX                                             |                            |           |                   |             |            |             |             |             |             |            |            |            |                                          |                                                                                                 |                                                                                                    |                                                                                                                       |
| Benzene                                          | µg/g                       | 0.02      | <                 | <           | <          | <           | <           | <           | <           | <          | <          | <          | <                                        | 0.02                                                                                            | 0.02                                                                                               | 0.02                                                                                                                  |
| Ethylbenzene                                     | µg/g                       | 0.05      | <                 | <           | <          | <           | <           | <           | <           | <          | <          | <          | <                                        | 0.05                                                                                            | 0.05                                                                                               | 0.05                                                                                                                  |
| Toluene                                          | µg/g                       | 0.05      | <                 | <           | <          | <           | <           | <           | <           | <          | <          | <          | <                                        | 0.2                                                                                             | 0.2                                                                                                | 0.2                                                                                                                   |
| Xylene m,p-                                      | µg/g                       | 0.05      | <                 | <           | <          | <           | <           | <           | <           | <          | <          | <          | <                                        |                                                                                                 |                                                                                                    |                                                                                                                       |
| Xylene, o-                                       | µg/g                       | 0.05      | <                 | <           | <          | <           | <           | <           | <           | <          | <          | <          | <                                        |                                                                                                 |                                                                                                    |                                                                                                                       |
| Xylene (total)                                   | µg/g                       | 0.05      | <                 | <           | <          | <           | <           | <           | <           | <          | <          | <          | <                                        | 0.091                                                                                           | 0.091                                                                                              | 0.05                                                                                                                  |
| Metals and Inorganics                            |                            |           |                   |             |            |             |             |             |             |            |            |            |                                          |                                                                                                 |                                                                                                    |                                                                                                                       |
| Antimony                                         | µg/g                       | 1.0       | <                 | <           | <          | <           | <           | <           | <           | <          | <          | <          | <                                        | 40                                                                                              | a                                                                                                  | 1.3                                                                                                                   |
| Arsenic                                          | µg/g                       | 1.0       | 1.3               | 1.8         | 1.9        | 1.8         | 2.8         | 2.3         | 1.8         | 1.8        | 2.0        | 1.3        | 1.8                                      | 18                                                                                              |                                                                                                    | 18                                                                                                                    |
| Barium                                           | µg/g                       | 1.0       | 31.6              | 49.5        | 81.6       | 72.2        | 59.6        | 121         | 66.7        | 66.5       | 37.6       | 32.1       | 47.7                                     | 670                                                                                             | a                                                                                                  | 220                                                                                                                   |
| Beryllium                                        | µg/g                       | 0.5       | <                 | <           | <          | <           | <           | <           | <           | <          | <          | <          | <                                        | 8                                                                                               | a                                                                                                  | 2.5                                                                                                                   |
| Boron (Hot Water Soluble)                        | µg/g                       | 0.5       | <                 | <           | <          | <           | <           | <           | <           | <          | <          | <          | <                                        | 2                                                                                               |                                                                                                    | NA                                                                                                                    |
| Boron (total)                                    | µg/g                       | 5.0       | <                 | <           | 10.8       | 5.6         | 5.7         | 10.6        | 7.6         | 6.8        | <          | <          | <                                        | 120                                                                                             | a                                                                                                  | 36                                                                                                                    |
| Cadmium                                          | µg/g                       | 0.5       | <                 | <           | <          | <           | 0.8         | <           | <           | <          | <          | <          | <                                        | 1.9                                                                                             | a                                                                                                  | 1.2                                                                                                                   |
| Chromium (total)                                 | µg/g                       | 5.0       | 9.9               | 15.7        | 20.3       | 17.2        | 17.1        | 26.9        | 16.9        | 17.8       | 14.9       | 8.9        | 14.7                                     | 160                                                                                             | a                                                                                                  | 70                                                                                                                    |
| Chromium VI                                      | µg/g                       | 0.2       | <                 | <           | <          | <           | <           | <           | <           | <          | 0.2        | <          | <                                        | 8                                                                                               |                                                                                                    | 0.66                                                                                                                  |
| Cobalt                                           | µg/g                       | 1.0       | 3.3               | 4.6         | 5.7        | 5.0         | 5.4         | 7.2         | 4.8         | 4.6        | 3.7        | 3.1        | 4.4                                      | 80                                                                                              | a                                                                                                  | 21                                                                                                                    |
| Copper                                           | µg/g                       | 5.0       | 7.0               | 9.7         | 11.4       | 10.0        | 12.5        | 14.9        | 9.9         | 9.3        | 9.6        | 6.8        | 9.0                                      | 230                                                                                             | a                                                                                                  | 92                                                                                                                    |
| Lead                                             | µg/g                       | 1.0       | 1.9               | 3.1         | 3.5        | 3.1         | 3.8         | 4.3         | 2.9         | 3.7        | 3.1        | 1.7        | 2.9                                      | 120                                                                                             |                                                                                                    | 120                                                                                                                   |
| Mercury                                          | µg/g                       | 0.1       | <                 | <           | <          | <           | <           | <           | <           | <          | <          | <          | <                                        | 0.27                                                                                            |                                                                                                    | 0.27                                                                                                                  |
| Molybdenum                                       | µg/g                       | 1.0       | <                 | <           | <          | <           | 1.2         | <           | <           | <          | <          | <          | <                                        | 40                                                                                              | a                                                                                                  | 2                                                                                                                     |
| Nickel                                           | µg/g                       | 5.0       | 5.3               | 8.3         | 11.1       | 9.6         | 10.7        | 15.4        | 9.4         | 9.0        | 7.9        | 5.0        | 7.9                                      | 270                                                                                             | a                                                                                                  | 82                                                                                                                    |
| Selenium                                         | µg/g                       | 1.0       | <                 | <           | <          | <           | <           | <           | <           | <          | <          | <          | <                                        | 5.5                                                                                             | a                                                                                                  | 1.5                                                                                                                   |
| Silver                                           | µg/g                       | 0.3       | <                 | <           | <          | <           | 0.8         | <           | <           | <          | <          | <          | <                                        | 40                                                                                              | a                                                                                                  | 0.5                                                                                                                   |
| Thallium                                         | µg/g                       | 1.0       | <                 | <           | <          | <           | <           | <           | <           | <          | <          | <          | <                                        | 3.3                                                                                             | a                                                                                                  | 1                                                                                                                     |
| Uranium                                          | µg/g                       | 1.0       | <                 | <           | <          | <           | 1.2         | <           | <           | <          | <          | <          | <                                        | 33                                                                                              | a                                                                                                  | 2.5                                                                                                                   |
| Vanadium                                         | µg/g                       | 10.0      | 18.4              | 25.0        | 28.2       | 26.1        | 29.3        | 38.1        | 25.8        | 26.8       | 23.2       | 15.3       | 24.0                                     | 86                                                                                              |                                                                                                    | 86                                                                                                                    |
| Zinc                                             | µg/g                       | 20.0      | <                 | <           | 20.5       | 21.2        | 21.2        | 30.9        | <           | 21.2       | <          | <          | <                                        | 340                                                                                             | a                                                                                                  | 290                                                                                                                   |
| General Chemistry                                |                            |           |                   |             |            |             |             |             |             |            |            |            |                                          |                                                                                                 |                                                                                                    |                                                                                                                       |
| Conductivity                                     | mS/cm                      | 0.005     | 0.135             | 0.143       | 0.160      | 0.135       | 0.129       | 0.175       | 0.158       | 0.165      | 0.160      | 0.103      | 0.139                                    | 1.4                                                                                             | 0.7                                                                                                | 0.57                                                                                                                  |
| pH                                               | pH Units                   | 0.05      | -                 | 7.34        | -          | -           | -           | -           | -           | -          | 7.28       | -          | 7.33                                     |                                                                                                 |                                                                                                    |                                                                                                                       |
| Sodium Adsorption Ratio (SAR)                    | N/A                        | 0.01      | 0.36              | 0.09        | 0.09       | 0.09        | 0.09        | 0.09        | 0.09        | 0.10       | 0.11       | 0.11       | 0.09                                     | 12                                                                                              | 5                                                                                                  | 2.4                                                                                                                   |
| Pesticides, PCBs, and Other Regulated Parameters |                            |           |                   |             |            |             |             |             |             |            |            |            |                                          |                                                                                                 |                                                                                                    |                                                                                                                       |
| Aldrin                                           | µg/g                       | 0.01      | -                 | <           | -          | -           | -           | <           | -           | -          | <          | -          | <                                        | 0.088                                                                                           | 0.05                                                                                               | 0.05                                                                                                                  |
| Gamma-BHC (Lindane)                              | µg/g                       | 0.01      | -                 | <           | -          | -           | -           | <           | -           | -          | <          | -          | <                                        |                                                                                                 |                                                                                                    |                                                                                                                       |
| Alpha-Chlordane                                  | µg/g                       | 0.01      | -                 | <           | -          | -           | -           | <           | -           | -          | <          | -          | <                                        |                                                                                                 |                                                                                                    |                                                                                                                       |
| Gamma-Chlordane                                  | µg/g                       | 0.01      | -                 | <           | -          | -           | -           | <           | -           | -          | <          | -          | <                                        |                                                                                                 |                                                                                                    |                                                                                                                       |
| Chlordane                                        | µg/g                       | 0.01      | -                 | <           | -          | -           | -           | <           | -           | -          | <          | -          | <                                        | 0.05                                                                                            | 0.05                                                                                               | 0.05                                                                                                                  |
| DDD, 2,4-                                        | µg/g                       | 0.01      | -                 | <           | -          | -           | -           | <           | -           | -          | <          | -          | <                                        |                                                                                                 |                                                                                                    |                                                                                                                       |
| DDD, 4,4-                                        | µg/g                       | 0.02      | -                 | <           | -          | -           | -           | <           | -           | -          | <          | -          | <                                        |                                                                                                 |                                                                                                    |                                                                                                                       |
| DDD, (total)                                     | µg/g                       | 0.02      | -                 | <           | -          | -           | -           | <           | -           | -          | <          | -          | <                                        | 4.6                                                                                             | 3.3                                                                                                | 0.05                                                                                                                  |
| DDE, 2,4-                                        | µg/g                       | 0.01      | -                 | <           | -          | -           | -           | <           | -           | -          | <          | -          | <                                        |                                                                                                 |                                                                                                    |                                                                                                                       |
| DDE, 4,4-                                        | µg/g                       | 0.01      | -                 | <           | -          | -           | -           | <           | -           | -          | <          | -          | <                                        |                                                                                                 |                                                                                                    |                                                                                                                       |
| DDE, (total)                                     | µg/g                       | 0.01      | -                 | <           | -          | -           | -           | <           | -           | -          | <          | -          | <                                        | 0.52                                                                                            | 0.26                                                                                               | 0.05                                                                                                                  |
| DDT, 2,4-                                        | µg/g                       | 0.01      | -                 | <           | -          | -           | -           | <           | -           | -          | <          | -          | <                                        |                                                                                                 |                                                                                                    |                                                                                                                       |
| DDT, 4,4-                                        | µg/g                       | 0.01      | -                 | <           | -          | -           | -           | <           | -           | -          | <          | -          | <                                        |                                                                                                 |                                                                                                    |                                                                                                                       |
| DDT, (total)                                     | µg/g                       | 0.01      | -                 | <           | -          | -           | -           | <           | -           | -          | <          | -          | <                                        | 1.4                                                                                             | 1.4                                                                                                | 1.4                                                                                                                   |
| Dieldrin                                         | µg/g                       | 0.02      | -                 | <           | -          | -           | -           | <           | -           | -          | <          | -          | <                                        | 0.088                                                                                           | a                                                                                                  | 0.05                                                                                                                  |
| Endosulfan I                                     | µg/g                       | 0.01      | -                 | <           | -          | -           | -           | <           | -           | -          | <          | -          | <                                        |                                                                                                 | a                                                                                                  |                                                                                                                       |
| Endosulfan II                                    | µg/g                       | 0.02      | -                 | <           | -          | -           | -           | <           | -           | -          | <          | -          | <                                        |                                                                                                 |                                                                                                    |                                                                                                                       |
| Endrin                                           | µg/g                       | 0.02      | -                 | <           | -          | -           | -           | <           | -           | -          | <          | -          | <                                        | 0.04                                                                                            | a                                                                                                  | 0.04                                                                                                                  |
| Heptachlor                                       | µg/g                       | 0.01      | -                 | <           | -          | -           | -           | <           | -           | -          | <          | -          | <                                        | 0.072                                                                                           |                                                                                                    | 0.05                                                                                                                  |
| Heptachlor Epoxide                               | µg/g                       | 0.01      | -                 | <           | -          | -           | -           | <           | -           | -          | <          | -          | <                                        | 0.05                                                                                            | a                                                                                                  | 0.05                                                                                                                  |
| Hexachlorobenzene                                | µg/g                       | 0.01      | -                 | <           | -          | -           | -           | <           | -           | -          | <          | -          | <                                        | 0.034                                                                                           |                                                                                                    | 0.01                                                                                                                  |
| Hexachlorobutadiene                              | µg/g                       | 0.01      | -                 | <           | -          | -           | -           | <           | -           | -          | <          | -          | <                                        | 0.01                                                                                            |                                                                                                    | 0.01                                                                                                                  |
| Hexachloroethane                                 | µg/g                       | 0.01      | -                 | <           | -          | -           | -           | <           | -           | -          | <          | -          | <                                        |                                                                                                 |                                                                                                    |                                                                                                                       |
| Methoxychlor                                     | µg/g                       | 0.01      | -                 | <           | -          | -           | -           | <           | -           | -          | <          | -          | <                                        | 0.19                                                                                            | 0.13                                                                                               | 0.05                                                                                                                  |

Notes: BH###-S## borehole and sample ID  
MDL method detection limit  
< result below method detection limit  
- not analyzed  
na not applicable  
a sample analyzed between December 12, 2022 to December 15, 2022  
Red, bold and underline indicates value exceeds MECP (2022) Table 2.1 volume independent standards for Industrial/Commercial/Community Property Use  
Orange, bold and underline indicates value exceeds MECP (2022) Table 2.1 volume independent standards for Residential/Parkland/Institutional Property Use  
Blue, bold and underline indicates value exceeds MECP (2022) Table 1 standards for Residential/Parkland/Institutional/Industrial/Commercial/Community Property Use

Data Input: AS  
Data Check: SG

## Certificate of Analysis

### Malroz Engineering Incorporated

308 Wellington Street 2 nd Floor  
Kingston, ON K7K 7 A8  
Attn: Dave Hodgson

Client PO:  
Project: 1703  
Custody: 139664, 665

Report Date: 16-Dec-2022  
Order Date: 9-Dec-2022

**Order #: 2250463**

This Certificate of Analysis contains analytical data applicable to the following samples as submitted :

| Paracel ID | Client ID |
|------------|-----------|
| 2250463-01 | BH102 S01 |
| 2250463-02 | BH103 S01 |
| 2250463-03 | BH103 S02 |
| 2250463-04 | BH106 S01 |
| 2250463-05 | BH110 S01 |
| 2250463-06 | BH111 S01 |
| 2250463-07 | BH112 S01 |
| 2250463-08 | BH117 S01 |
| 2250463-09 | BH118 S01 |
| 2250463-10 | BH121 S01 |
| 2250463-11 | BH126 S01 |

Approved By:



Dale Robertson, BSc  
Laboratory Director

Certificate of Analysis

Report Date: 16-Dec-2022

Client: **Malroz Engineering Incorporated**

Order Date: 9-Dec-2022

Client PO:

Project Description: **1703**

## Analysis Summary Table

| Analysis                        | Method Reference/Description                     | Extraction Date | Analysis Date |
|---------------------------------|--------------------------------------------------|-----------------|---------------|
| Boron, available                | MOE (HWE), EPA 200.8 - ICP-MS                    | 13-Dec-22       | 13-Dec-22     |
| BTEX by P&T GC-MS               | EPA 8260 - P&T GC-MS                             | 12-Dec-22       | 13-Dec-22     |
| Chromium, hexavalent - soil     | MOE E3056 - Extraction, colourimetric            | 13-Dec-22       | 13-Dec-22     |
| Conductivity                    | MOE E3138 - probe @25 °C, water ext              | 13-Dec-22       | 13-Dec-22     |
| Mercury by CVAA                 | EPA 7471B - CVAA, digestion                      | 13-Dec-22       | 13-Dec-22     |
| pH, soil                        | EPA 150.1 - pH probe @ 25 °C, CaCl buffered ext. | 12-Dec-22       | 12-Dec-22     |
| PHC F1                          | CWS Tier 1 - P&T GC-FID                          | 12-Dec-22       | 13-Dec-22     |
| PHCs F2 to F4                   | CWS Tier 1 - GC-FID, extraction                  | 12-Dec-22       | 14-Dec-22     |
| REG 153: Metals by ICP/MS, soil | EPA 6020 - Digestion - ICP-MS                    | 13-Dec-22       | 13-Dec-22     |
| REG 153: Pesticides, OC         | EPA 8081B - GC-ECD                               | 9-Dec-22        | 15-Dec-22     |
| SAR                             | Calculated                                       | 13-Dec-22       | 13-Dec-22     |
| Solids, %                       | CWS Tier 1 - Gravimetric                         | 12-Dec-22       | 12-Dec-22     |

Certificate of Analysis

Report Date: 16-Dec-2022

Client: Malroz Engineering Incorporated

Order Date: 9-Dec-2022

Client PO:

Project Description: 1703

| Client ID:   | BH102 S01       | BH103 S01       | BH103 S02       | BH106 S01       |
|--------------|-----------------|-----------------|-----------------|-----------------|
| Sample Date: | 07-Dec-22 11:00 | 07-Dec-22 10:00 | 07-Dec-22 10:00 | 06-Dec-22 09:00 |
| Sample ID:   | 2250463-01      | 2250463-02      | 2250463-03      | 2250463-04      |
| MDL/Units    | Soil            | Soil            | Soil            | Soil            |

**Physical Characteristics**

|          |              |      |      |      |      |
|----------|--------------|------|------|------|------|
| % Solids | 0.1 % by Wt. | 90.2 | 80.7 | 80.8 | 80.0 |
|----------|--------------|------|------|------|------|

**General Inorganics**

|              |               |      |      |      |      |
|--------------|---------------|------|------|------|------|
| SAR          | 0.01 N/A      | 0.36 | 0.09 | 0.09 | 0.09 |
| Conductivity | 5 uS/cm       | 135  | 143  | 139  | 160  |
| pH           | 0.05 pH Units | -    | 7.34 | 7.33 | -    |

**Metals**

|                  |               |       |       |       |      |
|------------------|---------------|-------|-------|-------|------|
| Antimony         | 1.0 ug/g dry  | <1.0  | <1.0  | <1.0  | <1.0 |
| Arsenic          | 1.0 ug/g dry  | 1.3   | 1.8   | 1.8   | 1.9  |
| Barium           | 1.0 ug/g dry  | 31.6  | 49.5  | 47.7  | 81.6 |
| Beryllium        | 0.5 ug/g dry  | <0.5  | <0.5  | <0.5  | <0.5 |
| Boron            | 5.0 ug/g dry  | <5.0  | <5.0  | <5.0  | 10.8 |
| Boron, available | 0.5 ug/g dry  | <0.5  | <0.5  | <0.5  | <0.5 |
| Cadmium          | 0.5 ug/g dry  | <0.5  | <0.5  | <0.5  | <0.5 |
| Chromium         | 5.0 ug/g dry  | 9.9   | 15.7  | 14.7  | 20.3 |
| Chromium (VI)    | 0.2 ug/g dry  | <0.2  | <0.2  | <0.2  | <0.2 |
| Cobalt           | 1.0 ug/g dry  | 3.3   | 4.6   | 4.4   | 5.7  |
| Copper           | 5.0 ug/g dry  | 7.0   | 9.7   | 9.0   | 11.4 |
| Lead             | 1.0 ug/g dry  | 1.9   | 3.1   | 2.9   | 3.5  |
| Mercury          | 0.1 ug/g dry  | <0.1  | <0.1  | <0.1  | <0.1 |
| Molybdenum       | 1.0 ug/g dry  | <1.0  | <1.0  | <1.0  | <1.0 |
| Nickel           | 5.0 ug/g dry  | 5.3   | 8.3   | 7.9   | 11.1 |
| Selenium         | 1.0 ug/g dry  | <1.0  | <1.0  | <1.0  | <1.0 |
| Silver           | 0.3 ug/g dry  | <0.3  | <0.3  | <0.3  | <0.3 |
| Thallium         | 1.0 ug/g dry  | <1.0  | <1.0  | <1.0  | <1.0 |
| Uranium          | 1.0 ug/g dry  | <1.0  | <1.0  | <1.0  | <1.0 |
| Vanadium         | 10.0 ug/g dry | 18.4  | 25.0  | 24.0  | 28.2 |
| Zinc             | 20.0 ug/g dry | <20.0 | <20.0 | <20.0 | 20.5 |

**Volatiles**

|                |               |       |       |       |       |
|----------------|---------------|-------|-------|-------|-------|
| Benzene        | 0.02 ug/g dry | <0.02 | <0.02 | <0.02 | <0.02 |
| Ethylbenzene   | 0.05 ug/g dry | <0.05 | <0.05 | <0.05 | <0.05 |
| Toluene        | 0.05 ug/g dry | <0.05 | <0.05 | <0.05 | <0.05 |
| m,p-Xylenes    | 0.05 ug/g dry | <0.05 | <0.05 | <0.05 | <0.05 |
| o-Xylene       | 0.05 ug/g dry | <0.05 | <0.05 | <0.05 | <0.05 |
| Xylenes, total | 0.05 ug/g dry | <0.05 | <0.05 | <0.05 | <0.05 |
| Toluene-d8     | Surrogate     | 112%  | 118%  | 117%  | 118%  |

Certificate of Analysis

Report Date: 16-Dec-2022

Client: Malroz Engineering Incorporated

Order Date: 9-Dec-2022

Client PO:

Project Description: 1703

| Client ID:   | BH102 S01       | BH103 S01       | BH103 S02       | BH106 S01       |
|--------------|-----------------|-----------------|-----------------|-----------------|
| Sample Date: | 07-Dec-22 11:00 | 07-Dec-22 10:00 | 07-Dec-22 10:00 | 06-Dec-22 09:00 |
| Sample ID:   | 2250463-01      | 2250463-02      | 2250463-03      | 2250463-04      |
| MDL/Units    | Soil            | Soil            | Soil            | Soil            |

#### Hydrocarbons

|                   |            |    |    |    |    |
|-------------------|------------|----|----|----|----|
| F1 PHCs (C6-C10)  | 7 ug/g dry | <7 | <7 | <7 | <7 |
| F2 PHCs (C10-C16) | 4 ug/g dry | <4 | <4 | <4 | <4 |
| F3 PHCs (C16-C34) | 8 ug/g dry | <8 | <8 | <8 | <8 |
| F4 PHCs (C34-C50) | 6 ug/g dry | <6 | <6 | <6 | <6 |

#### Pesticides, OC

|                     |               |   |       |       |   |
|---------------------|---------------|---|-------|-------|---|
| Aldrin              | 0.01 ug/g dry | - | <0.01 | <0.01 | - |
| gamma-BHC (Lindane) | 0.01 ug/g dry | - | <0.01 | <0.01 | - |
| alpha-Chlordane     | 0.01 ug/g dry | - | <0.01 | <0.01 | - |
| gamma-Chlordane     | 0.01 ug/g dry | - | <0.01 | <0.01 | - |
| Chlordane           | 0.01 ug/g dry | - | <0.01 | <0.01 | - |
| o,p'-DDD            | 0.01 ug/g dry | - | <0.01 | <0.01 | - |
| p,p'-DDD            | 0.02 ug/g dry | - | <0.02 | <0.02 | - |
| DDD                 | 0.02 ug/g dry | - | <0.02 | <0.02 | - |
| o,p'-DDE            | 0.01 ug/g dry | - | <0.01 | <0.01 | - |
| p,p'-DDE            | 0.01 ug/g dry | - | <0.01 | <0.01 | - |
| DDE                 | 0.01 ug/g dry | - | <0.01 | <0.01 | - |
| o,p'-DDT            | 0.01 ug/g dry | - | <0.01 | <0.01 | - |
| p,p'-DDT            | 0.01 ug/g dry | - | <0.01 | <0.01 | - |
| DDT                 | 0.01 ug/g dry | - | <0.01 | <0.01 | - |
| Dieldrin            | 0.02 ug/g dry | - | <0.02 | <0.02 | - |
| Endrin              | 0.02 ug/g dry | - | <0.02 | <0.02 | - |
| Endosulfan I        | 0.01 ug/g dry | - | <0.01 | <0.01 | - |
| Endosulfan II       | 0.02 ug/g dry | - | <0.02 | <0.02 | - |
| Heptachlor          | 0.01 ug/g dry | - | <0.01 | <0.01 | - |
| Heptachlor epoxide  | 0.01 ug/g dry | - | <0.01 | <0.01 | - |
| Hexachlorobenzene   | 0.01 ug/g dry | - | <0.01 | <0.01 | - |
| Hexachlorobutadiene | 0.01 ug/g dry | - | <0.01 | <0.01 | - |
| Hexachloroethane    | 0.01 ug/g dry | - | <0.01 | <0.01 | - |
| Methoxychlor        | 0.01 ug/g dry | - | <0.01 | <0.01 | - |
| Decachlorobiphenyl  | Surrogate     | - | 109%  | 91.3% | - |

Certificate of Analysis

Report Date: 16-Dec-2022

Client: Malroz Engineering Incorporated

Order Date: 9-Dec-2022

Client PO:

Project Description: 1703

| Client ID:   | BH110 S01       | BH111 S01       | BH112 S01       | BH117 S01       |
|--------------|-----------------|-----------------|-----------------|-----------------|
| Sample Date: | 06-Dec-22 13:00 | 06-Dec-22 14:00 | 05-Dec-22 15:00 | 05-Dec-22 09:00 |
| Sample ID:   | 2250463-05      | 2250463-06      | 2250463-07      | 2250463-08      |
| MDL/Units    | Soil            | Soil            | Soil            | Soil            |

**Physical Characteristics**

|          |              |      |      |      |      |
|----------|--------------|------|------|------|------|
| % Solids | 0.1 % by Wt. | 82.5 | 84.0 | 80.0 | 82.6 |
|----------|--------------|------|------|------|------|

**General Inorganics**

|              |          |      |      |      |      |
|--------------|----------|------|------|------|------|
| SAR          | 0.01 N/A | 0.09 | 0.09 | 0.09 | 0.09 |
| Conductivity | 5 uS/cm  | 135  | 129  | 175  | 158  |

**Metals**

|                  |               |      |      |      |       |
|------------------|---------------|------|------|------|-------|
| Antimony         | 1.0 ug/g dry  | <1.0 | <1.0 | <1.0 | <1.0  |
| Arsenic          | 1.0 ug/g dry  | 1.8  | 2.8  | 2.3  | 1.8   |
| Barium           | 1.0 ug/g dry  | 72.2 | 59.6 | 121  | 66.7  |
| Beryllium        | 0.5 ug/g dry  | <0.5 | <0.5 | <0.5 | <0.5  |
| Boron            | 5.0 ug/g dry  | 5.6  | 5.7  | 10.6 | 7.6   |
| Boron, available | 0.5 ug/g dry  | <0.5 | <0.5 | <0.5 | <0.5  |
| Cadmium          | 0.5 ug/g dry  | <0.5 | 0.8  | <0.5 | <0.5  |
| Chromium         | 5.0 ug/g dry  | 17.2 | 17.1 | 26.9 | 16.9  |
| Chromium (VI)    | 0.2 ug/g dry  | <0.2 | <0.2 | <0.2 | <0.2  |
| Cobalt           | 1.0 ug/g dry  | 5.0  | 5.4  | 7.2  | 4.8   |
| Copper           | 5.0 ug/g dry  | 10.0 | 12.5 | 14.9 | 9.9   |
| Lead             | 1.0 ug/g dry  | 3.1  | 3.8  | 4.3  | 2.9   |
| Mercury          | 0.1 ug/g dry  | <0.1 | <0.1 | <0.1 | <0.1  |
| Molybdenum       | 1.0 ug/g dry  | <1.0 | 1.2  | <1.0 | <1.0  |
| Nickel           | 5.0 ug/g dry  | 9.6  | 10.7 | 15.4 | 9.4   |
| Selenium         | 1.0 ug/g dry  | <1.0 | <1.0 | <1.0 | <1.0  |
| Silver           | 0.3 ug/g dry  | <0.3 | 0.8  | <0.3 | <0.3  |
| Thallium         | 1.0 ug/g dry  | <1.0 | <1.0 | <1.0 | <1.0  |
| Uranium          | 1.0 ug/g dry  | <1.0 | 1.2  | <1.0 | <1.0  |
| Vanadium         | 10.0 ug/g dry | 26.1 | 29.3 | 38.1 | 25.8  |
| Zinc             | 20.0 ug/g dry | 21.2 | 21.2 | 30.9 | <20.0 |

**Volatiles**

|                |               |       |       |       |       |
|----------------|---------------|-------|-------|-------|-------|
| Benzene        | 0.02 ug/g dry | <0.02 | <0.02 | <0.02 | <0.02 |
| Ethylbenzene   | 0.05 ug/g dry | <0.05 | <0.05 | <0.05 | <0.05 |
| Toluene        | 0.05 ug/g dry | <0.05 | <0.05 | <0.05 | <0.05 |
| m,p-Xylenes    | 0.05 ug/g dry | <0.05 | <0.05 | <0.05 | <0.05 |
| o-Xylene       | 0.05 ug/g dry | <0.05 | <0.05 | <0.05 | <0.05 |
| Xylenes, total | 0.05 ug/g dry | <0.05 | <0.05 | <0.05 | <0.05 |
| Toluene-d8     | Surrogate     | 117%  | 116%  | 117%  | 116%  |

**Hydrocarbons**

Certificate of Analysis

Report Date: 16-Dec-2022

Client: Malroz Engineering Incorporated

Order Date: 9-Dec-2022

Client PO:

Project Description: 1703

|                   | Client ID:   | BH110 S01       | BH111 S01       | BH112 S01       | BH117 S01       |
|-------------------|--------------|-----------------|-----------------|-----------------|-----------------|
|                   | Sample Date: | 06-Dec-22 13:00 | 06-Dec-22 14:00 | 05-Dec-22 15:00 | 05-Dec-22 09:00 |
|                   | Sample ID:   | 2250463-05      | 2250463-06      | 2250463-07      | 2250463-08      |
|                   | MDL/Units    | Soil            | Soil            | Soil            | Soil            |
| F1 PHCs (C6-C10)  | 7 ug/g dry   | <7              | <7              | <7              | <7              |
| F2 PHCs (C10-C16) | 4 ug/g dry   | <4              | <4              | <4              | <4              |
| F3 PHCs (C16-C34) | 8 ug/g dry   | <8              | <8              | <8              | <8              |
| F4 PHCs (C34-C50) | 6 ug/g dry   | <6              | <6              | <6              | <6              |

**Pesticides, OC**

|                     |               |   |   |       |   |
|---------------------|---------------|---|---|-------|---|
| Aldrin              | 0.01 ug/g dry | - | - | <0.01 | - |
| gamma-BHC (Lindane) | 0.01 ug/g dry | - | - | <0.01 | - |
| alpha-Chlordane     | 0.01 ug/g dry | - | - | <0.01 | - |
| gamma-Chlordane     | 0.01 ug/g dry | - | - | <0.01 | - |
| Chlordane           | 0.01 ug/g dry | - | - | <0.01 | - |
| o,p'-DDD            | 0.01 ug/g dry | - | - | <0.01 | - |
| p,p'-DDD            | 0.02 ug/g dry | - | - | <0.02 | - |
| DDD                 | 0.02 ug/g dry | - | - | <0.02 | - |
| o,p'-DDE            | 0.01 ug/g dry | - | - | <0.01 | - |
| p,p'-DDE            | 0.01 ug/g dry | - | - | <0.01 | - |
| DDE                 | 0.01 ug/g dry | - | - | <0.01 | - |
| o,p'-DDT            | 0.01 ug/g dry | - | - | <0.01 | - |
| p,p'-DDT            | 0.01 ug/g dry | - | - | <0.01 | - |
| DDT                 | 0.01 ug/g dry | - | - | <0.01 | - |
| Dieldrin            | 0.02 ug/g dry | - | - | <0.02 | - |
| Endrin              | 0.02 ug/g dry | - | - | <0.02 | - |
| Endosulfan I        | 0.01 ug/g dry | - | - | <0.01 | - |
| Endosulfan II       | 0.02 ug/g dry | - | - | <0.02 | - |
| Heptachlor          | 0.01 ug/g dry | - | - | <0.01 | - |
| Heptachlor epoxide  | 0.01 ug/g dry | - | - | <0.01 | - |
| Hexachlorobenzene   | 0.01 ug/g dry | - | - | <0.01 | - |
| Hexachlorobutadiene | 0.01 ug/g dry | - | - | <0.01 | - |
| Hexachloroethane    | 0.01 ug/g dry | - | - | <0.01 | - |
| Methoxychlor        | 0.01 ug/g dry | - | - | <0.01 | - |
| Decachlorobiphenyl  | Surrogate     | - | - | 112%  | - |

Certificate of Analysis

Report Date: 16-Dec-2022

Client: Malroz Engineering Incorporated

Order Date: 9-Dec-2022

Client PO:

Project Description: 1703

|              |                 |                 |                 |   |
|--------------|-----------------|-----------------|-----------------|---|
| Client ID:   | BH118 S01       | BH121 S01       | BH126 S01       | - |
| Sample Date: | 05-Dec-22 13:00 | 01-Dec-22 14:00 | 01-Dec-22 10:00 | - |
| Sample ID:   | 2250463-09      | 2250463-10      | 2250463-11      | - |
| MDL/Units    | Soil            | Soil            | Soil            | - |

**Physical Characteristics**

|          |              |      |      |      |   |
|----------|--------------|------|------|------|---|
| % Solids | 0.1 % by Wt. | 82.6 | 83.5 | 88.9 | - |
|----------|--------------|------|------|------|---|

**General Inorganics**

|              |               |      |      |      |   |
|--------------|---------------|------|------|------|---|
| SAR          | 0.01 N/A      | 0.10 | 0.11 | 0.11 | - |
| Conductivity | 5 uS/cm       | 165  | 160  | 103  | - |
| pH           | 0.05 pH Units | -    | 7.28 | -    | - |

**Metals**

|                  |               |      |       |       |   |
|------------------|---------------|------|-------|-------|---|
| Antimony         | 1.0 ug/g dry  | <1.0 | <1.0  | <1.0  | - |
| Arsenic          | 1.0 ug/g dry  | 1.8  | 2.0   | 1.3   | - |
| Barium           | 1.0 ug/g dry  | 66.5 | 37.6  | 32.1  | - |
| Beryllium        | 0.5 ug/g dry  | <0.5 | <0.5  | <0.5  | - |
| Boron            | 5.0 ug/g dry  | 6.8  | <5.0  | <5.0  | - |
| Boron, available | 0.5 ug/g dry  | <0.5 | <0.5  | <0.5  | - |
| Cadmium          | 0.5 ug/g dry  | <0.5 | <0.5  | <0.5  | - |
| Chromium         | 5.0 ug/g dry  | 17.8 | 14.9  | 8.9   | - |
| Chromium (VI)    | 0.2 ug/g dry  | <0.2 | 0.2   | <0.2  | - |
| Cobalt           | 1.0 ug/g dry  | 4.6  | 3.7   | 3.1   | - |
| Copper           | 5.0 ug/g dry  | 9.3  | 9.6   | 6.8   | - |
| Lead             | 1.0 ug/g dry  | 3.7  | 3.1   | 1.7   | - |
| Mercury          | 0.1 ug/g dry  | <0.1 | <0.1  | <0.1  | - |
| Molybdenum       | 1.0 ug/g dry  | <1.0 | <1.0  | <1.0  | - |
| Nickel           | 5.0 ug/g dry  | 9.0  | 7.9   | 5.0   | - |
| Selenium         | 1.0 ug/g dry  | <1.0 | <1.0  | <1.0  | - |
| Silver           | 0.3 ug/g dry  | <0.3 | <0.3  | <0.3  | - |
| Thallium         | 1.0 ug/g dry  | <1.0 | <1.0  | <1.0  | - |
| Uranium          | 1.0 ug/g dry  | <1.0 | <1.0  | <1.0  | - |
| Vanadium         | 10.0 ug/g dry | 26.8 | 23.2  | 15.3  | - |
| Zinc             | 20.0 ug/g dry | 21.2 | <20.0 | <20.0 | - |

**Volatiles**

|                |               |       |       |       |   |
|----------------|---------------|-------|-------|-------|---|
| Benzene        | 0.02 ug/g dry | <0.02 | <0.02 | <0.02 | - |
| Ethylbenzene   | 0.05 ug/g dry | <0.05 | <0.05 | <0.05 | - |
| Toluene        | 0.05 ug/g dry | <0.05 | <0.05 | <0.05 | - |
| m,p-Xylenes    | 0.05 ug/g dry | <0.05 | <0.05 | <0.05 | - |
| o-Xylene       | 0.05 ug/g dry | <0.05 | <0.05 | <0.05 | - |
| Xylenes, total | 0.05 ug/g dry | <0.05 | <0.05 | <0.05 | - |
| Toluene-d8     | Surrogate     | 115%  | 117%  | 112%  | - |

Certificate of Analysis

Report Date: 16-Dec-2022

Client: Malroz Engineering Incorporated

Order Date: 9-Dec-2022

Client PO:

Project Description: 1703

|                     |                 |                 |                 |   |
|---------------------|-----------------|-----------------|-----------------|---|
| <b>Client ID:</b>   | BH118 S01       | BH121 S01       | BH126 S01       | - |
| <b>Sample Date:</b> | 05-Dec-22 13:00 | 01-Dec-22 14:00 | 01-Dec-22 10:00 | - |
| <b>Sample ID:</b>   | 2250463-09      | 2250463-10      | 2250463-11      | - |
| <b>MDL/Units</b>    | Soil            | Soil            | Soil            | - |

**Hydrocarbons**

|                   |            |    |    |    |   |
|-------------------|------------|----|----|----|---|
| F1 PHCs (C6-C10)  | 7 ug/g dry | <7 | <7 | <7 | - |
| F2 PHCs (C10-C16) | 4 ug/g dry | <4 | <4 | <4 | - |
| F3 PHCs (C16-C34) | 8 ug/g dry | <8 | <8 | <8 | - |
| F4 PHCs (C34-C50) | 6 ug/g dry | <6 | <6 | <6 | - |

**Pesticides, OC**

|                     |               |   |       |   |   |
|---------------------|---------------|---|-------|---|---|
| Aldrin              | 0.01 ug/g dry | - | <0.01 | - | - |
| gamma-BHC (Lindane) | 0.01 ug/g dry | - | <0.01 | - | - |
| alpha-Chlordane     | 0.01 ug/g dry | - | <0.01 | - | - |
| gamma-Chlordane     | 0.01 ug/g dry | - | <0.01 | - | - |
| Chlordane           | 0.01 ug/g dry | - | <0.01 | - | - |
| o,p'-DDD            | 0.01 ug/g dry | - | <0.01 | - | - |
| p,p'-DDD            | 0.02 ug/g dry | - | <0.02 | - | - |
| DDD                 | 0.02 ug/g dry | - | <0.02 | - | - |
| o,p'-DDE            | 0.01 ug/g dry | - | <0.01 | - | - |
| p,p'-DDE            | 0.01 ug/g dry | - | <0.01 | - | - |
| DDE                 | 0.01 ug/g dry | - | <0.01 | - | - |
| o,p'-DDT            | 0.01 ug/g dry | - | <0.01 | - | - |
| p,p'-DDT            | 0.01 ug/g dry | - | <0.01 | - | - |
| DDT                 | 0.01 ug/g dry | - | <0.01 | - | - |
| Dieldrin            | 0.02 ug/g dry | - | <0.02 | - | - |
| Endrin              | 0.02 ug/g dry | - | <0.02 | - | - |
| Endosulfan I        | 0.01 ug/g dry | - | <0.01 | - | - |
| Endosulfan II       | 0.02 ug/g dry | - | <0.02 | - | - |
| Heptachlor          | 0.01 ug/g dry | - | <0.01 | - | - |
| Heptachlor epoxide  | 0.01 ug/g dry | - | <0.01 | - | - |
| Hexachlorobenzene   | 0.01 ug/g dry | - | <0.01 | - | - |
| Hexachlorobutadiene | 0.01 ug/g dry | - | <0.01 | - | - |
| Hexachloroethane    | 0.01 ug/g dry | - | <0.01 | - | - |
| Methoxychlor        | 0.01 ug/g dry | - | <0.01 | - | - |
| Decachlorobiphenyl  | Surrogate     | - | 109%  | - | - |

Certificate of Analysis

Report Date: 16-Dec-2022

Client: Malroz Engineering Incorporated

Order Date: 9-Dec-2022

Client PO:

Project Description: 1703

## Method Quality Control: Blank

| Analyte                       | Result | Reporting Limit | Units | Source Result | %REC | %REC Limit | RPD | RPD Limit | Notes |
|-------------------------------|--------|-----------------|-------|---------------|------|------------|-----|-----------|-------|
| <b>General Inorganics</b>     |        |                 |       |               |      |            |     |           |       |
| Conductivity                  | ND     | 5               | uS/cm |               |      |            |     |           |       |
| <b>Hydrocarbons</b>           |        |                 |       |               |      |            |     |           |       |
| F1 PHCs (C6-C10)              | ND     | 7               | ug/g  |               |      |            |     |           |       |
| F2 PHCs (C10-C16)             | ND     | 4               | ug/g  |               |      |            |     |           |       |
| F3 PHCs (C16-C34)             | ND     | 8               | ug/g  |               |      |            |     |           |       |
| F4 PHCs (C34-C50)             | ND     | 6               | ug/g  |               |      |            |     |           |       |
| <b>Metals</b>                 |        |                 |       |               |      |            |     |           |       |
| Antimony                      | ND     | 1.0             | ug/g  |               |      |            |     |           |       |
| Arsenic                       | ND     | 1.0             | ug/g  |               |      |            |     |           |       |
| Barium                        | ND     | 1.0             | ug/g  |               |      |            |     |           |       |
| Beryllium                     | ND     | 0.5             | ug/g  |               |      |            |     |           |       |
| Boron, available              | ND     | 0.5             | ug/g  |               |      |            |     |           |       |
| Boron                         | ND     | 5.0             | ug/g  |               |      |            |     |           |       |
| Cadmium                       | ND     | 0.5             | ug/g  |               |      |            |     |           |       |
| Chromium (VI)                 | ND     | 0.2             | ug/g  |               |      |            |     |           |       |
| Chromium                      | ND     | 5.0             | ug/g  |               |      |            |     |           |       |
| Cobalt                        | ND     | 1.0             | ug/g  |               |      |            |     |           |       |
| Copper                        | ND     | 5.0             | ug/g  |               |      |            |     |           |       |
| Lead                          | ND     | 1.0             | ug/g  |               |      |            |     |           |       |
| Mercury                       | ND     | 0.1             | ug/g  |               |      |            |     |           |       |
| Molybdenum                    | ND     | 1.0             | ug/g  |               |      |            |     |           |       |
| Nickel                        | ND     | 5.0             | ug/g  |               |      |            |     |           |       |
| Selenium                      | ND     | 1.0             | ug/g  |               |      |            |     |           |       |
| Silver                        | ND     | 0.3             | ug/g  |               |      |            |     |           |       |
| Thallium                      | ND     | 1.0             | ug/g  |               |      |            |     |           |       |
| Uranium                       | ND     | 1.0             | ug/g  |               |      |            |     |           |       |
| Vanadium                      | ND     | 10.0            | ug/g  |               |      |            |     |           |       |
| Zinc                          | ND     | 20.0            | ug/g  |               |      |            |     |           |       |
| <b>Pesticides, OC</b>         |        |                 |       |               |      |            |     |           |       |
| Aldrin                        | ND     | 0.01            | ug/g  |               |      |            |     |           |       |
| gamma-BHC (Lindane)           | ND     | 0.01            | ug/g  |               |      |            |     |           |       |
| alpha-Chlordane               | ND     | 0.01            | ug/g  |               |      |            |     |           |       |
| gamma-Chlordane               | ND     | 0.01            | ug/g  |               |      |            |     |           |       |
| Chlordane                     | ND     | 0.01            | ug/g  |               |      |            |     |           |       |
| o,p'-DDD                      | ND     | 0.01            | ug/g  |               |      |            |     |           |       |
| p,p'-DDD                      | ND     | 0.02            | ug/g  |               |      |            |     |           |       |
| DDD                           | ND     | 0.02            | ug/g  |               |      |            |     |           |       |
| o,p'-DDE                      | ND     | 0.01            | ug/g  |               |      |            |     |           |       |
| p,p'-DDE                      | ND     | 0.01            | ug/g  |               |      |            |     |           |       |
| DDE                           | ND     | 0.01            | ug/g  |               |      |            |     |           |       |
| o,p'-DDT                      | ND     | 0.01            | ug/g  |               |      |            |     |           |       |
| p,p'-DDT                      | ND     | 0.01            | ug/g  |               |      |            |     |           |       |
| DDT                           | ND     | 0.01            | ug/g  |               |      |            |     |           |       |
| Dieldrin                      | ND     | 0.02            | ug/g  |               |      |            |     |           |       |
| Endrin                        | ND     | 0.02            | ug/g  |               |      |            |     |           |       |
| Endosulfan I                  | ND     | 0.01            | ug/g  |               |      |            |     |           |       |
| Endosulfan II                 | ND     | 0.02            | ug/g  |               |      |            |     |           |       |
| Heptachlor                    | ND     | 0.01            | ug/g  |               |      |            |     |           |       |
| Heptachlor epoxide            | ND     | 0.01            | ug/g  |               |      |            |     |           |       |
| Hexachlorobenzene             | ND     | 0.01            | ug/g  |               |      |            |     |           |       |
| Hexachlorobutadiene           | ND     | 0.01            | ug/g  |               |      |            |     |           |       |
| Hexachloroethane              | ND     | 0.01            | ug/g  |               |      |            |     |           |       |
| Methoxychlor                  | ND     | 0.01            | ug/g  |               |      |            |     |           |       |
| Surrogate: Decachlorobiphenyl | 0.139  |                 | ug/g  |               | 139  | 50-140     |     |           |       |
| <b>Volatiles</b>              |        |                 |       |               |      |            |     |           |       |
| Benzene                       | ND     | 0.02            | ug/g  |               |      |            |     |           |       |
| Ethylbenzene                  | ND     | 0.05            | ug/g  |               |      |            |     |           |       |

Certificate of Analysis

Report Date: 16-Dec-2022

Client: Malroz Engineering Incorporated

Order Date: 9-Dec-2022

Client PO:

Project Description: 1703

### Method Quality Control: Blank

| Analyte               | Result | Reporting Limit | Units | Source Result | %REC | %REC Limit | RPD | RPD Limit | Notes |
|-----------------------|--------|-----------------|-------|---------------|------|------------|-----|-----------|-------|
| Toluene               | ND     | 0.05            | ug/g  |               |      |            |     |           |       |
| m,p-Xylenes           | ND     | 0.05            | ug/g  |               |      |            |     |           |       |
| o-Xylene              | ND     | 0.05            | ug/g  |               |      |            |     |           |       |
| Xylenes, total        | ND     | 0.05            | ug/g  |               |      |            |     |           |       |
| Surrogate: Toluene-d8 | 7.27   |                 | ug/g  |               | 90.9 | 50-140     |     |           |       |

Certificate of Analysis

Report Date: 16-Dec-2022

Client: Malroz Engineering Incorporated

Order Date: 9-Dec-2022

Client PO:

Project Description: 1703

## Method Quality Control: Duplicate

| Analyte                         | Result | Reporting Limit | Units    | Source Result | %REC | %REC Limit | RPD | RPD Limit | Notes |
|---------------------------------|--------|-----------------|----------|---------------|------|------------|-----|-----------|-------|
| <b>General Inorganics</b>       |        |                 |          |               |      |            |     |           |       |
| SAR                             | 0.34   | 0.01            | N/A      | 0.36          |      |            | 5.7 | 30        |       |
| Conductivity                    | 138    | 5               | uS/cm    | 135           |      |            | 2.2 | 5         |       |
| pH                              | 7.34   | 0.05            | pH Units | 7.34          |      |            | 0.0 | 2.3       |       |
| <b>Hydrocarbons</b>             |        |                 |          |               |      |            |     |           |       |
| F1 PHCs (C6-C10)                | ND     | 7               | ug/g     | ND            |      |            | NC  | 40        |       |
| F2 PHCs (C10-C16)               | ND     | 4               | ug/g     | ND            |      |            | NC  | 30        |       |
| F3 PHCs (C16-C34)               | ND     | 8               | ug/g     | ND            |      |            | NC  | 30        |       |
| F4 PHCs (C34-C50)               | ND     | 6               | ug/g     | ND            |      |            | NC  | 30        |       |
| <b>Metals</b>                   |        |                 |          |               |      |            |     |           |       |
| Antimony                        | ND     | 1.0             | ug/g     | ND            |      |            | NC  | 30        |       |
| Arsenic                         | 1.2    | 1.0             | ug/g     | 1.3           |      |            | 7.5 | 30        |       |
| Barium                          | 29.0   | 1.0             | ug/g     | 31.6          |      |            | 8.7 | 30        |       |
| Beryllium                       | ND     | 0.5             | ug/g     | ND            |      |            | NC  | 30        |       |
| Boron, available                | ND     | 0.5             | ug/g     | ND            |      |            | NC  | 35        |       |
| Boron                           | ND     | 5.0             | ug/g     | ND            |      |            | NC  | 30        |       |
| Cadmium                         | ND     | 0.5             | ug/g     | ND            |      |            | NC  | 30        |       |
| Chromium (VI)                   | ND     | 0.2             | ug/g     | ND            |      |            | NC  | 35        |       |
| Chromium                        | 9.4    | 5.0             | ug/g     | 9.9           |      |            | 5.0 | 30        |       |
| Cobalt                          | 3.2    | 1.0             | ug/g     | 3.3           |      |            | 3.2 | 30        |       |
| Copper                          | 6.6    | 5.0             | ug/g     | 7.0           |      |            | 5.9 | 30        |       |
| Lead                            | 1.8    | 1.0             | ug/g     | 1.9           |      |            | 8.6 | 30        |       |
| Mercury                         | ND     | 0.1             | ug/g     | ND            |      |            | NC  | 30        |       |
| Molybdenum                      | ND     | 1.0             | ug/g     | ND            |      |            | NC  | 30        |       |
| Nickel                          | ND     | 5.0             | ug/g     | 5.3           |      |            | NC  | 30        |       |
| Selenium                        | ND     | 1.0             | ug/g     | ND            |      |            | NC  | 30        |       |
| Silver                          | ND     | 0.3             | ug/g     | ND            |      |            | NC  | 30        |       |
| Thallium                        | ND     | 1.0             | ug/g     | ND            |      |            | NC  | 30        |       |
| Uranium                         | ND     | 1.0             | ug/g     | ND            |      |            | NC  | 30        |       |
| Vanadium                        | 17.7   | 10.0            | ug/g     | 18.4          |      |            | 3.6 | 30        |       |
| Zinc                            | ND     | 20.0            | ug/g     | ND            |      |            | NC  | 30        |       |
| <b>Pesticides, OC</b>           |        |                 |          |               |      |            |     |           |       |
| Aldrin                          | ND     | 0.01            | ug/g     | ND            |      |            | NC  | 40        |       |
| gamma-BHC (Lindane)             | ND     | 0.01            | ug/g     | ND            |      |            | NC  | 40        |       |
| alpha-Chlordane                 | ND     | 0.01            | ug/g     | ND            |      |            | NC  | 40        |       |
| gamma-Chlordane                 | ND     | 0.01            | ug/g     | ND            |      |            | NC  | 40        |       |
| o,p'-DDD                        | ND     | 0.01            | ug/g     | ND            |      |            | NC  | 40        |       |
| p,p'-DDD                        | ND     | 0.02            | ug/g     | ND            |      |            | NC  | 40        |       |
| o,p'-DDE                        | ND     | 0.01            | ug/g     | ND            |      |            | NC  | 40        |       |
| p,p'-DDE                        | ND     | 0.01            | ug/g     | ND            |      |            | NC  | 40        |       |
| o,p'-DDT                        | ND     | 0.01            | ug/g     | ND            |      |            | NC  | 40        |       |
| p,p'-DDT                        | ND     | 0.01            | ug/g     | ND            |      |            | NC  | 40        |       |
| Dieldrin                        | ND     | 0.02            | ug/g     | ND            |      |            | NC  | 40        |       |
| Endrin                          | ND     | 0.02            | ug/g     | ND            |      |            | NC  | 40        |       |
| Endosulfan I                    | ND     | 0.01            | ug/g     | ND            |      |            | NC  | 40        |       |
| Endosulfan II                   | ND     | 0.02            | ug/g     | ND            |      |            | NC  | 40        |       |
| Heptachlor                      | ND     | 0.01            | ug/g     | ND            |      |            | NC  | 40        |       |
| Heptachlor epoxide              | ND     | 0.01            | ug/g     | ND            |      |            | NC  | 40        |       |
| Hexachlorobenzene               | ND     | 0.01            | ug/g     | ND            |      |            | NC  | 40        |       |
| Hexachlorobutadiene             | ND     | 0.01            | ug/g     | ND            |      |            | NC  | 40        |       |
| Hexachloroethane                | ND     | 0.01            | ug/g     | ND            |      |            | NC  | 40        |       |
| Methoxychlor                    | 1.22   | 0.01            | ug/g     | ND            |      |            | NC  | 40        |       |
| Surrogate: Decachlorobiphenyl   | 0.146  |                 | ug/g     |               | 128  | 50-140     |     |           |       |
| <b>Physical Characteristics</b> |        |                 |          |               |      |            |     |           |       |
| % Solids                        | 85.3   | 0.1             | % by Wt. | 84.7          |      |            | 0.7 | 25        |       |
| <b>Volatiles</b>                |        |                 |          |               |      |            |     |           |       |

Certificate of Analysis

Report Date: 16-Dec-2022

Client: Malroz Engineering Incorporated

Order Date: 9-Dec-2022

Client PO:

Project Description: 1703

### Method Quality Control: Duplicate

| Analyte               | Result | Reporting Limit | Units | Source Result | %REC | %REC Limit | RPD | RPD Limit | Notes |
|-----------------------|--------|-----------------|-------|---------------|------|------------|-----|-----------|-------|
| Benzene               | ND     | 0.02            | ug/g  | ND            |      |            | NC  | 50        |       |
| Ethylbenzene          | ND     | 0.05            | ug/g  | ND            |      |            | NC  | 50        |       |
| Toluene               | ND     | 0.05            | ug/g  | ND            |      |            | NC  | 50        |       |
| m,p-Xylenes           | ND     | 0.05            | ug/g  | ND            |      |            | NC  | 50        |       |
| o-Xylene              | ND     | 0.05            | ug/g  | ND            |      |            | NC  | 50        |       |
| Surrogate: Toluene-d8 | 10.4   |                 | ug/g  |               | 111  | 50-140     |     |           |       |

Certificate of Analysis

Report Date: 16-Dec-2022

Client: Malroz Engineering Incorporated

Order Date: 9-Dec-2022

Client PO:

Project Description: 1703

## Method Quality Control: Spike

| Analyte                       | Result | Reporting Limit | Units | Source Result | %REC | %REC Limit | RPD | RPD Limit | Notes |
|-------------------------------|--------|-----------------|-------|---------------|------|------------|-----|-----------|-------|
| <b>Hydrocarbons</b>           |        |                 |       |               |      |            |     |           |       |
| F1 PHCs (C6-C10)              | 167    | 7               | ug/g  | ND            | 83.6 | 80-120     |     |           |       |
| F2 PHCs (C10-C16)             | 102    | 4               | ug/g  | ND            | 115  | 60-140     |     |           |       |
| F3 PHCs (C16-C34)             | 254    | 8               | ug/g  | ND            | 117  | 60-140     |     |           |       |
| F4 PHCs (C34-C50)             | 176    | 6               | ug/g  | ND            | 128  | 60-140     |     |           |       |
| <b>Metals</b>                 |        |                 |       |               |      |            |     |           |       |
| Arsenic                       | 43.2   | 1.0             | ug/g  | ND            | 85.3 | 70-130     |     |           |       |
| Barium                        | 51.2   | 1.0             | ug/g  | 12.6          | 77.1 | 70-130     |     |           |       |
| Beryllium                     | 41.3   | 0.5             | ug/g  | ND            | 82.5 | 70-130     |     |           |       |
| Boron, available              | 3.97   | 0.5             | ug/g  | ND            | 79.4 | 70-122     |     |           |       |
| Boron                         | 41.7   | 5.0             | ug/g  | ND            | 80.4 | 70-130     |     |           |       |
| Cadmium                       | 39.5   | 0.5             | ug/g  | ND            | 78.9 | 70-130     |     |           |       |
| Chromium (VI)                 | 4.4    | 0.2             | ug/g  | ND            | 79.5 | 70-130     |     |           |       |
| Chromium                      | 47.9   | 5.0             | ug/g  | ND            | 88.0 | 70-130     |     |           |       |
| Cobalt                        | 43.6   | 1.0             | ug/g  | 1.3           | 84.4 | 70-130     |     |           |       |
| Copper                        | 44.2   | 5.0             | ug/g  | ND            | 82.8 | 70-130     |     |           |       |
| Lead                          | 38.6   | 1.0             | ug/g  | ND            | 75.6 | 70-130     |     |           |       |
| Mercury                       | 1.30   | 0.1             | ug/g  | ND            | 86.7 | 70-130     |     |           |       |
| Molybdenum                    | 42.1   | 1.0             | ug/g  | ND            | 84.0 | 70-130     |     |           |       |
| Nickel                        | 43.1   | 5.0             | ug/g  | ND            | 81.9 | 70-130     |     |           |       |
| Selenium                      | 39.5   | 1.0             | ug/g  | ND            | 78.9 | 70-130     |     |           |       |
| Silver                        | 38.7   | 0.3             | ug/g  | ND            | 77.5 | 70-130     |     |           |       |
| Thallium                      | 39.0   | 1.0             | ug/g  | ND            | 78.1 | 70-130     |     |           |       |
| Uranium                       | 41.7   | 1.0             | ug/g  | ND            | 83.2 | 70-130     |     |           |       |
| Vanadium                      | 51.7   | 10.0            | ug/g  | ND            | 88.6 | 70-130     |     |           |       |
| Zinc                          | 43.8   | 20.0            | ug/g  | ND            | 78.5 | 70-130     |     |           |       |
| <b>Pesticides, OC</b>         |        |                 |       |               |      |            |     |           |       |
| Aldrin                        | 0.29   | 0.01            | ug/g  | ND            | 128  | 50-140     |     |           |       |
| gamma-BHC (Lindane)           | 0.27   | 0.01            | ug/g  | ND            | 119  | 50-140     |     |           |       |
| alpha-Chlordane               | 0.27   | 0.01            | ug/g  | ND            | 117  | 50-140     |     |           |       |
| gamma-Chlordane               | 0.25   | 0.01            | ug/g  | ND            | 111  | 50-140     |     |           |       |
| o,p'-DDD                      | 0.24   | 0.01            | ug/g  | ND            | 104  | 50-140     |     |           |       |
| p,p'-DDD                      | 0.31   | 0.02            | ug/g  | ND            | 137  | 50-140     |     |           |       |
| o,p'-DDE                      | 0.27   | 0.01            | ug/g  | ND            | 119  | 50-140     |     |           |       |
| p,p'-DDE                      | 0.29   | 0.01            | ug/g  | ND            | 129  | 50-140     |     |           |       |
| o,p'-DDT                      | 0.30   | 0.01            | ug/g  | ND            | 132  | 50-140     |     |           |       |
| p,p'-DDT                      | 0.29   | 0.01            | ug/g  | ND            | 128  | 50-140     |     |           |       |
| Dieldrin                      | 0.31   | 0.02            | ug/g  | ND            | 135  | 50-140     |     |           |       |
| Endrin                        | 0.29   | 0.02            | ug/g  | ND            | 125  | 50-140     |     |           |       |
| Endosulfan I                  | 0.31   | 0.01            | ug/g  | ND            | 137  | 50-140     |     |           |       |
| Endosulfan II                 | 0.26   | 0.02            | ug/g  | ND            | 112  | 50-140     |     |           |       |
| Heptachlor                    | 0.30   | 0.01            | ug/g  | ND            | 133  | 50-140     |     |           |       |
| Heptachlor epoxide            | 0.27   | 0.01            | ug/g  | ND            | 118  | 50-140     |     |           |       |
| Hexachlorobenzene             | 0.28   | 0.01            | ug/g  | ND            | 123  | 50-140     |     |           |       |
| Hexachlorobutadiene           | 0.30   | 0.01            | ug/g  | ND            | 132  | 50-140     |     |           |       |
| Hexachloroethane              | 0.29   | 0.01            | ug/g  | ND            | 128  | 50-140     |     |           |       |
| Methoxychlor                  | 0.27   | 0.01            | ug/g  | ND            | 118  | 50-140     |     |           |       |
| Surrogate: Decachlorobiphenyl | 0.114  |                 | ug/g  |               | 100  | 50-140     |     |           |       |
| <b>Volatiles</b>              |        |                 |       |               |      |            |     |           |       |

Certificate of Analysis

Report Date: 16-Dec-2022

Client: Malroz Engineering Incorporated

Order Date: 9-Dec-2022

Client PO:

Project Description: 1703

### Method Quality Control: Spike

| Analyte               | Result | Reporting Limit | Units | Source Result | %REC | %REC Limit | RPD | RPD Limit | Notes |
|-----------------------|--------|-----------------|-------|---------------|------|------------|-----|-----------|-------|
| Benzene               | 4.22   | 0.02            | ug/g  | ND            | 106  | 60-130     |     |           |       |
| Ethylbenzene          | 4.12   | 0.05            | ug/g  | ND            | 103  | 60-130     |     |           |       |
| Toluene               | 4.25   | 0.05            | ug/g  | ND            | 106  | 60-130     |     |           |       |
| m,p-Xylenes           | 8.09   | 0.05            | ug/g  | ND            | 101  | 60-130     |     |           |       |
| o-Xylene              | 4.20   | 0.05            | ug/g  | ND            | 105  | 60-130     |     |           |       |
| Surrogate: Toluene-d8 | 8.11   |                 | ug/g  |               | 101  | 50-140     |     |           |       |

Certificate of Analysis

Client: Malroz Engineering Incorporated

Client PO:

Report Date: 16-Dec-2022

Order Date: 9-Dec-2022

Project Description: 1703

**Qualifier Notes:*****Login Qualifiers :***

Container and COC sample IDs don't match - CoC and jar label read BH103 S01. Jar lid reads BH103 S02.

*Applies to samples: BH103 S02*

Container and COC sample IDs don't match - CoC and jar label read BH103 S02. Jar lid reads BH103 S01.

*Applies to samples: BH103 S01***Sample Data Revisions**

None

**Work Order Revisions / Comments:**

None

**Other Report Notes:**

n/a: not applicable

ND: Not Detected

MDL: Method Detection Limit

Source Result: Data used as source for matrix and duplicate samples

%REC: Percent recovery.

RPD: Relative percent difference.

NC: Not Calculated

Soil results are reported on a dry weight basis when the units are denoted with 'dry'.

Where %Solids is reported, moisture loss includes the loss of volatile hydrocarbons.

***CCME PHC additional information:***

- The method for the analysis of PHCs complies with the Reference Method for the CWS PHC and is validated for use in the laboratory. All prescribed quality criteria identified in the method has been met.
- F1 range corrected for BTEX.
- F2 to F3 ranges corrected for appropriate PAHs where available.
- The gravimetric heavy hydrocarbons (F4G) are not to be added to C6 to C50 hydrocarbons.
- In the case where F4 and F4G are both reported, the greater of the two results is to be used for comparison to CWS PHC criteria.
- When reported, data for F4G has been processed using a silica gel cleanup.

Paracel ID: 2250463



Paracel Order Number  
(Lab Use Only)

2250463

Chain Of Custody  
(Lab Use Only)

Nº 139664

Client Name: MAHAR  
Contact Name: DAVID HOGANSON, NICOLA BELLISSIMO  
Address: 308 Wellington St, Kingston On.  
Telephone: \_\_\_\_\_

Project Ref: \_\_\_\_\_  
Quote #: \_\_\_\_\_  
PO #: 1703  
E-mail: Hogson@melro.com bellissimo@melro.com

Page 1 of 2

Turnaround Time

☐ 1 day ☐ 3 day  
☐ 2 day ☐ Regular

Date Required: \_\_\_\_\_

☐ REG 153/04 ☒ REG 406/19  
Other Regulation  
☐ Table 1 ☐ Res/Park ☐ Med/Fine  
☐ Table 2 ☐ Ind/Comm ☐ Coarse  
☐ Table 3 ☐ Agri/Other  
☐ Table 2.1 RPI  
For RSC: ☐ Yes ☐ No

Matrix Type: S (Soil/Sed.) GW (Ground Water)  
SW (Surface Water) SS (Storm/Sanitary Sewer)  
P (Paint) A (Air) O (Other)

Required Analysis

| Sample ID/Location Name | Matrix | Air Volume | # of Containers | Sample Taken |       | PHCs F1-F4+BTEX | VOCs | PAHs | Metals by ICP | Hg | CrVI | B (HWS) | EC | SAR | OC PESTICIDE | PH |
|-------------------------|--------|------------|-----------------|--------------|-------|-----------------|------|------|---------------|----|------|---------|----|-----|--------------|----|
|                         |        |            |                 | Date         | Time  |                 |      |      |               |    |      |         |    |     |              |    |
| 1 BH102 501             | S      |            | 2               | 7 Dec 2022   | 11:00 | X               |      |      | X             | X  | X    | X       | X  | X   |              |    |
| 2 BH103 501             |        |            |                 | "            | 10:00 | X               |      |      | X             | X  | X    | X       | X  | X   | X            |    |
| 3 BH103 502             |        |            |                 | "            | 10:10 | X               |      |      | X             | X  | X    | X       | X  | X   | X            |    |
| 4 BH106 501             |        |            |                 | 6 Dec "      | 9:00  | X               |      |      | X             | X  | X    | X       | X  | X   | X            |    |
| 5 BH110 501             |        |            |                 | "            | 13:00 | X               |      |      | X             | X  | X    | X       | X  | X   |              |    |
| 6 BH111 501             |        |            |                 | "            | 14:00 | X               |      |      | X             | X  | X    | X       | X  | X   |              |    |
| 7 BH112 501             |        |            |                 | 5 Dec "      | 15:00 | X               |      |      | X             | X  | X    | X       | X  | X   | X            |    |
| 8 BH117 501             |        |            |                 | "            | 9:00  | X               |      |      | X             | X  | X    | X       | X  | X   |              |    |
| 9 BH118 501             |        |            |                 | "            | 13:00 | X               |      |      | X             | X  | X    | X       | X  | X   |              |    |
| 10 BH121 501            |        |            |                 | 1 Dec 2022   | 14:00 | X               |      |      | X             | X  | X    | X       | X  | X   | X            |    |

Comments:

Method of Delivery:

Drop Box

Relinquished By (Sign): Nicola Bellissimo

Received By Driver/Depot:

Chris Payne

Received at Lab:

Catherine Wainman

Verified By:

Racem Cooper

Relinquished By (Print): NICOLA BELLISSIMO

Date/Time:

Dec 09 22 8:43

Date/Time:

Dec 10/22 14:03

Date/Time:

Dec 09 15:35

Date/Time: 8 Dec 2022

Temperature:

8.5 °C

Temperature:

9.1 °C

pH Verified: ☐

By:

Chain of Custody (Env) xlsx



2250463

Nº 139665

|               |  |              |  |                                                                                                                                                         |  |
|---------------|--|--------------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Client Name:  |  | Project Ref: |  | Page <u>2</u> of <u>2</u>                                                                                                                               |  |
| Contact Name: |  | Quote #:     |  | Turnaround Time<br><br><input type="checkbox"/> 1 day <input type="checkbox"/> 3 day<br><input type="checkbox"/> 2 day <input type="checkbox"/> Regular |  |
| Address:      |  | PO #:        |  |                                                                                                                                                         |  |
| Telephone:    |  | E-mail:      |  |                                                                                                                                                         |  |
|               |  |              |  | Date Required: _____                                                                                                                                    |  |

| <input type="checkbox"/> REG 153/04 <input checked="" type="checkbox"/> REG 406/19                                                                                                                                                                                                                                                                                                                                  |            | Other Regulation                                                                                                                                                                                                                                               |  | Matrix Type: S (Soil/Sed.) GW (Ground Water)<br>SW (Surface Water) SS (Storm/Sanitary Sewer)<br>P (Paint) A (Air) O (Other) |            | Required Analysis |              |       |                 |      |      |               |    |      |         |    |     |             |    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------------------------------------------------|------------|-------------------|--------------|-------|-----------------|------|------|---------------|----|------|---------|----|-----|-------------|----|
| <input type="checkbox"/> Table 1 <input type="checkbox"/> Res/Park <input type="checkbox"/> Med/Fine<br><input type="checkbox"/> Table 2 <input type="checkbox"/> Ind/Comm <input type="checkbox"/> Coarse<br><input type="checkbox"/> Table 3 <input type="checkbox"/> Agri/Other<br><input checked="" type="checkbox"/> Table <u>2.1 API</u><br>For RSC: <input type="checkbox"/> Yes <input type="checkbox"/> No |            | <input type="checkbox"/> REG 558 <input type="checkbox"/> PWQO<br><input type="checkbox"/> CCME <input type="checkbox"/> MISA<br><input type="checkbox"/> SU - Sani <input type="checkbox"/> SU - Storm<br>Mun: _____<br><input type="checkbox"/> Other: _____ |  |                                                                                                                             |            |                   |              |       |                 |      |      |               |    |      |         |    |     |             |    |
| Sample ID/Location Name                                                                                                                                                                                                                                                                                                                                                                                             |            |                                                                                                                                                                                                                                                                |  | Matrix                                                                                                                      | Air Volume | # of Containers   | Sample Taken |       | PHCs F1-F4+BTEX | VOCs | PAHs | Metals by ICP | Hg | CrVI | B (HWS) | EC | SAR | OC postcode | PH |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                   | BH 126 S01 |                                                                                                                                                                                                                                                                |  | S                                                                                                                           |            | 2                 | 1 Dec 2022   | 10:00 | X               |      |      | X             | X  | X    | X       | X  | X   |             |    |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                   |            |                                                                                                                                                                                                                                                                |  |                                                                                                                             |            |                   |              |       |                 |      |      |               |    |      |         |    |     |             |    |
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| 6                                                                                                                                                                                                                                                                                                                                                                                                                   |            |                                                                                                                                                                                                                                                                |  |                                                                                                                             |            |                   |              |       |                 |      |      |               |    |      |         |    |     |             |    |
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| 8                                                                                                                                                                                                                                                                                                                                                                                                                   |            |                                                                                                                                                                                                                                                                |  |                                                                                                                             |            |                   |              |       |                 |      |      |               |    |      |         |    |     |             |    |
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| Comments:                |                           | Method of Delivery: |                                           |
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| Relinquished By (Print): | Date/Time:                | Date/Time:          | Date/Time:                                |
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**Appendix E**  
**Geophysical Investigation Report**



# FRONTWAVE G E O P H Y S I C S

February 3, 2023

File No. F-23095

Dylan Hill, P.Eng.  
Geotechnical Engineer  
**Malroz Engineering Inc.**  
308 Wellington Street, 2<sup>nd</sup> Floor  
Kingston, Ontario  
K7K 7A8

Email: [Hill@malroz.com](mailto:Hill@malroz.com)

**Re: Shear wave velocity testing for seismic site classification at 175 Talbot Street, Picton, Ontario.**

Dear Mr. Hill:

Frontwave Geophysics Inc. was retained by Malroz Engineering Inc. to carry out a geophysical investigation at the site of the proposed development at 175 Talbot Street in Picton, Ontario.

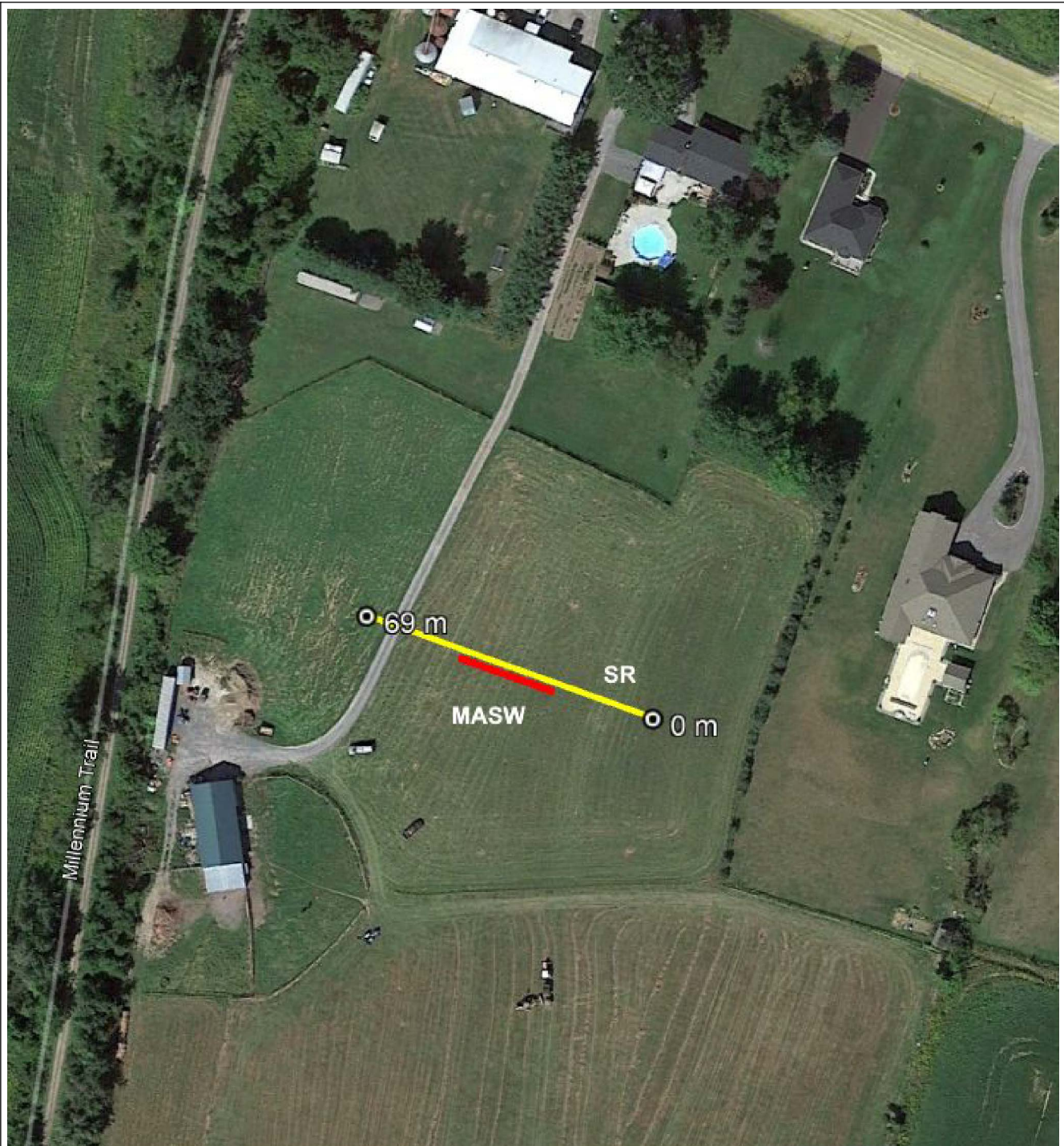
The objective of the survey was to determine site class for seismic site response based on the average shear wave velocity value measured over the upper 30 m ( $V_{s30}$ ). The seismic refraction and multi-channel analysis of surface waves (MASW) methods were used to obtain shear wave velocity profile.

The fieldwork was conducted on February 2<sup>nd</sup>, 2023. The location of the seismic survey line is shown in Figure 1. The line coordinates are provided in Table 1.

**Table 1.** UTM coordinates of the MASW survey line.

|                     | Easting  | Northing  |
|---------------------|----------|-----------|
| Start of Line (0 m) | 326760.2 | 4874765.7 |
| End of Line (69 m)  | 326696.4 | 4874790.4 |

This report describes basic principles of the seismic refraction and MASW methods, survey design, interpretation method, and presents the results of the investigation in chart and table format.



### Legend



Location of 23 m, 24-geophone MASW spread

Location of 69 m, 24-geophone seismic refraction spread

Image: Google Earth 2015

Date: 2023-02-03

File No: F-23095



Title: Survey Location Plan

Location: 175 Talbot St,  
Picton, ON

Figure:  
**1**

## MASW Survey

### Overview

The Multi-channel Analysis of Surface Waves (MASW) is a seismic method widely applied to produce shear wave velocity ( $V_s$ ) profiles. It is based on the dispersive nature of Rayleigh or Love surface waves in layered media. Surface waves with longer wavelengths propagate deeper in the subsurface, hence, their phase velocity is more influenced by the elastic properties of deeper layers. The velocity of surface waves depends mainly on the shear wave velocity of the medium. Distribution of surface waves phase velocities as a function of wavelength (or frequency) can be visualized as a dispersion curve. The inverse problem is then solved by modelling the experimental data with a theoretical dispersion curve; the model parameters are typically limited to layer thickness and shear wave velocity with an assumption of horizontally layered strata. As a result of the inversion, a shear wave velocity depth profile is obtained. Figure 2 illustrates the overall procedure of the MASW method.

### Survey Design

The acquisition layout consisted of 24 receivers in a linear array (spread), connected with two 12-channel cables to P.A.S.I. Gea-24 seismograph. 4.5 Hz natural frequency vertical geophones were used for this survey. The measurements were conducted with a spread length of 23 m (1 m spacing between geophones).

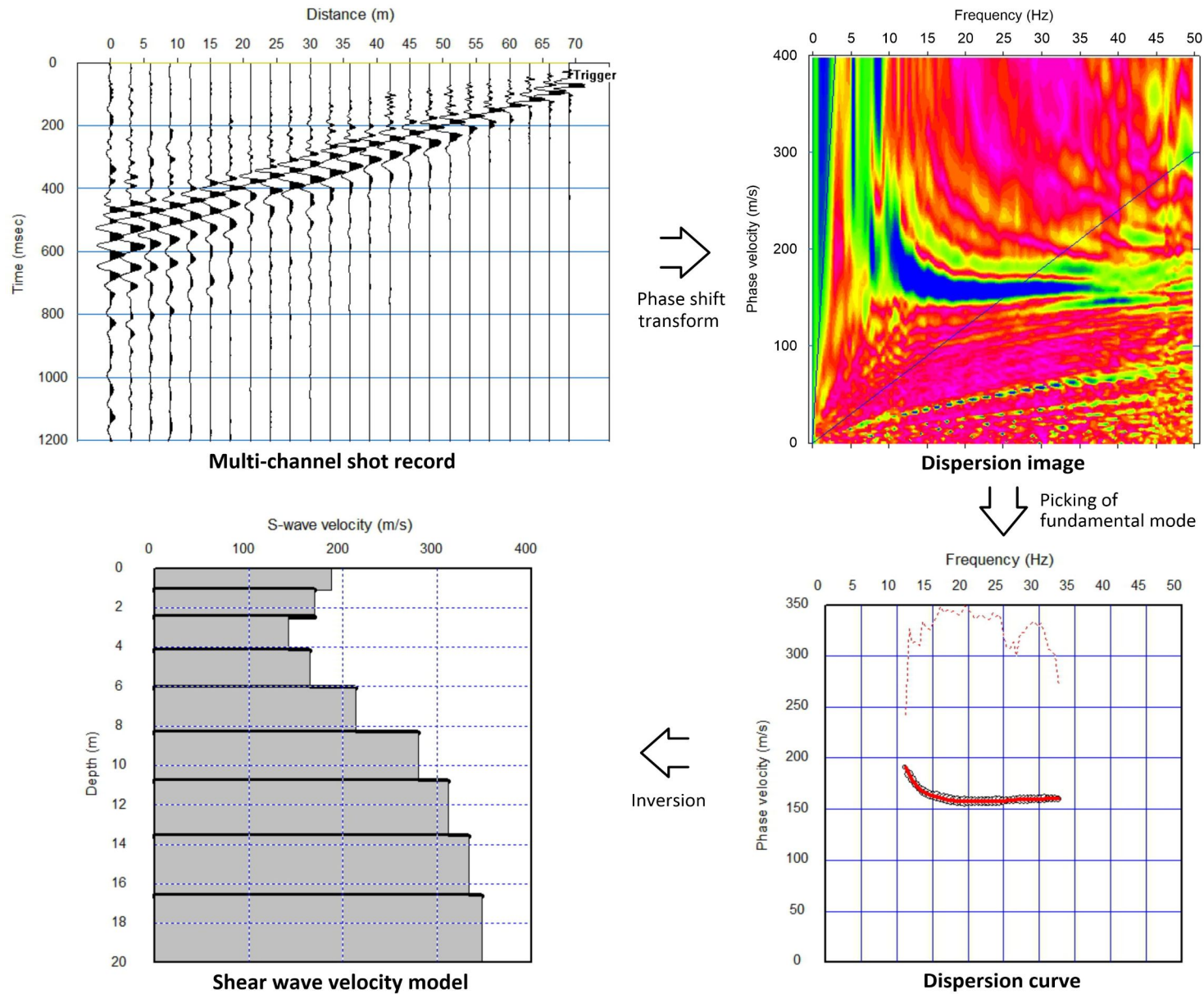
8-kg sledgehammer was used as an energy source. Shots were executed at five locations: two shots close to the ends of the spread, one shot in the middle, and two shots with an offset of 23 m from the ends of the spread. The record length was set to 1500 ms with a 0.1 ms sampling interval.

### Interpretation

A dispersion curve is obtained from each field record by converting the shot gather into a dispersion image and then identifying and picking the fundamental mode. A shear wave velocity profile is obtained through inversion of the dispersion curve by modelling the subsurface as a horizontally layered medium with the model parameters limited to the number of layers, their thickness and shear-wave velocity.

ZondST2D software package was used for processing, picking and inversion of the MASW data.

Some variability among the dispersion curves and resulting models obtained from different shot records is always observed due to lateral velocity variations, near and far field effects, different signal-to-noise ratio, etc. Combining independent inversion results from multiple shot records improves the estimation of the actual shear wave velocity and provides an assessment of uncertainty. The results of the interpretation are presented in the form of the average shear wave velocity profile; the observed variability of the MASW data is reported as upper and lower bound velocity profiles.



**Figure 2.** The procedure of MASW data processing using the SeisImager SW software package.

### Accuracy of the results

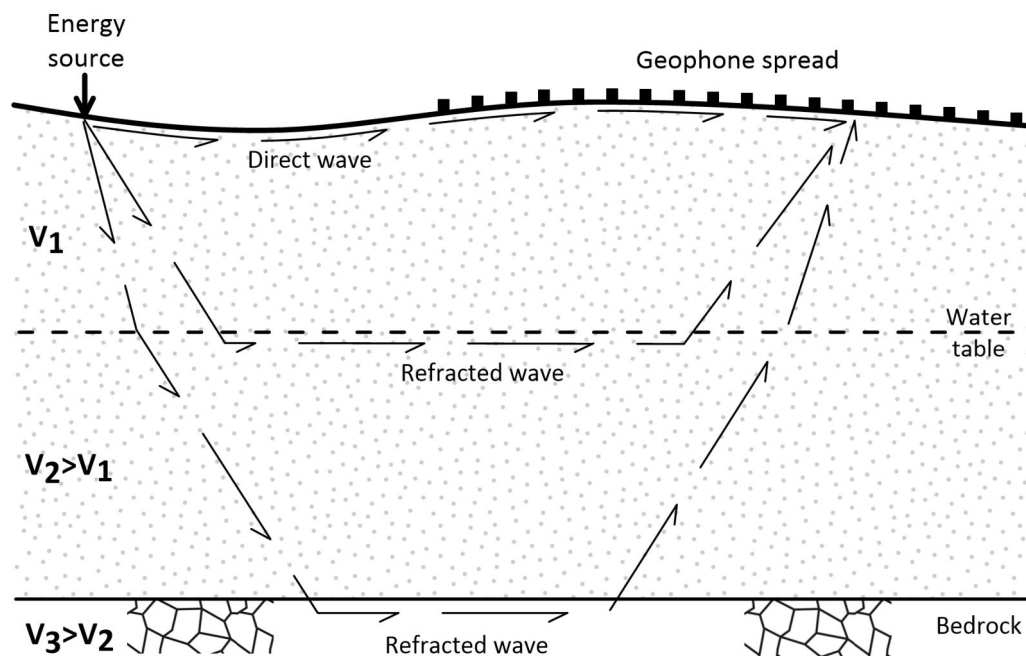
The accuracy of MASW generally depends on the complexity of the subsurface and specific site conditions (noise levels, topography, etc.). Lateral velocity variations and steeper bedrock topography increase the dispersion uncertainty. The presence of high velocity contrast layers such as bedrock will require the use of a-priori information to optimize model parameters for more accurate results. Hence, if the a-priori information is not available (e.g., when the data are overly noisy to carry out refraction analysis), the accuracy decreases.

At bedrock sites or at sites with very shallow overburden overlying bedrock the MASW method performs poorly. Very strong velocity contrast between layers at shallow depths results in superposition of fundamental and higher Rayleigh wave modes that often cannot be distinguished. At sites where the thickness of the overburden is sufficient to obtain a coherent dispersion, the inversion would significantly underestimate the S-wave velocity within rock.

## Seismic Refraction Survey

### Overview

The seismic refraction method is based on the measurement of arrival times of seismic waves refracted at interfaces between geological layers. The method allows to obtain a velocity depth model and to map interfaces between layers with significant velocity contrast such as water table and bedrock surface. Compressional (P) wave or shear (S) wave refracted arrivals can be recorded using vertically or horizontally oriented sensors and sources, respectively. Figure 3 is a schematic of a simplified seismic model showing the basic principle of the refraction method.



**Figure 3.** Seismic model showing the basic principle of refraction method.

### Survey Design

The acquisition layout consisted of 24 receivers in a linear array (spread), connected with two 12-channel cables to P.A.S.I. Gea-24 seismograph. 10 Hz natural frequency horizontal geophones were used for this survey. The measurements were conducted with a spread length of 69 m (3 m spacing between geophones).

5-kg sledgehammer was used as an energy source. Shots were executed at four locations per spread: two shots at the ends of the spread and two shots with an offset of 20 m from the ends of the spread. Preferential S-wave energy was generated by horizontally striking a metal bar in a direction perpendicular to the survey line. Shots in two opposite directions were recorded at each shot location to record S-wave arrivals of opposite polarity. The record length was set to 1000 ms with a 0.05 ms sampling interval.

### Interpretation

The reciprocal (plus-minus) method was used for the interpretation of the seismic refraction data. The method assumes the subsurface as a series of discrete layers (refractors) with simple velocity distributions. It allows to calculate the depth and velocity of a continuous undulating refractor, providing the target layer is of sufficient thickness and the dip angles are moderate.

ZondST2D software package was used for processing of the refraction data. The processing involved stacking of shot records obtained with opposite source directions, identification and picking of S-wave first arrivals.

### Accuracy of the results

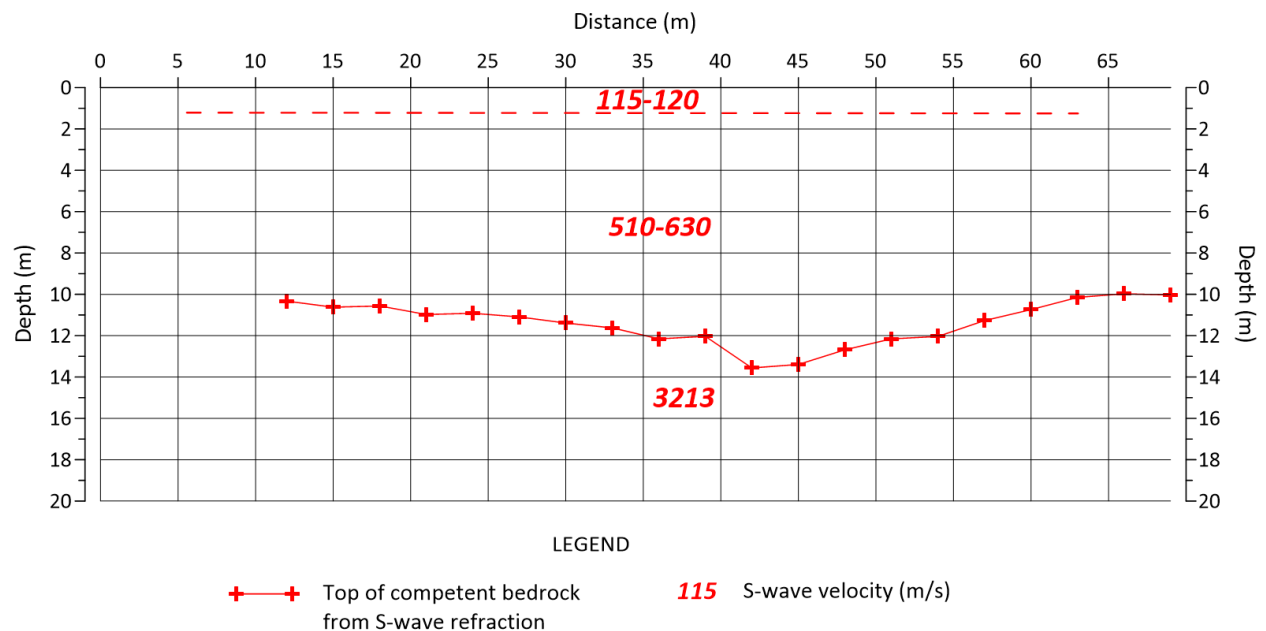
Typically, the depth to bedrock can be determined to within  $\pm 10\%$  of the true depth. Larger errors are observed when bedrock depth is in the order of only several meters. The accuracy of bedrock velocity determinations is estimated to be within 5%.

## RESULTS

The quality of seismic records was good; first arrivals of refracted waves were well defined. Examples of the S-wave refraction shot record and MASW dispersion curve obtained at this site are presented in Figure 6 at the end of this report.

The results of the interpretation of refraction data are presented in Figure 4 in the form of a layered S-wave velocity model. The velocity model consists of three layers. The top layer, with an apparent velocity range of 115 to 120 m/s, represents soft soils. The second layer, with measured apparent velocity range of 510 to 630 m/s, represents stiff soils. The third layer, with an S-wave velocity of 3213 m/s, represents limestone bedrock. The interpreted depth to competent bedrock ranged from approximately 10 m to 13 m below ground surface.

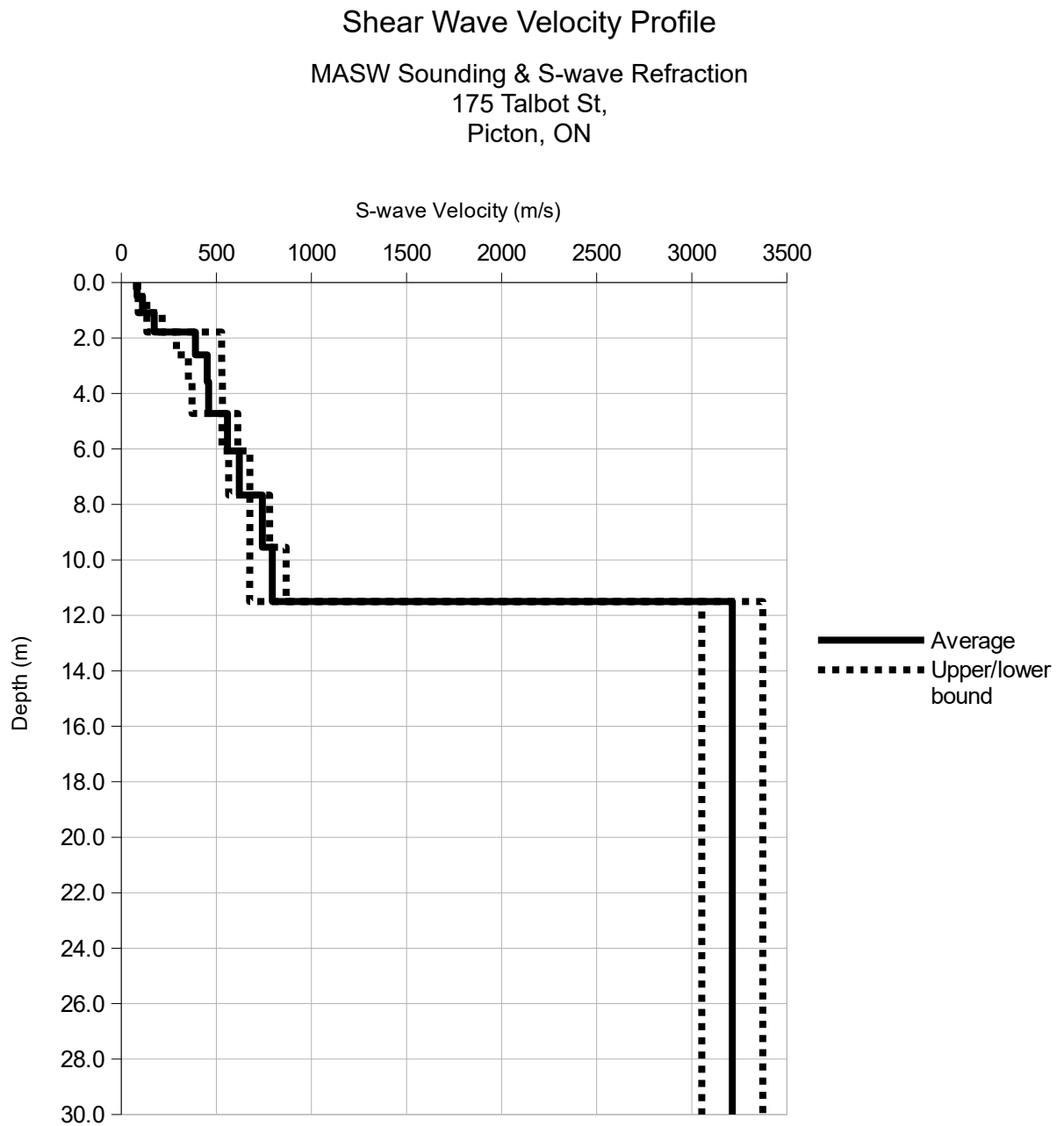
Refraction analysis performed on the MASW data set produced a compressional (P) wave velocity of 4120 m/s for rock. This velocity is lower than could be expected based on the measured S-wave velocity, which probably indicates the presence of a layer of weathered limestone. The top of the weathered rock could be approximately 8 m deep.



**Figure 4.** Interpreted S-wave velocity model showing the depth to competent bedrock.

Refraction data were used for parameterization of the initial MASW inversion model. The measured shear wave velocity for the bedrock is representative of the top of the rock. According to Commentary J (Paragraph 96) of the National Building Code of Canada 2015 (NBC), the measured value may be extrapolated if the rock conditions are known to be continuous to a depth of 30 m.

The resulting shear wave velocity depth profile is presented in Figure 5.



**Figure 5.** Shear wave velocity profile from MASW sounding and S-wave refraction.

The tabulated shear wave velocity model is presented in Table 2.

**Table 2.** Shear wave velocities from MASW sounding and S-wave refraction.

| Depth Interval (m) |      | S-wave Velocity<br>(m/s) |
|--------------------|------|--------------------------|
| From               | To   |                          |
| 0.0                | 0.5  | 83                       |
| 0.5                | 1.1  | 112                      |
| 1.1                | 1.8  | 173                      |
| 1.8                | 2.6  | 390                      |
| 2.6                | 3.6  | 452                      |
| 3.6                | 4.6  | 459                      |
| 4.6                | 6.1  | 559                      |
| 6.1                | 7.7  | 621                      |
| 7.7                | 9.5  | 742                      |
| 9.5                | 11.5 | 794                      |
| 11.5               | 30.0 | 3213                     |

The average shear wave velocity within the upper 30 meters ( $V_{s30}$ ) is defined as the travel-time weighted average velocity from surface to a depth of 30 m and calculated using the following formula:

$$V_{s30} = 30 / \Sigma (d/V_s),$$

where  $d$  is the thickness of any layer and  $V_s$  is the layer S-wave velocity. In other words,  $V_{s30}$  is calculated as 30 m divided by the sum of the S-wave travel times for each layer within the topmost 30 m.

The calculated  $V_{s30}$  values are presented in Table 3.

**Table 3.**  $V_{s30}$  values from MASW sounding and S-wave refraction.

| Depth Range<br>(m) | Minimum $V_{s30}$<br>(m/s) | Average $V_{s30}$<br>(m/s) | Maximum $V_{s30}$<br>(m/s) | NBC 2015<br>Seismic Site Class |
|--------------------|----------------------------|----------------------------|----------------------------|--------------------------------|
| 0 to 30            | 685                        | <b>794</b>                 | 892                        | <b>C / B</b>                   |

The  $V_{s30}$  values obtained from the S-wave sounding varied from 685 m/s to 892 m/s with an average of 794 m/s.

Based on the Site Classification for Seismic Site Response (Table 4.1.8.4.-A) of the National Building Code of Canada 2015 (NBC), the average  $V_{s30}$  value marginally falls in the Site Class B range ( $760 < V_{s30} \leq 1500$  m/s).

According to the requirements of Table 4.1.8.4.-A of NBC 2015, site classes A and B are not to be used if there is more than 3 m of soil between the rock surface and the bottom of the spread footing, pile cap or mat foundation of the building. If there is more than 3 m of soil between the building foundation and bedrock surface, the investigated site must be assigned Site Class “C”.

Taking into consideration the expected presence of more than 3 m of overburden, **the applicable Site Class is C.**

We hope you find this report satisfactory. Should you have any questions or require additional information, please do not hesitate to contact the undersigned.

**Frontwave Geophysics Inc.**



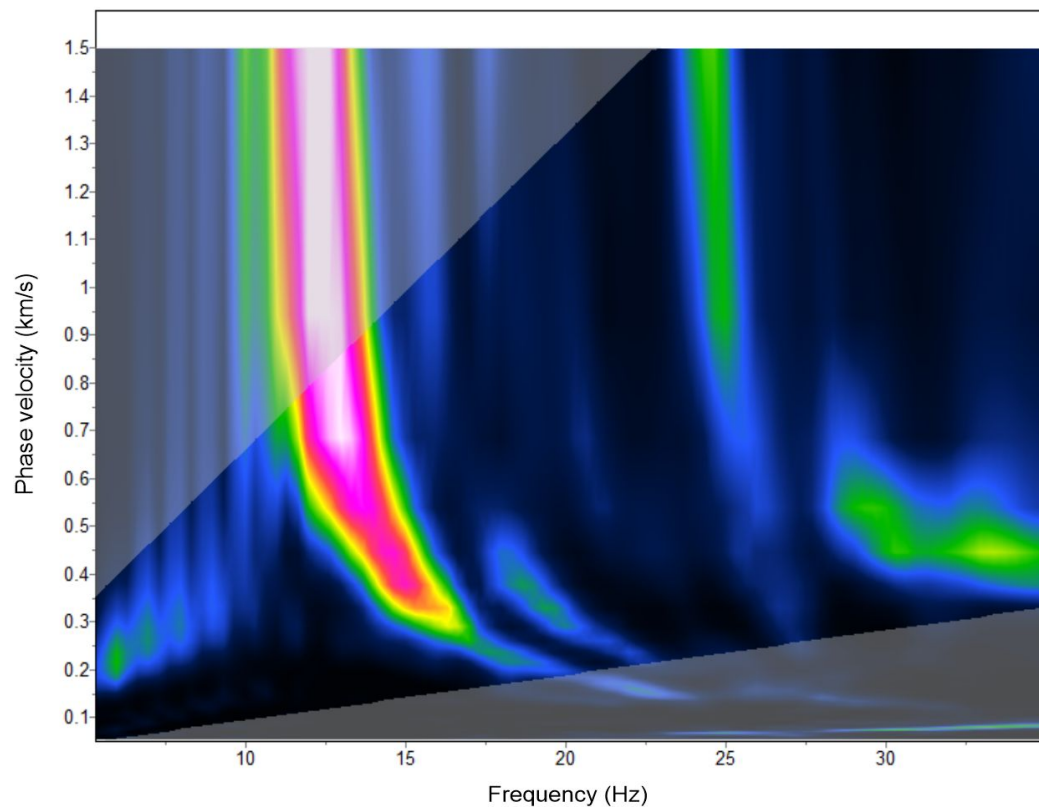
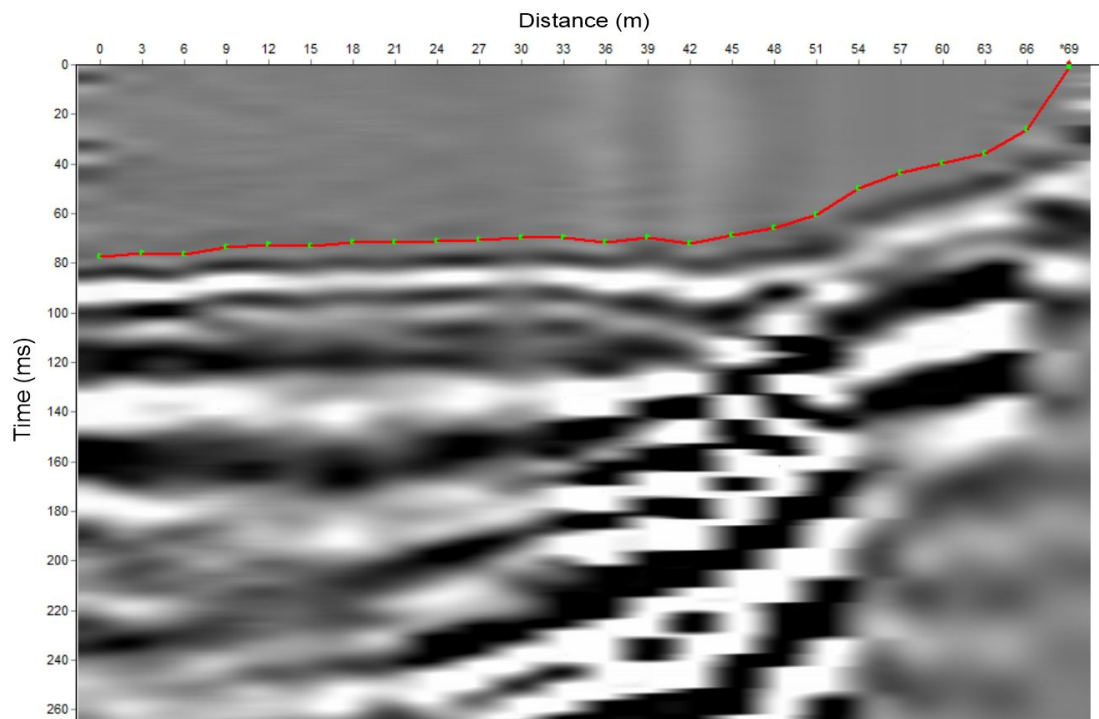
**Ilia Gusakov, P.Geo.**

Geophysicist

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**Figure 6.** Data examples displaying a stacked S-wave refraction shot record (top) and MASW dispersion curve (bottom).